

Immunoglobulin Molecules

Immunoglobulins are the effector molecules of the immune system. They occur either as membrane-bound cell surface receptors or as free antibodies. The enormous diversity of their variable regions enables them to bind many very different antigens. Although they differ in details of their function, they share a relatively simple basic pattern, which is derived from a common ancestral molecule (p. 312).

A. Immunoglobulin G (IgG)

Immunoglobulin G is the prototype of antibody molecules produced by derivatives of B lymphocytes, the plasma cells. The molecule has two H chains and two L chains, held together by disulfide bonds. Each H chain has three constant regions (C_H1 , C_H2 , and C_H3) and one variable region (V_H). Each H chain has a total of 446 amino acids, of which the first 109 belong to the variable region at the N-terminal end. Each L chain has one variable (V_L) and one constant (C_L) domain and consists of 214 amino acids. In the L chains also, the first 109 amino acids form the variable region. The variable domains of the H chains and the L chains form the antigen-binding sites. The three hypervariable regions within the V region of each chain are also called complementarity determining regions (CDR) because the actual physical contact of molecules based on their complementary structure occurs in these regions. Each domain consists of about 110 amino acid residues. A jointlike area (hinge) between constant region 1 (C_H1) and constant region 2 (C_H2) of the heavy chain allows some flexibility of the molecule. The H chains are bound to each other and the H to the L chains by disulfide bridges ($-S-S-$). Furthermore, there are disulfide bridges within the constant and variable domains. The L chains are of one of two types, κ or λ . In addition to immunoglobulin G, there are other types of immunoglobulins, which differ from each other in the constant part of the H chain: IgA (C_α), IgD (C_δ), and IgE (C_ϵ). A very large immunoglobulin, IgM, is made up of five IgG subunits. The different types of H chains are referred to as isotypes.

B. T-cell receptor (TCR)

Antigen receptors on the surfaces of T cells are heterodimers of covalently bound polypeptide chains, one α and one β chain. The basic struc-

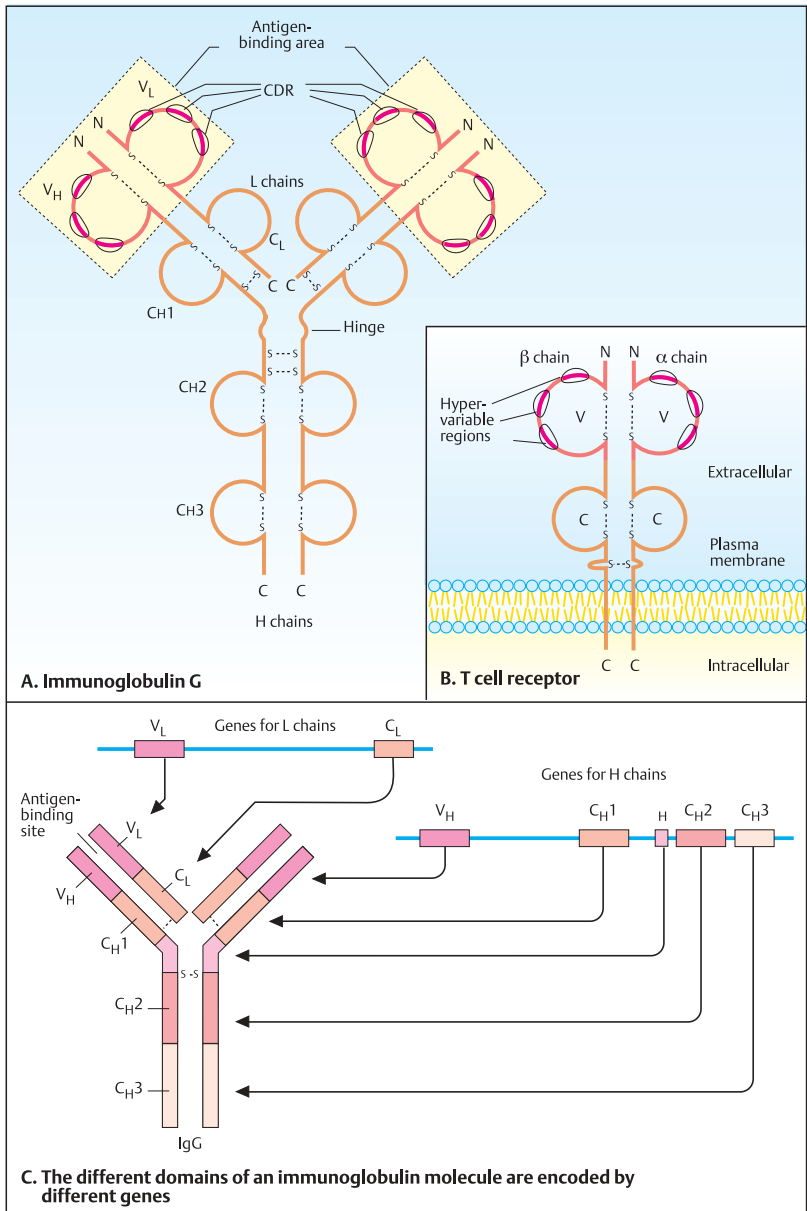
ture of a cell surface antigen receptor is similar to that of the secreted immunoglobulins except that the cell surface receptors contain just one constant region. The β chain is the slightly larger chain. The constant regions of the α and β chains (C) each consist of 140–180 amino acids. The variable regions (V) consist of 102–109 amino acids and contain three hypervariable regions, like the immunoglobulin molecules. In addition to those for α and β , genes for a T-cell-receptor γ and δ chain exist. The loci for the TCR α , γ , and δ chains in man lie together on chromosome 14; the locus for TCR β lies on chromosome 13.

C. The different domains of an immunoglobulin molecule are encoded by different genes

Each immunoglobulin and receptor molecule is encoded by different DNA sequences, which belong to a large series of genes for the L chains (types κ and λ) and the H chain. Genes for the H chain are located on chromosome 14q32 in humans and chromosome 14 in mice. The genes for the κ light chain are located on the short arm of chromosome 2 (2p12) in humans and chromosome 6 in mice, and for the λ light chain on the long arm of chromosome 22 (22q11) in humans and on chromosome 16 in mice. The genes are rearranged in the developing B and T cells in a pattern that is different and specific in each cell, as seen in the next plate.

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Genetic Diversity Generated by Somatic Recombination

If each of the many different immunoglobulin molecules with their variable regions were coded for by separate genes, many millions of genes would be required. This is not the case. Rather, during lymphocyte differentiation, a practically unlimited number of different cells are produced by recombination of a large but limited number of genes. This occurs by somatic recombination of genes during the differentiation of B cells and T cells. Antibody diversity arises by the following genetic mechanisms: (1) multiple DNA sequences of the germ-line genes for H and L chains code for Ig molecules with different specificities; (2) somatic recombination of the various DNA segments greatly increases the number of possible combinations; (3) somatic mutations occur in the hypervariable regions, leading to further genetic differences.

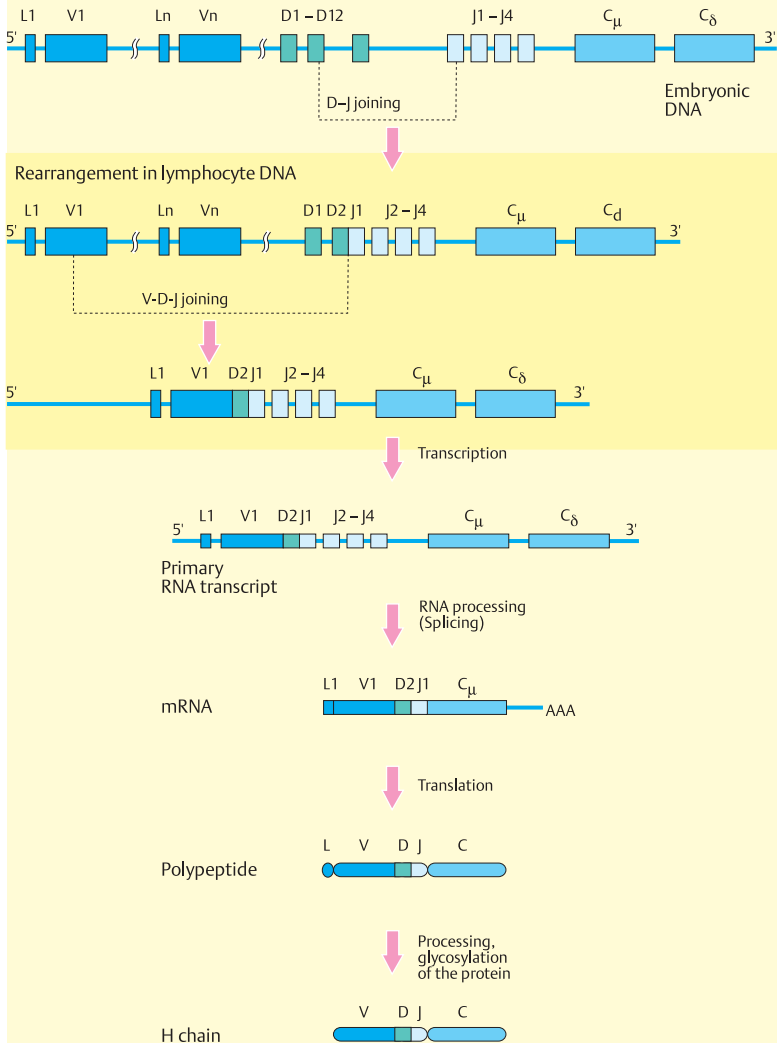
A. Somatic recombination during the formation of lymphocytes

Somatic recombination occurs within the genes for the L and the H chains during the maturation of B cells and within the genes for the four T-cell receptor chains in the maturing T cells. By means of this process, different coding DNA sequences (exons) of the various domains of the particular Ig molecule are in unique combinations in each cell. This provides each molecule with an antigen-binding specificity that differs from that of all other cells. Here, an example of the genetic processes during formation of an immunoglobulin H chain is shown. The exons (V1 to Vn) of the variable region (V) lie at the 5' end of the IgH locus. They are separated from each other by different lengths of noncoding DNA. A small exon (60–90 bp) that codes for a signal to initiate translation (leader or signal peptide L) lies more than 90 base pairs (bp) in the 5' direction of the V-region exons. Signal peptides guide the growing polypeptide into the lumen of the endoplasmic reticulum before they are cleaved off. The D genes of the constant region (C) lie at different distances in the 3' direction from the V genes. Each C segment consists of different exons, corresponding to the domains of the complete C region and different isotypes (C_{μ} , C_{δ}), etc.

First, a D gene segment and a J segment are joined (D–J joining). Next, this D–J segment is joined to one of the V segments (V–D–J joining). The combination of V, D, and J segments is different in each cell, with at least 25000 different possible combinations due to the number of different segments present (100–125 V segments, 12 D segments, and 4 J segments). The joined VDJ segments form the primary RNA transcript. At this stage, noncoding segments (introns) are still present. In the example shown, a D2 is joined to a J1, but not to J2–4. J segments not directly connected to a D segment (here, J2–4) are removed. After the RNA is processed by splicing, the mature messenger RNA (mRNA) is formed as template for the translation of an H chain polypeptide. By further processing, such as removal of the leader segment (L) and glycosylation of the protein at certain sites, the definitive H chain is finally produced (of type μ in the example shown). Unlike the H chains, the L chains have no diversity (D) genes, so that a J and a V gene are directly joined by somatic recombination during DNA rearrangement in the lymphocytes. Thus, rearrangement of the genes for the H chain and two types of L chains in the lymphocyte DNA leads to a new combination of genes in each cell.

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A. Somatic recombination during synthesis of immunoglobulin molecules

Mechanisms in Immunoglobulin Gene Rearrangement

The rearrangement of genes for immunoglobulin molecules in immature B cells and for the T cell receptor in immature T cells involves an excision mechanism. This requires precise recognition to ensure formation of the correct coding information. Noncoding DNA between genes for different regions of the molecule is excised and the remaining DNA is subsequently rejoined (ligation). Unlike recombination during meiosis, IgG rearrangements represent a special type of recombination, since nonhomologous DNA sequences are recombined.

A. DNA recognition sequences for recombination

Recombination between the DNA segments coding for an immunoglobulin molecule is mediated by a system of enzymes called recombinases. Their activities are varied and comprise lymphocyte-specific and more general activities including exonuclease and ligase functions. The enzymes are controlled by specific DNA recognition sequences. These are located in adjacent noncoding DNA segments at the 3' end of each V exon (exons for the variable region) and at the 5' end of each J segment. The D segments are flanked on both sides by recognition sequences. Recognition sequences are noncoding but highly conserved DNA segments of seven base pairs (heptamer) or nine base pairs (nonamer). They are separated by precisely defined intervals, produced by spacers of 23 or 12 base pairs (bp). Upstream (5' direction) and downstream (3' direction) from a D segment, the spacers are 12 base pairs long. The sequences of the nucleotide base pairs of the spacers are not conserved. The characteristic rearrangement between neighboring signal sequences for Ig molecules and TCR receptors requires spacers of different lengths, i.e., 12 and 23 base pairs (so-called 12/23 rule). When an H chain is formed, nonhomologous pairing of the heptamer of a D segment and of a J segment occurs. These D and J segments are then joined (D–J joining) by means of recombination: the spacer of 12 or 23 base pairs and all of the intervening DNA forms a loop. This is excised, and the D and J segments are joined. By pairing and recombination of the recognition sequences at

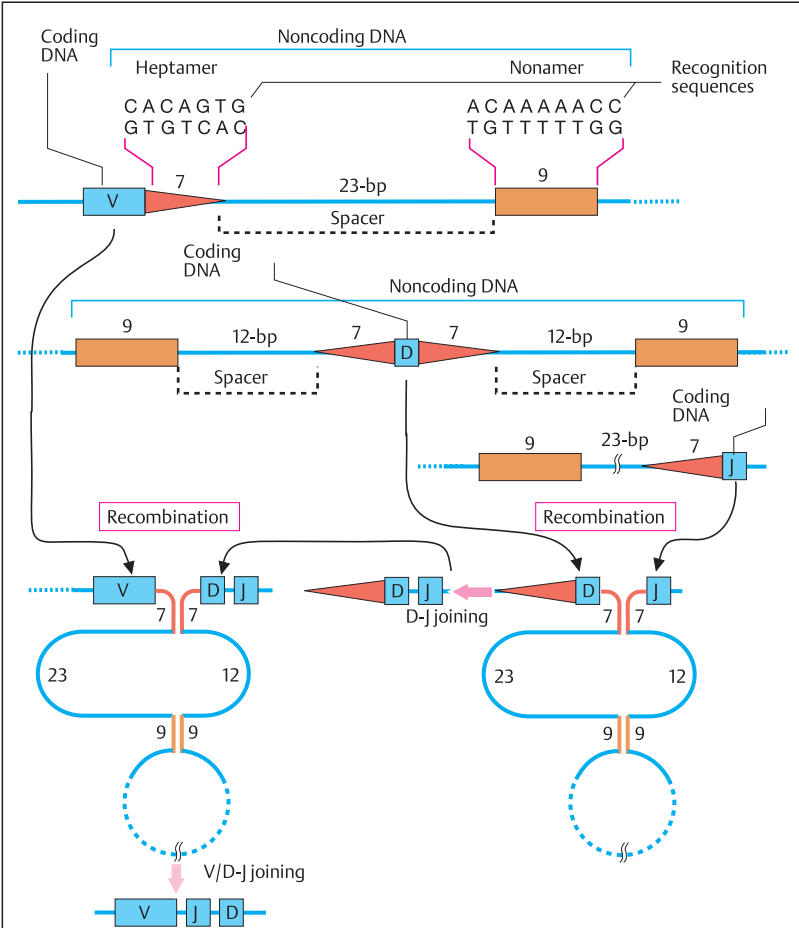
the 5' end of a DJ segment and the recognition sequence at the 3' end of a V gene, a V segment is joined to the DJ segment. The recombination of genes coding for T-cell receptors (see p. 308) proceeds in a similar manner. Two genes, recombination-activating genes 1 and 2 (*RAG-1* and *RAG-2*) have been identified as stimulating Ig gene recombination in pre-B cells and immature T cells. Mutations in these genes cause severe combined immune deficiency (see p. 314).

B. Genetic diversity in immunoglobulin and T-cell receptor genes

The total diversity, about 10^{18} possible combinations for all types of genes for immunoglobulins and T-cell receptors, is the result of different mechanisms. To begin with, different numbers of variable DNA segments are available for different chains (250–1000 for the H chain, 250 for the L chains, 75 for the α chain of the T-cell receptor TCR α , etc.). The different D and J segments also multiply the number of possible combinations. Finally, DNA sequence changes (somatic mutations) occur regularly in the hypervariable regions, further increasing the total number of possible combinations.

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A. DNA recognition sequences for recombination

Mechanism	Immunoglobulin		TCRαβ		TCRγδ	
	H chain	L chains	α	β	γ	δ
Variable domain	250 – 1000	250	75	25	7	10
Number of D segments	12	0	0	2	0	2
Number of J segments	4	4	50	12	2	2
Variable segment combinations	65 000 – 250 000		1825		70	
Total diversity	10 ¹¹		10 ¹⁶		10 ¹⁸	

B. Genetic diversity in immunoglobulin and T-cell receptor genes

Genes of the MHC Region

The MHC (major histocompatibility complex) region is a region of highly polymorphic genes (about 10–50 alleles per locus). It spans about 3500 kb on the short arm of chromosome 6 in humans and on chromosome 17 in mice. Collectively, these genes are called immune response (Ir) genes. They are expressed on the surface of various cells. Their products can be demonstrated serologically or by cellular reactions in a mixed lymphocyte test. MHC genes control the immune response to antigen proteins by specific binding to T cells.

A. Basic structure of the MHC gene complex in humans and in mice

The gene loci of the MHC region are grouped into three classes (I–III). Class I in humans includes HLA-A, HLA-B, and HLA-C; in mice, D, L, and K of the H2 system. Many other loci also belong to this class, such as HLA-E to -J. Class II includes HLA-DP, -DQ, and -DR in humans and I-A, and I-E (the letter I, not the Roman numeral) in mice. Alleles are designated according to a numerical system, e.g., HLA-A2, -B5, -DR4, etc. The gene products of the alleles of the HLA system (human leukocyte antigens; also said by some to refer to Los Angeles, where some of the first basic discoveries were made) can be demonstrated by the toxicity of serum of defined specificities to other leukocytes (serological cytotoxicity). Cytolysis occurs unless the specificities of the serum and of the cells being tested are identical. The gene products of the alleles of the HLA-D system are distinguished by the mixed lymphocyte test, based on lymphocyte proliferation as a reaction to foreign T cells. Strictly speaking, the class III genes do not belong to the MHC loci. They contain genes for different complement proteins and a few other genes.

B. Genomic organization of the MHC loci

The class II loci are located closest to the centromere. Each consists of a series of genomic subunits. The HLA-D region is about 900 kb long and contains a far greater number of loci than shown here. In addition, some have been renamed. For example HLA-DP has two α and two β genes, now called A1, A2 and B1, B2, respectively. The class III loci contain genes for com-

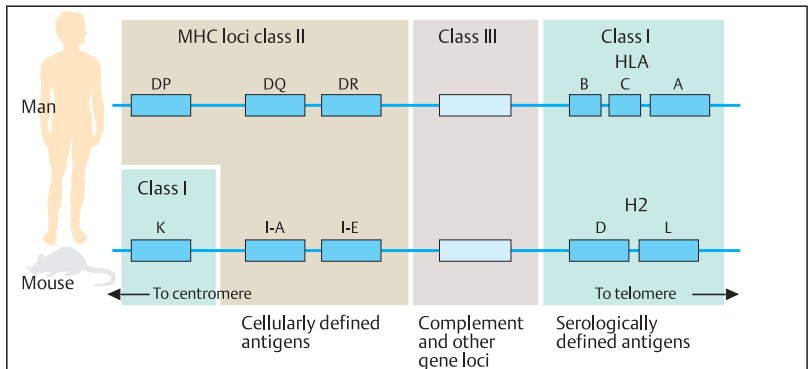
plement factors C2 and C4, steroid 21-hydroxylase, and cytokines (tumor necrosis factor, TNFA; and lymphotoxin, LT, new designation TNFB). Several other class III genes are located here.

C. Class I and class II MHC molecules

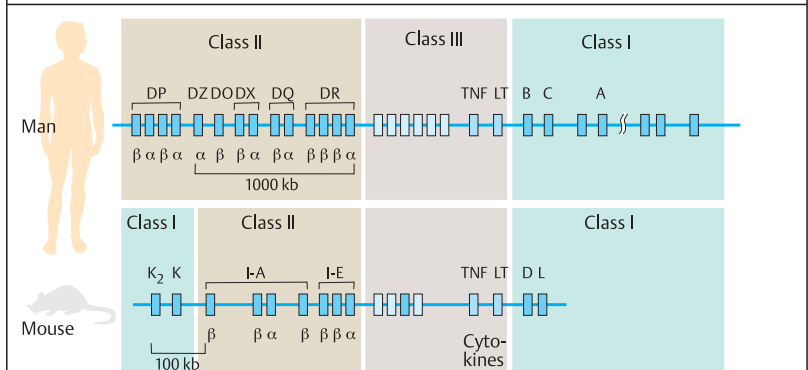
Corresponding to the general organization of the individual class I and class II genes, the class I and class II molecules are basically similar. Both consist of two different polypeptide chains. In class I molecules, an MHC-coded α chain is associated with a non-MHC-coded β chain (β_2 -microglobulin). The extracellular portion of the α chain consists of three domains, $\alpha 3$, $\alpha 2$, and $\alpha 1$, each with about 90 amino acids. $\alpha 1$ and $\alpha 2$ form the highly polymorphic peptide-binding region; $\alpha 3$ and β_2 -microglobulin structurally correspond to an immunoglobulin-like region. Elucidation of the crystalline structure of the class I MHC molecules showed that $\alpha 1$ and $\alpha 2$ interact to form a type of platform of eight-stranded β -folded proteins. The cleft formed between $\alpha 1$ and $\alpha 2$ ($25 \text{ \AA} \times 10 \text{ \AA} \times 11 \text{ \AA}$) can bind a protein fragment consisting of 10–20 amino acids. Class II MHC molecules consist of two polypeptide chains, α and β , each with two domains, i.e., $\alpha 1$, $\alpha 2$ and $\beta 1$, $\beta 2$, each with about 90 amino acids and a transmembrane region of about 25 amino acids. As with the class I molecules, the peptide-binding regions ($\alpha 1$ and $\beta 1$) are highly polymorphic. Unlike the $\beta 1$ domain, $\alpha 1$ does not contain a disulfide bridge.

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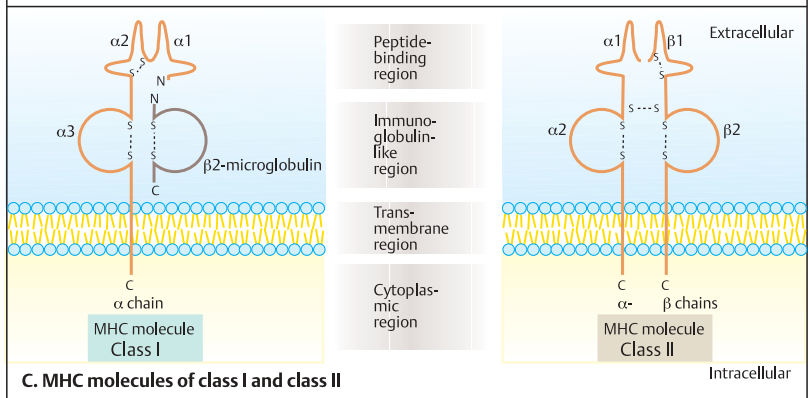
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A. Basic structure of the MHC gene complex in man and mouse



B. Genomic organization of the MHC loci



C. MHC molecules of class I and class II

T-Cell Receptors

The cell surface of a T lymphocyte contains receptor molecules that with high specificity recognize foreign antigens and cell surface molecules of the MHC complex. The T-cell antigen receptor (TCR) consists of a complex of several integral plasma membrane proteins. Unlike B cells, T cells recognize only fragments of foreign antigen proteins. In addition, they bind physically to the MHC complex of an antigen-presenting cell. During maturation of the T cells in the thymus, T-cell gene segments are rearranged in a defined order by somatic recombination, similarly to the formation of immunoglobulins.

A. T-cell receptor genes (TCR) of man

In the germ line, the genes for the β chain of the T-cell receptor consist of 75–100 variable segments (V_β), two D segments ($D_\beta 1$, $D_\beta 2$), two joining segments ($J_\beta 1$, $J_\beta 2$), and two constant segments ($C_\beta 1$, $C_\beta 2$). The genomic organization of the genes for the α , δ , and γ chains is similar. During T-lymphocyte maturation, different segments are joined together by somatic recombination as during B-cell maturation. In a given T cell, only one of the two α -chain loci and only one of the two β -chain loci become functionally rearranged and expressed (allelic exclusion). As in the rearrangement of immunoglobulin genes, different mechanisms help produce diversity of the T-cell receptor genes. The genomic organization in humans and mice is very similar. The β -chain genes are located on chromosome 7 in humans and on chromosome 6 in mice. The α - and δ -chain gene loci are located on chromosome 14 in both humans and mice. The γ -chain genes lie on chromosome 7 in humans and on chromosome 13 in mice.

B. T-cell receptor binding to antigens and MHC surface proteins

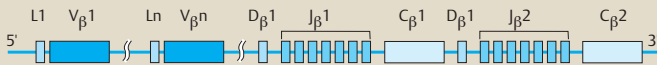
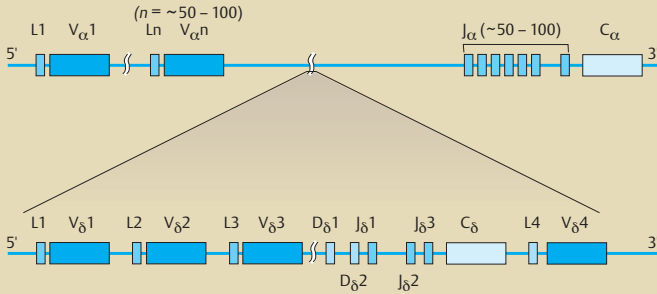
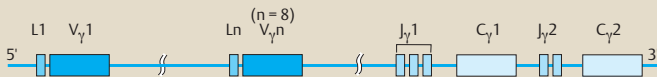
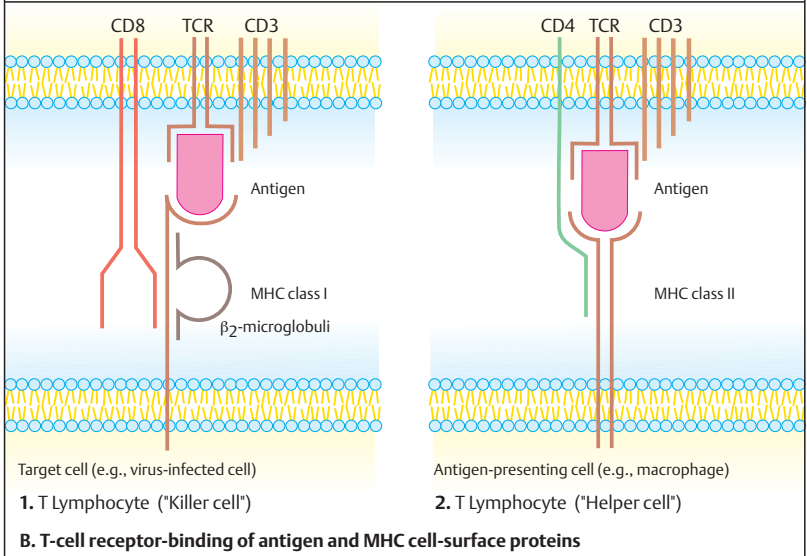
Unlike B cells, T cells recognize and react to foreign protein antigens only when the antigens are attached to the surface of other cells. Two different classes of T lymphocytes recognize different types of MHC gene products. T cells with the ability to destroy other cells by cytotoxicity (cytolytic T lymphocytes, CTLs, or “killer cells”) recognize class I MHC molecules by means of a coreceptor, CD8 (formerly T8). CD8 is a membrane-bound glycoprotein of two dimers (α and β), whose extracellular domains are held

together by two disulfide bridges (either α -chain homodimers or α - β heterodimers). The extracellular aminoterminal domains are immunoglobulinlike. In general, the structure of CD8 resembles a box with rounded corners. The other class of T lymphocytes (“helper cells”) specifically bind class II MHC molecules of the antigen-presenting cells. The antigen consists of a relatively small peptide fragment of foreign protein. With antigen presentation, a series of surface molecules lend physical help, e.g., CD3, CD4, and CD8. CD4 is a coreceptor, a rod-shaped single polypeptide with an extracellular part consisting of four immunoglobulinlike domains. The gp120 protein of the HIV virus reacts with the second domain of CD4. A number of accessory membrane proteins of the CD3 system participate in the specific binding between TCR and MHC (TCR–CD3 complex). (Drawings adapted from Abbas et al., 1997.)

The figures in B are highly schematic and do not show the real three-dimensional structures.

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TCR β -chain locus (chromosome 7)TCR α , δ -chain locus (chromosome 14)TCR γ -chain locus (chromosome 7)**A. T-cell-receptor genes (TCR) of man**

Evolution of the Immunoglobulin Supergene Family

The many cell surface and soluble molecules of the immune system that mediate different functions such as recognition, binding, or adhesion of specific molecules show many structural similarities. Some parts are found outside the immune system. As a group, they constitute a gene superfamily, derived from an ancestral gene common to all members. The homologies of the domains of their gene products and of their gene sequences can be explained by evolutionary origin from a common ancestral gene. The Ig gene family members code for immunoglobulin domains, usually of about 70–110 amino acids homologous with an Ig variable (V) or constant (C) domain. Each Ig domain is derived from conserved DNA sequences.

A. Basic structure of proteins of the immunoglobulin supergene family

The immunoglobulin molecules of the T-cell receptors (TCR) and the class I and class II MHC molecules (1) are basically similar. They consist of variable Ig-like domains (V), constant Ig-like domains (C), or primordial Ig-like domains (H). Although their genes are located on different chromosomes, the gene products form functional complexes with each other. Others, such as the V, D, and J gene segments of all antigen receptors and their genes for the C domain lie close together in gene clusters. In addition, genes of the MHC loci and for the two CD8 chains lie together. The basic structures of accessory molecules (2) such as CD2, CD3, CD4, CD8, and thymosine 1 (Th-1) are relatively simple. Other members of the Ig superfamily (3) are the Fc receptor II (FcRII); polyimmunoglobulin receptor (pIgR), which transports antibodies through the membranes of epithelial cells; NCAM (neural cell adhesion molecules); and PDGFR (platelet-derived growth factor receptor) (3). (After Hunkapillar and Hood, 1989.)

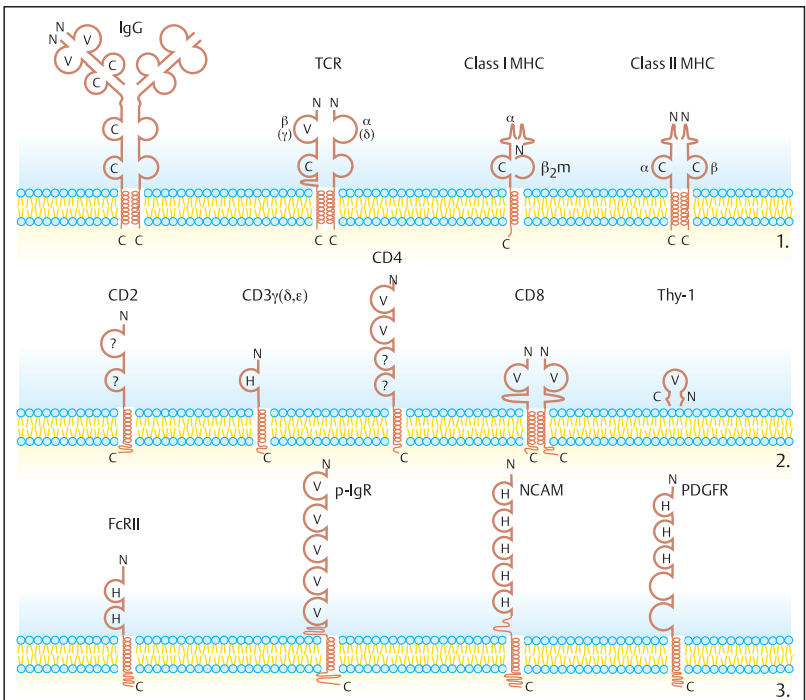
B. Evolution of genes of the immunoglobulin supergene family

Distinct evolutionary relationships can be recognized in the homology of genes for Ig-like molecules. A precursor gene for a variant (V) and a constant (C) region must have arisen by

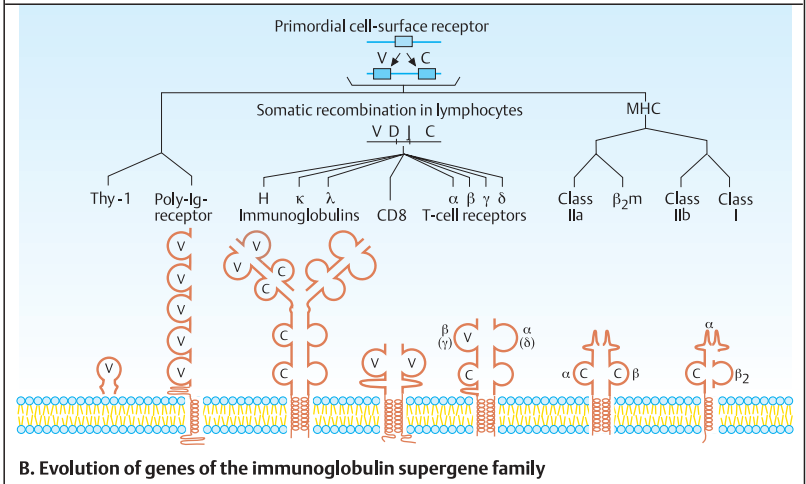
duplication and subsequent diversification of a gene for a primordial cell surface receptor. Such a primordial gene could have looked like the gene for thymosine (Thy-1) or poly-Ig receptor. No somatic recombination occurs in these gene families or in the genes of the MHC complex. In contrast, the rearrangement of lymphocyte germ-line genes by somatic recombination during the maturation of B cells and T cells is the basis for the formation of immunoglobulins, T-cell receptors, and CD8. Somatic recombination of the genes for antigen-binding molecules was an enormous evolutionary advantage. Consequently, this is found even in early vertebrates. (Figure adapted from Hood et al., 1985).

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A. Basic structure of proteins of the immunoglobulin supergene family



B. Evolution of genes of the immunoglobulin supergene family

Hereditary and Acquired Immune Deficiencies

Numerous types of impairment of the immune system exist, congenital (hereditary) and acquired. All result in abnormal susceptibility to infections, often associated with lymphoreticular malignancies, and in some cases autoimmune disease.

A. Examples of hereditary immune deficiency diseases

Severe combined immune deficiency (SCID) is a heterogeneous group of genetic disorders due to various defects in both B cell and T differentiation.

X-linked agammaglobulinemia type Bruton (McKusick 300300) was the first hereditary immune deficiency described, in 1952 by Ogden Bruton. The first developmental step of B cell differentiation from pre-B to mature B cell is blocked by deficiency of Bruton tyrosine kinase due to mutations in the *BTk* gene on the X chromosome (Xq22). Other forms involve later steps of differentiation (variable immune deficiency) or isolated Ig isotype (subclass) deficiencies. Several T cell immune deficiency diseases exist. The most important is the DiGeorge syndrome (McKusick 188400), characterized by a broad spectrum of highly variable manifestations. The underlying defect involves the third and fourth brachial arch derivatives during embryonic development. A deletion in chromosome region 22q11 is found in most patients, usually as a *de novo* event. Other disorders involve T cell activation and function of one or both major subsets of T cells, CD4 or CD8.

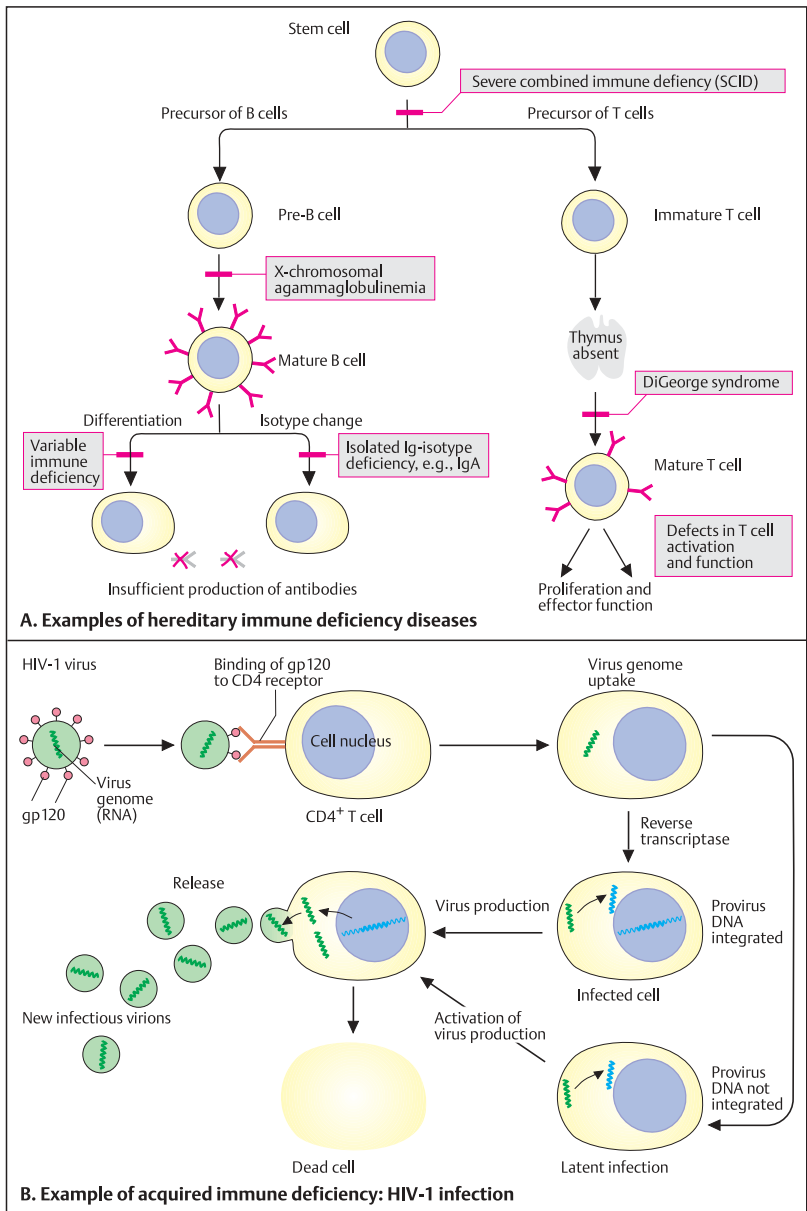
B. Example of acquired immune deficiency: HIV-1 infection

The global epidemic of the acquired immune deficiency syndrome (AIDS) due to the infection with human immunodeficiency virus type 1 (HIV-1) poses a major public health problem of unprecedented dimensions in modern times. Although it was first noted in 1981 in restricted populations, it now occurs in all populations at all ages throughout the world (United Nations Programme on HIV/AIDS at <http://www.UN-AIDS.org/hivaidsinfo/documents-html>). HIV-1 selectively infects T cells of the CD4 type. The first step of infection is specific binding of the

extracellular domain of the viral transmembrane glycoprotein gp120 to the CD4 receptor (gp41 is the transmembrane protein). Two chemokine receptors, CCR5 and CXR4, function as major coreceptors (not shown in the figure). Following virus genome uptake, a phase of viral replication by reverse transcription into double-stranded DNA occurs. Provirus DNA is integrated into the cellular DNA by the virus-encoded integrase enzyme. As the infected cell divides, the viral DNA (provirus) is also replicated. The HIV-1 provirus can remain in the cell without being transcribed for a long period (latent infection). Another phase of the HIV-1 life cycle involves transcription of viral DNA (provirus) into viral RNA. The full-length RNA transcripts are spliced and transported to the cytoplasm. Here the viral RNA is translated into viral proteins by cellular enzymes; the viral proteins and RNA are then processed and assembled into new virus particles (virus production). These leave the cell as free infectious virions. Usually the cell dies by lysis, but not always. In this case low-level chronic virus production may persist in infected individuals.

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Origin of Tumors

Influence of Growth Factors on Cell Division

Development, differentiation, and the maintenance of vital functions require exact regulation of the time and location of cell divisions. Rapidly multiplying cells in embryonic tissues must be controlled, just like those in stationary phases in adult tissues. Rapid response to injury or to foreign antigens requires controlled cell division. Multicellular organisms have an extensive repertoire of genetically regulated mechanisms at their disposal for controlling cell division and tissue proliferation. As a group, they are referred to as growth factors. Every growth factor has a specific cell surface receptor. Binding to the receptor initiates (or in some cases blocks) cell division. Most growth factors regulate only certain types of cells and tissues.

A. Control of cell division by growth factors

Basically, cell division (mitosis) can be controlled by stimulation or inhibition. In the absence of stimulation or with active inhibition, no mitosis occurs. Growth factors have an effect not only on specific types of cells, but also on defined phases of the cell cycle. The most frequently controlled phase of the cell cycle is the transition from G_0 to G_1 . The growth factor group includes growth factors for epidermal cells (EGF), for nerve cells (NGF), for connective tissue or mesenchymal cells (fibroblasts, FGF), and for thrombus-forming cells in the inner lining (endothelium) of blood vessels (PDGF). Their stimulating effect may be opposed by an antagonistic effect (e.g., TGF β , transforming growth factor, or TNF, tumor necrosis factor). The function of each growth factor is mediated by a specific receptor.

B. Activation of a growth factor receptor

A growth factor receptor becomes activated by specific extracellular binding to the growth factor. The activated receptor in turn activates a substrate protein.

C. PDGF-receptor kinases have an effect on numerous substrates

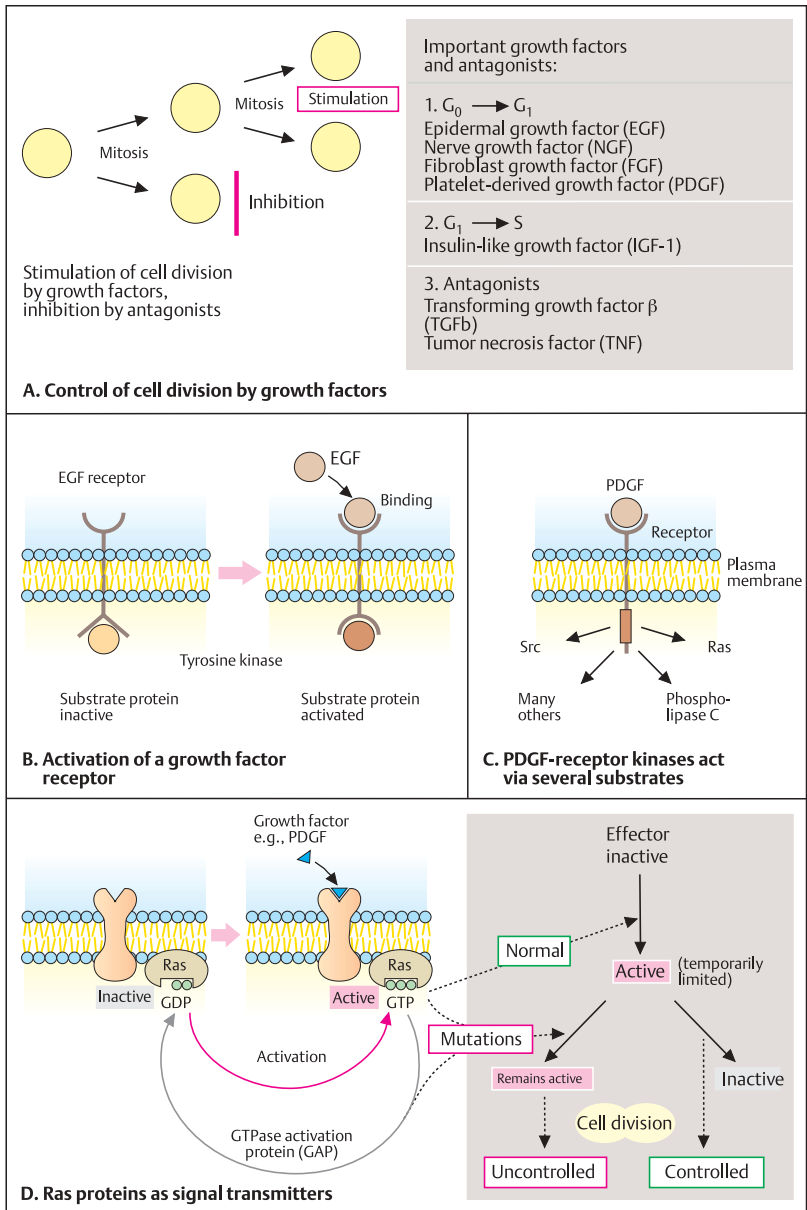
A receptor such as the PDGF (platelet-derived growth factor) receptor can have an effect on numerous substrates. Substrates of the PDGF receptors include the Ras protein (see D), the Src protein (the name is derived from the tumor, a sarcoma, in which it was first found), phospholipase C (a signal transmitter), and others.

D. Ras proteins as signal transmitters

The Ras proteins play a central role as signal transmitters. They belong to the group of G proteins (guanosine-residue-binding proteins with signal-transmitting functions, see p. 266). The binding of growth factor, e.g., PDGF, activates the Ras protein by stimulating the conversion of associated GDP (guanosine diphosphate) to GTP (guanosine triphosphate) and triggering a short time-limited signal that initiates cell division. The signal is terminated by inactivation of Ras by a GTPase-activating protein (GAP), which converts GTP into GDP. Mutation of the Ras protein or of GAP can remove the time limit of the cell-stimulating signals and result in an active condition with uncontrolled cell division. This can lead to a tumor with uncontrolled growth (malignancy). Several mutations have been defined in the pertinent genes. (Diagrams adapted from Watson et al., 1992.)

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Tumor Suppressor Genes

Malignant tumors arise as a result of mutations in three basic types of genes, DNA repair genes (see p. 80 DNA repair), tumor suppressor genes, and proto-oncogenes (see next plate). A single mutation does not cause cancer. Rather several mutations in different genes must accumulate in one or several cells, which eventually lose growth control in favor of aggressive growth properties. Most mutations are somatic, i.e., limited to the neoplastic cells. A relatively small subset of mutations are present in the germline (hereditary forms of cancer) and predispose the individual to certain types of cancer.

Tumor suppressor genes encode proteins with function in growth regulation or differentiation pathways. Their name is derived from the observation that one functional allele will suppress tumor development even in the presence of a mutation in the other allele (or its loss). Thus, two mutational events are required to release the growth-controlling function of a tumor suppressor gene (see retinoblastoma, p. 330). The two mutational events in a tumor suppressor gene often become manifest in loss of heterozygosity (LOH) in tumor cells (see B). Tumor suppressor genes can be compared to the brake of a car, cellular oncogenes to the accelerator (see next plate).

A. Tumor suppressor gene

In contrast to the cellular oncogenes, for which a change in one allele will alter normal function, both alleles of a tumor suppressor gene must lose their function before a tumor develops. The first event is usually a mutation by base exchange or deletion. The second event, affecting the other allele (allele 2), may also be a mutation, but the loss of function more often appears to be from loss of the chromosome after a faulty cell division (mitotic nondisjunction) or other mechanisms (e.g., mitotic recombination with gene conversion).

B. Loss of heterozygosity in tumor cells

Usually, in about half of the individuals who are heterozygous for DNA markers at the tumor suppressor gene locus of interest, the loss of one allele (event 2) can be demonstrated by Southern blot analysis. In contrast to normal somatic cells (blood), tumor cells contain only one allele (loss of heterozygosity, LOH). The remaining al-

lele carries the mutation. Thus, the mutant allele can be identified by demonstration of LOH. LOH is useful in diagnosis as an indication of the existence of a tumor suppressor gene.

C. Somatic and germinal mutation

The first mutation in a suppressor gene can either be present in the zygote (germinal mutation, i.e., germ cell mutation due to transmission from an affected parent or due to new mutation) or occur in a single cell of the corresponding tissue (somatic mutation). Loss of function of one allele (corresponding to event 1 in A) predisposes the cell to tumor development.

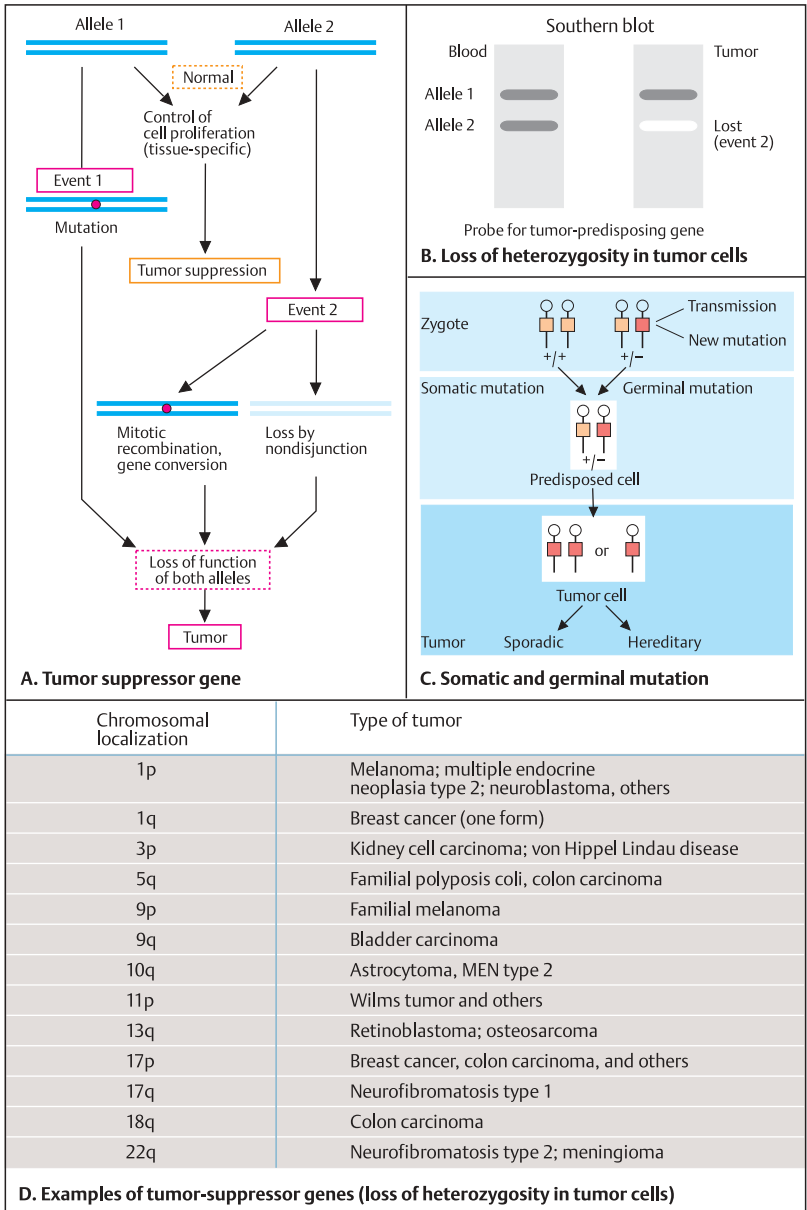
With a germinal mutation, all cells are predisposed. The tumor arises after loss of function of the second allele. When somatic mutation occurs in a single cell, loss of function of both alleles rarely affects the same cell. But with a germ cells mutation, loss of function of the second allele is frequent, since all cells carry the first mutation, i.e., are predisposed. With somatic mutation, the tumor occurs sporadically (is not hereditary) and arises unifocally from a single cell. In the hereditary form resulting from a germ cell mutation, several tumors may arise from different cells (multifocal tumor). The predisposition for the tumor in the hereditary form shows autosomal dominant inheritance.

D. Examples of tumor suppressor genes

Numerous types of tumors arise due to loss of function in both alleles of a tumor suppressor gene. Loss of heterozygosity (LOH) can be demonstrated in the tumor cells in about half of these patients.

References

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Cellular Oncogenes

Oncogenes (tumor-causing genes) were originally identified in RNA tumor viruses (retroviruses) as genes (*v-onc*) that could transform cells into an altered state of control of cell proliferation, often resulting in a tumor, mainly in chicken, mice, and rats. More than 20 different viral oncogenes are known to have a counterpart in normal cells (*c-onc*), called proto-oncogenes or cellular oncogenes. These cellular genes are highly conserved in evolution because they have important functions in all eukaryotic cells. They encode proteins that are required at defined sites throughout the cell where they regulate the ordered progression through the cell cycle, cell division, and differentiation.

A. Cellular and viral oncogenes

A typical retrovirus contains an RNA genome that codes for three genes or groups of genes: *gag* (group-specific antigen), *pol* (polymerase), and *env* (coat protein, envelope). As with all genes of higher organisms, a cellular oncogene (*c-onc*) consists of exons and introns with defined structure and sequence, as in the gene *src* (the name is derived from sarcoma, a tumor that is induced by a change in this gene). The virus may contain parts of the cellular oncogene (*c-scr*). This is designated viral oncogene (*v-src*) (Rous sarcoma virus). In chickens, it induces a malignant tumor (a sarcoma), first observed by Peyton Rous in 1911. Since many cellular oncogenes are also known in an altered, viral form, it is assumed that the viruses have integrated parts of the respective cellular oncogenes into their own genomes.

Virus-induced tumors are known especially in chickens, rodents, and cats. In man, they do not play a general role in the induction of tumors. Important exceptions are papilloma virus-induced carcinoma of the cervix and carcinoma of the liver secondary to hepatitis virus infection.

B. Mechanisms of oncogene activation

A cellular oncogene controls cell division. It controls the time and location of the orderly proliferation of cells and tissues (normal growth). Genetic changes can lead to disorders of the regulation of cell division, increased proliferation of cells, and formation of a tumor. This can be traced back to relatively few mechanisms.

A point mutation in a critical region of the gene can lead to disturbances in the regulation of cell division. Examples are mutations in codon 12 or 63 or the *H-ras* gene.

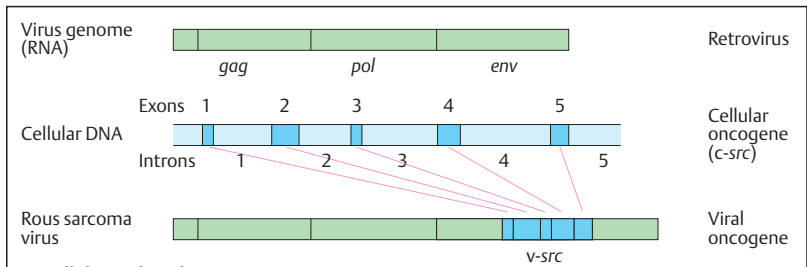
An inactive cellular oncogene may become activated when it is moved by chromosomal translocation to the vicinity of an active gene. In Burkitt lymphoma, an inactive gene is moved to the region of an active gene for the H or L chain of an immunoglobulin. In other cases, the breakpoint of the chromosome translocation may lie within a cellular oncogene and thereby affect its expression. An example is the Philadelphia translocation (see p. 332). Multiplication (amplification) of a gene is a further mechanism that can lead to altered (usually increased) expression.

C. Examples of cellular oncogenes and their proteins

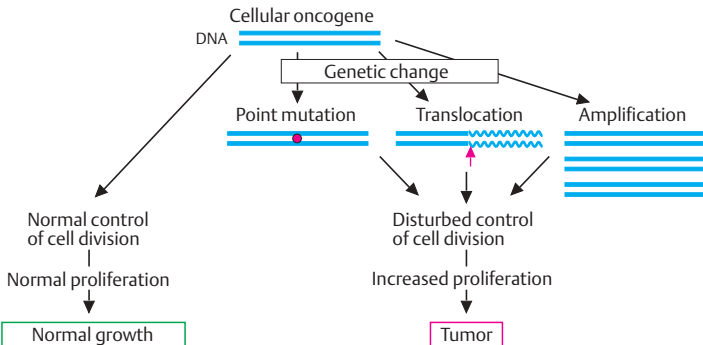
The table shows examples of the about 60 known cellular oncogenes, their basic functions, a few tumors induced in man by mutation of the cellular oncogene (*c-onc*), and tumors induced in vertebrates by the homologous viral oncogene (*v-onc*). (Data from Lodish et al., 2000; Cannon-Albright et al., 1992.)

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A. Cellular and viral oncogenes



B. Mechanisms of oncogene activation

Oncogene	Function	Origin of tumor by	
		<i>c-onc</i>	<i>v-onc</i>
<i>sis</i>	One form of PDGF		Simian sarcoma
<i>abl</i>	Tyrosine protein kinase	Chronic myelogenous leukemia	Abelson mouse leukemia
<i>fes</i>	Tyrosine protein kinase		Cat sarcoma
<i>erbB</i>	Epidermal growth factor	Mammary carcinoma Ovarian carcinoma	Chicken erythroblastosis
<i>src</i>	Membrane-bound protein kinase		Rous chicken sarcoma
<i>Ha-ras</i> <i>Ki-ras</i> <i>N-ras</i>	Membrane-bound G proteins with GTPase activity	Different carcinomas, neuroblastoma	Harvey mouse sarcoma Kirsten mouse sarcoma
<i>fos</i>	Transcription factor AP1		Osteosarcoma in chicken
<i>myb</i>	Nuclear protein	Leukemias	Myeloblastosis in chicken
<i>myc</i>	Nuclear protein	Leukemias	Myelocytosis in chicken
<i>N-myc</i>	Nuclear protein	Neuroblastoma	

C. Examples of cellular oncogenes and their proteins

The p53 Protein, a Guardian of the Genome

The p53 protein (named after its molecular weight of 53 kD), a nuclear phosphoprotein, is indispensable for genomic integrity and cell cycle control. It binds to specific DNA sequences and regulates the expression of different regulatory genes involved in growth. It interacts with other proteins in response to DNA damage and mediates apoptosis (cell death) of the cell when the damage is beyond repair. Its basic function is to control entry of the cell into the S phase (see cell cycle control, p. 112). Somatic mutations in the *p53* gene occur in about half of all tumors. Germline mutations lead to a familial form of multiple different cancers (Li–Fraumeni syndrome, see B).

A. The human p53 protein

The active form of the human p53 protein is a tetramer of four identical subunits. Each subunit has 393 amino acids and five highly conserved regions, I–V. Region I is part of the transcription-activation domain; regions II–V belong to sequence-specific DNA-binding domains. The carboxyl end beyond amino acid 300 consists of a nonspecific DNA interaction domain and the tetramerization domain. Proteins encoded by DNA tumor viruses bind to p53 and inhibit its activity. Mutations in the *p53* gene on human chromosome 17 at p13 (spanning 20 kb of DNA and yielding a 2.8 kb mRNA transcript from 11 exons) have the greatest effect when they occur in the conserved regions II–V in codons 129–146 (exon 4), 171–179 (exon 5), 234–260 (exon 7), and 270–287 (exon 8). Particularly vulnerable are the conserved amino acids arginine (R) in positions 175, 248, 249, 273, and 282 and glycine (G) in position 245. Mutations occur mainly as missense, resulting from base-pair substitutions, but some are insertions and deletions and exert a dominant negative effect. Knockout mice develop normally, but develop tumors at a high rate. Activated benzopyrene induces mutations at codons 175, 248, and 275 in cultured bronchial epithelial cells.

B. Germline mutations of p53

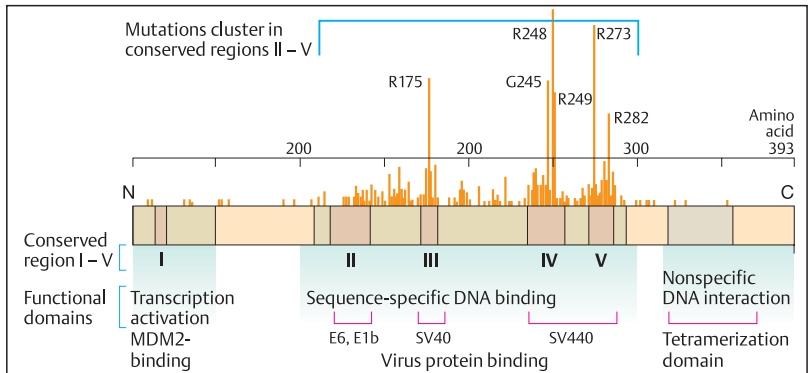
In 1969, Li and Fraumeni identified families in whom other members were affected with diverse types of tumors, mainly soft-tissue sarcomas, early-onset breast cancer, brain cancers, cancer of the bone (osteosarcoma) and bone marrow (leukemias), and carcinoma of the lung, pancreas, and adrenal cortex. Similar observations had been reported as “cancer family syndrome” by Lynch. This autosomal dominant cancer syndrome is called the Li–Fraumeni syndrome (McKusick 114480). In the pedigree shown in panel 1, four individuals (II-2, II-3, III-1, III-2) are affected by different types of tumors. A mutation in codon 248 of the *p53* gene (CGG arginine, to TGG, tryptophan) is present in these patients. The mutation is also present in individuals I-1 and III-5. This places these individuals at increased risk for one of the types of cancer mentioned above and shown in panel 2. In contrast, absence of the mutation in individuals III-3 and III-4 indicates that they do not have an increased risk of cancer (Data of D. Malkin). A subset of patients with Li–Fraumeni syndrome does not show p53 mutations.

C. Model of function of the p53 gene

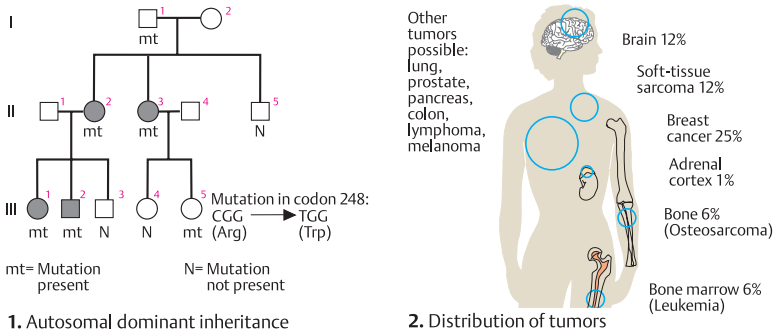
Normally, the *p53* gene is inactive (1). p53 plays an important role in regulating growth in damaged cells (2). DNA damage in cells leads to increased expression of p53 and interruption of the cell cycle in G₁. If DNA repair is successful, the cell can continue its cycle. If repair is not successful, the cell dies (cell death, apoptosis). Damaged cells with p53 protein that is mutant are not arrested in G₁. (After Lane, D.P.: *Nature* 358:15–15, 1992.)

References

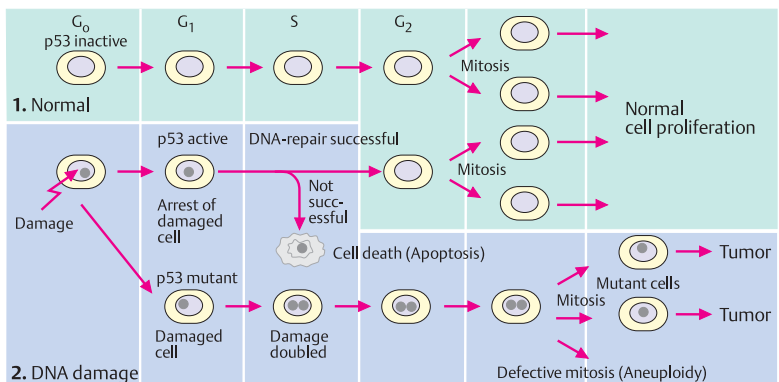
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A. The human p53 protein



B. Mutations of the p53 gene in familial multiple tumors (Li Fraumeni syndrome)



C. Model of the function of the p53 gene

Neurofibromatosis 1 and 2

The neurofibromatoses are clinically and genetically different autosomal dominant hereditary diseases that predispose to benign and malignant tumors of the nervous system. Numerous different forms are known. The most important are neurofibromatosis 1 (NF1, von Recklinghausen disease, MIM 162200) and neurofibromatosis 2 (NF2, MIM 101000).

A. The main signs of NF1

NF1 is very variable. Lisch nodules of the iris (1) in more than 90% of patients, café-au-lait spots (2) (more than five spots of more than 2 cm diameter are considered diagnostic) in more than 95%, and multiple neurofibromas (3) in more than 90% of patients are the most important signs.

B. Neurofibromatosis gene *NF1* on human chromosome 17 at q11.2

The localization of the *NF1* gene revealed the gene on a 600 kb *NruI* restriction fragment. A CpG island (CpG-1) and two translocation breakpoints at t(17;22) and t(1;17) served as important anchor points for gene identification. The *NF1* gene has 79 exons, which span about 335 kb of genomic DNA. Three unrelated genes, *OMGP*, *EV12B*, and *EV12A*, are embedded within the *NF1* gene in intron 35 on the opposite DNA strand. Mutation analysis of the *NF1* gene shows deletions, insertions, base substitutions, and splice mutations leading to truncated and presumably nonfunctional gene products. Currently mutations are found in about 60–70% of patients. (Figure adapted from Claudio and Rouleau, 1998).

C. *NF1* gene product (neurofibromin)

The *NF1* gene encodes a gene product with 2810 amino acids, called neurofibromin. Between amino acids 840 and 1200, this large protein contains a domain that corresponds to a GTPase-activating protein. The homology includes a gene product in yeast (*S. cerevisiae*), IRA1 (inhibitor of *ras* mutants). Mutations at the *NF1* locus interrupt a signal pathway to the *ras* genes. (After Xu et al., 1990).

D. Neurofibromatosis gene *NF2* on human chromosome 22 at q12.1

The *NF2* gene was identified in 1993 by Rouleau et al. and Trofatter et al. within a cosmid contig contained in YAC clones (yeast artificial chromosomes). Two deletions observed in unrelated patients aided in finding the almost 100 kb gene with 17 exons. Mutations can be detected in more than 50% of patients (large deletions including the entire gene or several exons and small deletions are frequent). The gene product, called schwannomin, is related to a family of cytoskeleton-membrane proteins (erythrocyte protein 4.1, see p. 374, and the ERM family ezrin, radixin, and moesin) and a family of protein tyrosine phosphatases. The basic function of these proteins is to maintain cellular integrity. (Figure adapted from Claudio and Rouleau, 1998).

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Neurofibromatosis 1 (NF1) (von Recklinghausen disease)

Autosomal dominant
Frequency 1 in 3000
Gene locus on 17q11.2
Café-au-lait spots
Lisch nodules in the iris
Multiple neurofibromas
Skeletal anomalies
Predisposition to tumors
of the nervous system
50% new mutations



1. Lisch nodule

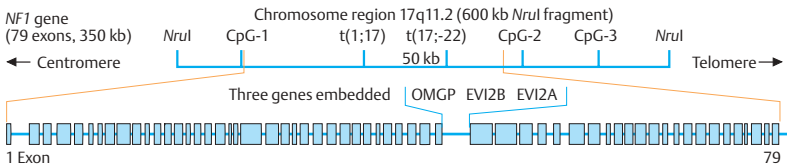


2. Café-au-lait spot

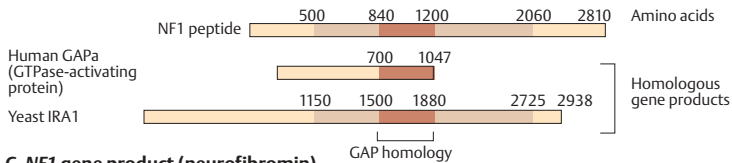


3. Neurofibromas

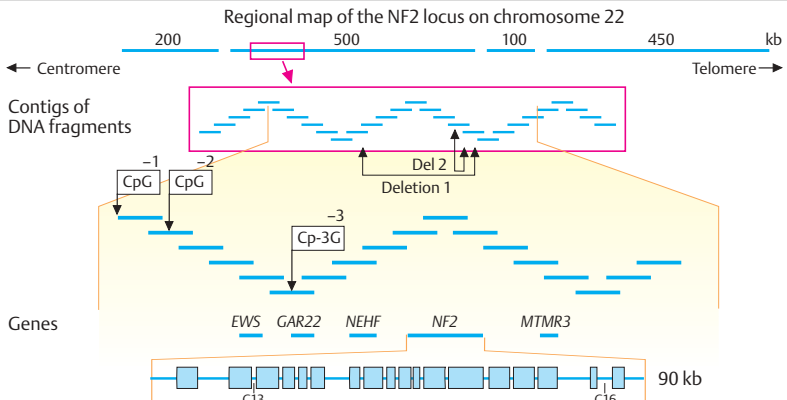
A. Main manifestations of neurofibromatosis 1



B. Neurofibromatosis gene *NF1* on human chromosome 17q11.2



C. *NF1* gene product (neurofibromin)



D. Neurofibromatosis gene *NF2* on chromosome 22q12.1

APC Gene in Familial Polyposis Coli

Cancer of the colon and rectum is the second leading cause of death from cancer. About 5% of the population can be expected to develop colorectal cancer. Most colorectal tumors arise from a series of somatic mutations in several genes.

A. Polyposis coli and colon carcinoma

Familial polyposis (FAP) is an autosomal dominant hereditary disease. In late childhood and early adulthood, up to 1000 and more polyps develop in the mucous membrane of the large intestine (colon) (1). Each polyp can develop into a carcinoma (2). In about 85% of affected persons, small hypertrophic areas that do not affect vision are present in the retina (3). Hereditary non-polyposis colorectal cancer (HNPCC) affects about one in 200–1000 individuals (3% of all colorectal cancers). It results from germline mutation in one of the DNA mismatch repair genes *hMSH1*, *hMLH2*, *hPMS1*, and *hPMS2* or related genes. Microsatellite instability is an important feature of HNPCC. (Photos 1 and 2 from U. Pfeifer, Institut für Pathologie der Universität Bonn; photo 3 from W. Friedl et al., 1991.)

B. Mutations at different gene loci in polyposis coli and carcinoma of the colon

At least six gene loci are involved in the development of carcinoma of the colon associated with polyposis coli. Somatic mutations may occur in two recessive oncogenes (Ras genes *KRAS1* and *KRAS2*) and in four dominant tumor suppressor genes. Most forms of carcinoma of the colon are not associated with polyposis coli.

C. The APC gene and distributions of mutations

The APC gene (adenoma polyposis coli) consists of 8538 bp in 15 exons encoding a 2843-amino-acid protein (not 8535 bp and 2844 amino acids as shown in C). Exon 15 is very large, 6579 base pairs. Over 95% of mutations result in a non-functional truncated protein due to nonsense mutations (40%), deletions (41%), insertions (12%), and splice site mutations (7%). The APC

gene is also involved in sporadic colorectal cancer.

D. Indirect DNA diagnosis in FAP

Linked DNA markers (RFLPs) near the APC locus (1) can be used for indirect DNA diagnosis. The alleles of three flanking marker pairs (K,k and E,e on the centromere side and A,a on the distal side) form the haplotypes, e.g., e–K–a and E–k–a in individual I-1 in the pedigree (2). The mutation-carrying haplotype must be e–K–a. Since individual III-2 has inherited this haplotype, he is at risk for the disease, whereas individual III-1 is not.

E. Several mutations in the production of colon carcinoma

Tumor formation goes through several stages. It starts with a somatic or germinal mutation in the APC gene. After loss of the other allele (LOH), an adenoma develops with less-differentiated cells and polyp formation. Mutations in other genes lead to malignant transformation and eventually to tumor development. (Diagram after Fearon and Vogelstein, 1990.)

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1.



2.

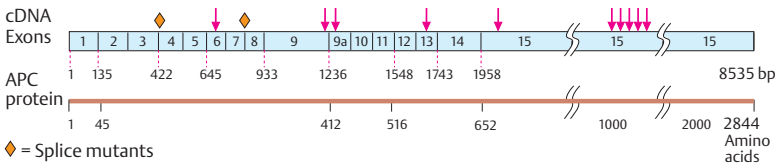


3.

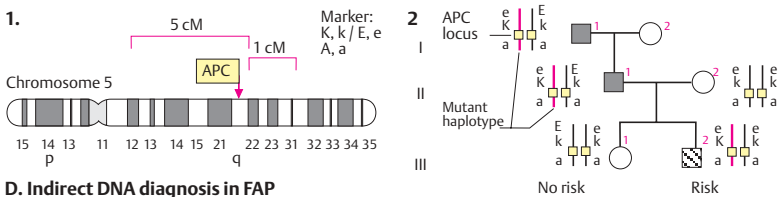
A. Polyposis coli and colon carcinoma

McKusick no.	Disease/gene involved	Gene locus
190110	Colorectal adenoma/carcinoma (<i>KRAS1</i>)	6p12-11
190070	Colorectal adenoma/carcinoma (<i>KRAS2</i>)	12p12
114500	Isolated colon carcinoma (somatic mutation in the <i>APC</i> gene)	5q21
175100	Familial adenomatous polyposis coli (germlinal mutation in the <i>APC</i> gene)	
175100	<i>APC</i> with other tumors: Gardner syndrome (allelic)	5q21-22 17p12-13 18q21.3
159350	DNA sequences mutated in colon carcinoma (<i>MCC</i>)	
191170	p53 gene (identical with McKusick * 120460)	
120470	Deletion in colorectal carcinoma (<i>DCC</i>)	

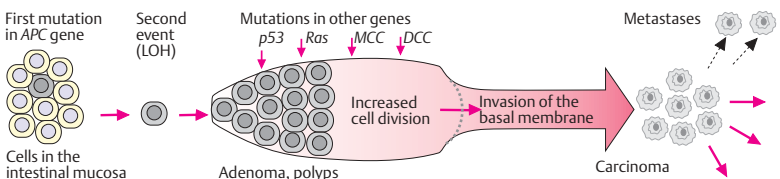
B. Mutations at different gene loci in polyposis coli and colon carcinoma



C. APC gene (scheme) and distribution of mutations (↓)



D. Indirect DNA diagnosis in FAP



E. Several mutations in the origin of colon carcinoma

Breast Cancer Susceptibility Genes

In 1994 and 1995 two genes were identified that confer susceptibility to breast and ovarian cancer when mutated, the breast cancer genes *BRCA1* and *BRCA2*. Both genes encode multifunctional proteins with important cellular functions in genomic stability, homologous recombination, and double-stranded and transcription-coupled DNA repair (see p. 80). The *BRCA1* and *BRCA2* proteins interact and play a role in cell cycle control (see p. 112) and in development. An autosomal dominant susceptibility mutant allele in one of these genes is considered the main cause of the cancer in about 5–10% of patients. Mutations in other genes are involved in some cases. The direct causative role of *BRCA1* and *BRCA2* mutations is difficult to assess in individual patients. Different mutations as well as polymorphic variants occur throughout the genes.

A. The breast cancer susceptibility gene *BRCA1*

The *BRCA1* gene on chromosome 17 at q21.1 consists of 24 exons spanning 80 kb of genomic DNA that encode a 7.8 kb mRNA transcript. The protein has 1863 amino acids. Exon 11 is quite large (3.4 kb). About 55% of all mutations occur in exon 11. Although some mutations occur relatively frequently in other exons, they tend to be evenly distributed throughout the gene (only some mutations are shown). The deletion of an adenine (A) and a guanine (G) in nucleotide position 185 (185delAG) and the insertion of a cytosine in position 5382 (5382insC) account for about 10% of mutations each. These mutations are particularly frequent in the Ashkenazi Jewish population.

The protein has five main functional domains. The RING finger region near the N-terminus at amino acids 1–112 defines a zinc-binding domain of conserved cysteine and histidine residues that mediate protein–protein or protein–DNA interactions. This region is also the site of heterodimerization of *BRCA1* and *BARD1* (*BRCA1*-associated RING domain 1). Other functional domains define the central part of the *BRCA1* protein. These are two nuclear localization signals (NLS) and two protein-binding domains, one for p53 protein, retinoblastoma

(RB) protein, and RAD50 and RAD51. RAD50 and 51 are proteins involved in recombination during mitosis and meiosis, and in recombinational repair of double-stranded DNA breaks. The C-terminus contains a region involved in transcriptional activation and DNA repair.

B. The breast cancer susceptibility gene *BRCA2*

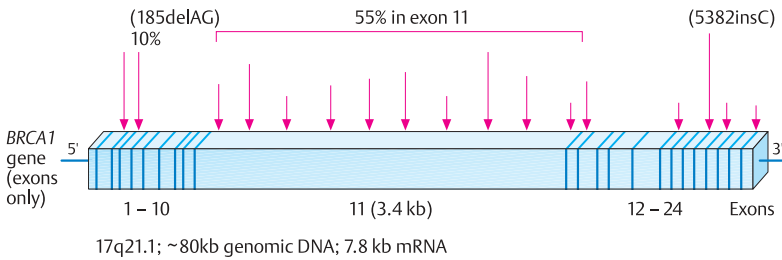
The *BRCA2* gene on 13q12 comprises 27 exons spanning 80 kb of genomic DNA that encode a 10.4 kb mRNA transcript. Its protein has 3418 amino acids. Exon 11 is large (11.5 kb), as in *BRCA1*. Mutations occur throughout the gene (only some are shown). A deletion of thymine at nucleotide position 6174 (6174delT) is relatively (1%) frequent in the Ashkenazi Jewish population.

The *BRCA2* protein has a transcriptional activation domain near the N-terminus and a nuclear location signal (NLS) near the C-terminus. A large central domain consists of eight copies of a 30–80-amino-acid repeat, which are conserved in all mammalian *BRCA2* proteins (BRC repeats). Four of these interact with the RAD51 protein.

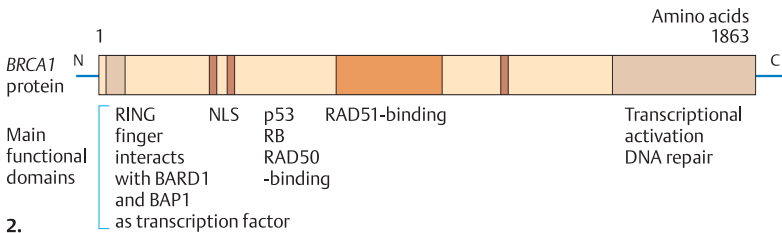
The *BRCA1* and *BRCA2* genes are expressed ubiquitously with the highest levels of expression in thymus and testis. The spatial and temporal expression patterns of *Brca1* and *Brca2* in the mouse fetal and adult tissues are essentially identical, with highest expression of both in rapidly dividing tissues during differentiation, especially in mammary epithelium. In the mammary gland both genes are expressed during puberty and pregnancy, and their expression is reduced during lactation. (Figures redrawn from Welch et al., 2000.)

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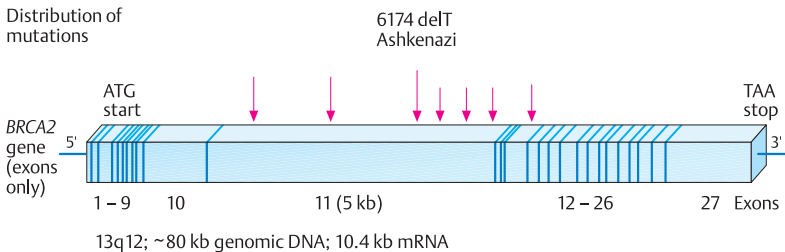


1. Distribution and relative frequency of mutations

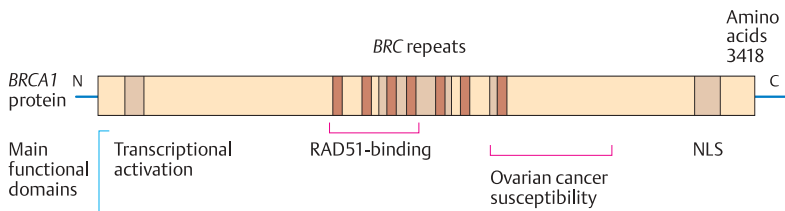


2.

A. The breast cancer susceptibility gene *BRCA1*



1.



2.

B. The breast and ovarian cancer susceptibility gene *BRCA2*

Retinoblastoma

Retinoblastoma (McKusick 180200) is the most frequent tumor of the eye in infancy and early childhood. It occurs in 1 of 15 000–18 000 live births. This tumor results from loss of function of both alleles of the retinoblastoma gene *RB1*. Tumor initiation is preceded by two steps as A. Knudson predicted in 1971 in his “two-hit” hypothesis (tumor suppressor gene, p.318). The first predisposing mutation in one allele may occur either in a retinoblast, an undifferentiated retinal cell in the developing embryo, or in the germline. The other allele is inactivated by a second mutation.

A. Phenotype

Retinoblastoma occurs in one eye or both eyes. An important early sign is the so-called “cat’s eye,” a white shimmer out of the affected eye (1) or the development of strabismus. One or several tumors originate from the retina (2). The tumor progresses rapidly (3). The relative proportions of the genetic types of retinoblastoma are about 60% somatic mutations (non-hereditary form) and 40% germline mutations, transmitted as an autosomal dominant trait (hereditary form, in about 10–15%, due to transmission from a parent; the remainder due to a new mutation). New mutations usually affect a paternal allele (about 10 : 1). In about 10% of carriers of a germline mutation no tumor develops (nonpenetrance).

B. Retinoblastoma locus on chromosome 13

The *RB1* locus at 13q14.2 was first defined with cytogenetically visible interstitial deletions.

C. Retinoblastoma gene *BR-1* and the pRB protein

The *RB1* gene is organized into 27 exons spanning 183 kb of genomic DNA (1). The *RB1* gene is ubiquitously expressed and transcribed into a 4.7 kb mRNA (2). The gene product (pRB protein) has 928 amino acids (3). It is a 100 kD phosphoprotein with important functions in the regulation of the cell cycle. It is activated by phosphorylation (P) during cell cycle progression from G₀ to G₁ (p. 112) at about 12 distinct serine and threonine residues. Three functional domains, A, B, and C, and a nuclear localization signal (NLS) can be distinguished.

D. Diagnostic principle

Molecular diagnosis of retinoblastoma greatly contributes to its early recognition and to the correct assessment of individual risks within families. In about 3–5% of patients an interstitial deletion 13q14 or a larger deletion is visible by chromosomal analysis (1). In familial retinoblastoma indirect DNA diagnosis can be achieved by segregation analysis using DNA markers at the *RB1* locus (2). In the example shown, the affected girl (II-1) has inherited haplotype **a** from the unaffected father and haplotype **c** from the unaffected mother. In tumor cells, obtained after one eye had to be removed, haplotype **a** only is present (loss of heterozygosity, LOH, see p. 318). This reveals that haplotype **a** represents the mutation-carrying *RB1* allele. In the family shown (3), I-2 and II-2 are affected (3). Sequence analysis reveals a C-to-T transversion in codon 575 in the two affected individuals (CAA glutamine to TAA stop codon). The mutational spectrum in hereditary retinoblastoma involves deletions (~26%), insertions (~9%), and point mutations (~65%), including splice-site mutations. (Illustrations courtesy of W. Höpping (A) and D. Lohmann (C and D).)

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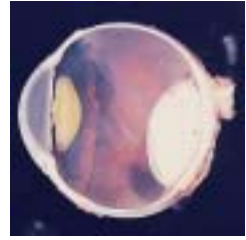
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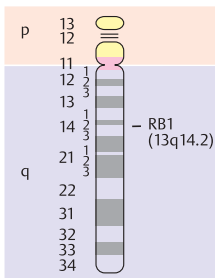
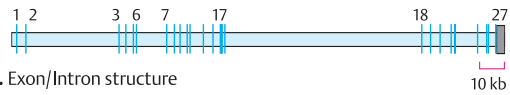
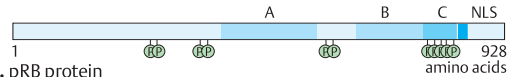
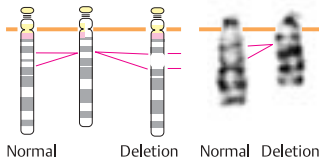
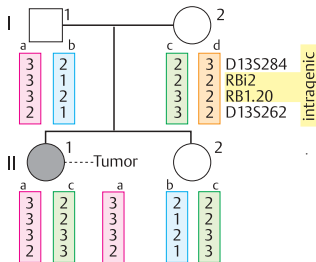
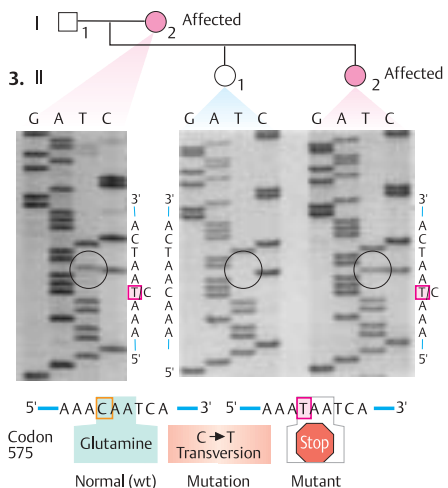
1. So-called cat's eye



2. Tumor in the retina



3. Large tumor in the eye

A. Phenotype**B. Retinoblastoma locus on chromosome 13****1. Exon/Intron structure****2. Coding regions (27 exons)****3. pRB protein****C. Retinoblastoma gene *RB1* and the pRB protein****1. Interstitial deletion****2. Haplotype analysis****D. Diagnostic principle****4. Sequence analysis**

Fusion Gene as Cause of Tumors: CML

Chronic myeloid leukemia (CML) is a malignant tumor that originates from a single cell of the bone marrow in adulthood. The number of myelocytes (white blood cells from the bone marrow) is greatly increased. The disease follows a chronic course. Acute crises develop intermittently and terminally. In about 90% of the patients, affected bone marrow cells contain a chromosome 22 with a shortened long arm (22q-, Philadelphia chromosome).

A. The Philadelphia chromosome (Ph¹) in different forms of leukemia

A Philadelphia chromosome is present in the bone marrow cells of most patients with the chronic form of the disease (CML). If it is not present, the illness progresses more rapidly than usual and has a poorer prognosis. In addition, the Philadelphia chromosome may be found in some acute leukemias (acute lymphocytic leukemia, ALL; acute myelocytic leukemia, AML) in adults and in children. Here, Ph¹ indicates a poor prognosis, whereas its absence is favorable.

B. The Ph¹ translocation [t(9;22)(q34;q11)]

The Philadelphia chromosome arises by reciprocal translocation between a chromosome 22 and a chromosome 9. The breakpoints are in 9q34 and 22q11. A good half of the long arm of a chromosome 22 is translocated to the long arm of a chromosome 9. A very small segment of the distal long arm of a chromosome 9 (9q34), not visible under the light microscope, is translocated to a chromosome 22. The Philadelphia chromosome (22q-) consists of the short arm and the proximal one-third of the long arm of a chromosome 22 and the small distal segment from the long arm of a chromosome 9. For demonstration of the Philadelphia translocation by in-situ hybridization, see p. 192).

C. The Ph¹ translocation leads to the fusion of two genes

The breakpoints of the Ph¹ translocation are located in the *BCR* gene of chromosome 22 and in the *ABL* gene of chromosome 9. The translocation leads to the fusion of these genes. The exact

locations of the breakpoints differ from patient to patient, but in the *BCR* gene they are limited to a small region of just 6 kb (thus the designation *BCR*, or breakpoint cluster region). In CML, the breakpoints lie in exons 10–12 of the *BCR* gene; in acute Ph¹-positive leukemias (e.g., ALL) they lie further in the 5' direction in exon 1 or 2. The breakpoint region in the *ABL* gene extends over 180 kb between exons 1a and 1b, which are separated by an intron.

D. The gene fusion leads to changes in transcription and gene products

The *ABL* gene codes for mRNA transcripts of 7 kb (exon 1b, 2–11) and 6 kb (exon 1a, 2–11) by differential splicing; these in turn code for a protein of about 145 000 Da (p145^{abl}). From the fusion of the two genes in CML, an 8.5 kb mRNA transcript results, which codes for a fusion protein of 210 000 Da (p210^{bc_rabl}). In the acute form of leukemia (ALL), a transcript results that codes for a fusion protein of 190 000 Da (p190^{bc_rabl}). In contrast to the normal protein, it has high tyrosinase activity. This results in uncontrolled cell division in the affected cells and tumor growth.

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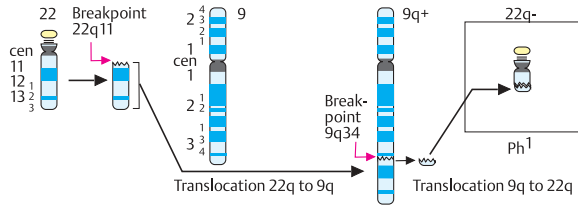
Chronic myelogenous leukemia (CML)

Myelocytes in peripheral blood increased
Early chronic course
Adults affected
Origin from one myeloid cell
In ~ 90% a Philadelphia chromosome (Ph¹)
(Translocation 22q to 9q)
Poor prognosis when Ph¹-negative

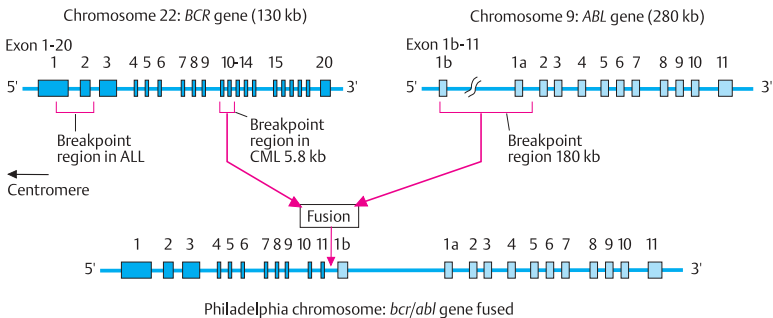
Other Ph¹-positive acute leukemias

Lymphocytes or myelocytes increased
Acute course
20% adults with ALL Ph¹-positive
2% adults with AML Ph¹-positive
5% children with ALL Ph¹-positive
Philadelphia translocation as in CML
Poor prognosis when Ph¹-positive

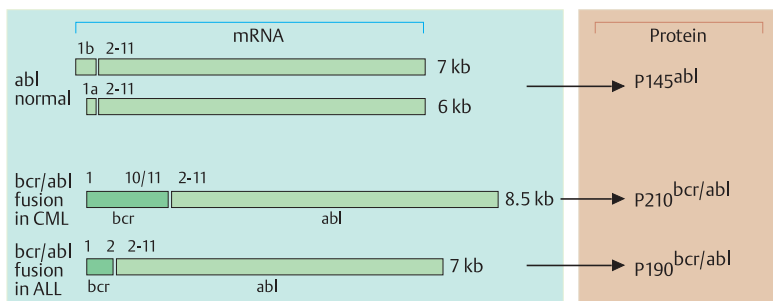
A. Philadelphia translocation (Ph¹) in different forms of leukemia



B. Ph¹ translocation [t(9;22) (q34;q11)]



C. Ph¹ translocation causes fusion of two genes



D. Gene fusion causes altered transcripts and gene products

Genomic Instability Syndromes

Genomic instability, visible by light microscopy as breaks and rearrangements in different chromosomes in a variable proportion of metaphase cells, is a hallmark of a group of characteristic hereditary diseases. The underlying genetic defect predisposes patients to different types of cancer. Three important examples are presented here.

A. Bloom syndrome (BS)

In Bloom syndrome (McKusick 210900) (1), prenatal and postnatal growth deficiency is pronounced (birth weight 2000 g, birth length ~40 cm, adult height around 150 cm). The phenotype (2) includes a narrow face. Usually, but not always, a sunlight-induced erythema develops on the cheeks, eyelids, mouth, ears, and back of the hands (a and b). The photograph on the right (c) shows a boy with Bloom syndrome and acute leukemia. Metaphase cells show about a tenfold increase in the rate of sister chromatid exchanges (SCE), ~60 instead of about 6 per metaphase in normal cells (3). (Sister chromatid exchanges are explained in the glossary, p. 423). Metaphases of patients contain increased breaks in one or both chromatids and exchanges between homologous chromosomes in about 1–2% of cells. In Bloom syndrome patients, different types of malignancies occur in a distribution comparable to that of the general population, but at a much earlier age (mean age 24.7, range 2–48 years). Some patients have multiple primary tumors, which underlines the striking susceptibility to cancer in Bloom syndrome. Chemotherapy is very poorly tolerated.

Homozygosity for mutations in the Bloom syndrome gene (*BLM*) results in an increased rate of somatic mutations, a manifestation of genomic instability. The *BLM* gene on chromosome 15 at q16.1 encodes a member of the RecQ family of DNA helicases. The 1417-amino-acid protein shows homology to the yeast *SGS1* gene product (slow growth suppressor) and the human *WRN* gene product (Werner syndrome, McKusick 277700). Allozygous nonsense mutations (two different mutations in the two alleles of the same gene) are frequent in the *BLM* gene. A characteristic homozygous 6 bp deletion/7 bp insertion at nucleotide 2281 occurs in Ashkenazi Jewish individuals as a result of a founder effect.

B. Fanconi anemia (FA)

Fanconi anemia (hereditary pancytopenia) (McKusick 227650) is a malformation syndrome (1) with variable clinical expression. Growth deficiency (2), hypoplastic or absent thumbs (3), and short or absent radii are characteristic physical signs.

FA cells are hypersensitive to DNA-crosslinking agents, such as diepoxybutane (DEB). Several complementation groups can be distinguished. Three FA genes have been identified, at chromosome 16q24.3 (FAA), 9q22.3 (FAC), and 3p22–26 (FAD). FAA is the most prevalent group in 60–65% of patients.

C. Ataxia telangiectasia

Ataxia telangiectasia (McKusick 208900) is a pleiotropic, variable disease due to mutations in the *ATM* gene on chromosome 11q23. Preferential reciprocal translocations between chromosomes 7 and 14 with breakpoints at 7p14, 7q14, 14q11, and 14q32 occur in a small proportion of metaphases. The *ATM* gene has 66 exons spanning more than 150 kb of genomic DNA. From its 13-kb transcript (and smaller alternatively spliced products), a 3056-amino-acid protein kinase ATM (350 kDa) is translated. ATM is activated in response to double-strand DNA breaks. It is part of a network of proteins that regulate cellular responses to DNA damage (p. 80). Clinically different disorders are related at the cellular level (Nijmegen breakage syndrome, McKusick 251260, and others).

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Bloom syndrome

Extreme intrauterine and postnatal growth retardation
 Chromosomal instability
 Predisposition to leukemias, lymphomas, and other tumors
 Immune defects
 Sunlight-induced erythema of the face
 Hypo- and hyper-pigmented skin areas
 Autosomal recessive
 Gene locus on chromosome 15q26.1



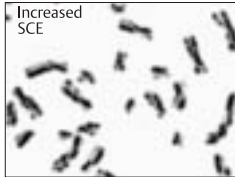
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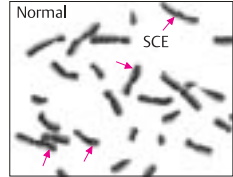
b



c



Bloom syndrome



Normal control

3. Increased rate of sister chromatid exchanges (SCE)

1. Main features

A. Bloom syndrome (BS)**Fanconi anemia**

Growth retardation
 Skeletal defects (e.g., radius and thumb)
 Bone marrow failure
 Skeletal and kidney malformation
 Localized pigment changes
 Autosomal recessive
 Several gene loci



2. Phenotype



3. Thumb hypoplasia

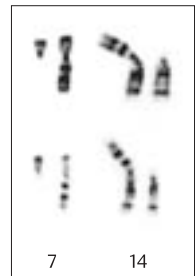
1. Main features

B. Fanconi anemia (FA)**Ataxia telangiectasia**

Cerebellar ataxia
 Immune defects
 Telangiectases of the conjunctivae
 Predisposition to tumors (lymphoma, leukemia)
 Extreme radiation sensitivity
 Autosomal recessive



2. Telangiectases



3. Translocation 7q;14p

1. Main features

C. Ataxia telangiectasia (AT)

Oxygen and Electron Transport

Hemoglobin

Hemoglobin and myoglobin are the oxygen-transporting proteins in vertebrates. Hemoglobin is found in red blood cells, myoglobin in muscle. Hemoglobin arose from myoglobin during the course of evolution. Hemoglobin has four oxygen-binding sites, myoglobin has one. Their genes and their three-dimensional protein structures are completely known in atomic detail. Different types of hemoglobin that are optimally adapted to prenatal and postnatal life have evolved in mammals from an ancestral gene.

A. Types of hemoglobin

Hemoglobin consists of four subunits (globin chains), two pairs of identical polypeptides, each polypeptide being attached to a heme group with an oxygen-binding site. The different kinds of hemoglobins, which are characteristic of different stages of development (embryonic, fetal, and postnatal), differ in the composition of their chains: the hemoglobin of adults (HbA) contains two α and two β chains ($\alpha_2\beta_2$). A small proportion of adult hemoglobin has two α and two δ chains (HbA₂: $\alpha_2\delta_2$). Hemoglobin formed during the fetal period (HbF) contains two α and two γ chains ($\alpha_2\gamma_2$). In the embryonic stage, ζ chains are joined to ϵ or γ chains (Hb Gower 1: $\zeta_2\epsilon_2$, and Hb Portland: $\zeta_2\gamma_2$); two α and two ϵ chains form Hb Gower 2 ($\alpha_2\epsilon_2$).

B. Hemoglobins in thalassemia

The thalassemias are a group of genetically determined disorders of hemoglobin synthesis. Thalassemia occurs due to the absence or reduced synthesis of a globin chain, which results in unstable hemoglobin. It affects either the β chain (β -thalassemia) or the α chain (α -thalassemia). Since the α chain is a component of both fetal and adult hemoglobins, the α -thalassemias (α -thal) are especially severe. Hemoglobins with four identical globin chains are completely unstable and incompatible with life (HbH with four β chains, Hb Bart's with four γ chains). Theoretically, a homotetramer of four δ chains could be formed but, because of the nor-

mally slow rate of synthesis of δ chains, these are clinically insignificant. The α -thalassemias occur in very different grades of severity because two neighboring genes code for the α chain (see hemoglobin genes, p. 338).

C. Evolution of hemoglobin

Since hemoglobin has four polypeptide chains, it is a much more efficient oxygen carrier than the single-chained myoglobin molecule. Furthermore, the existence of different globin chains confers a selective advantage in evolution because their slight functional differences result in optimal adaptation to the differences in oxygen concentration before and after birth. The genes for individual hemoglobin chains arose from myoglobin by a series of gene duplications during evolution. The evolutionary age of the individual Hb chains can be estimated from their differences in relation to the rate of mutation. When mammals began to evolve about 100 million years ago, the genes for α and β chains were present, whereas β -like chains (ϵ , γ , δ) evolved later.

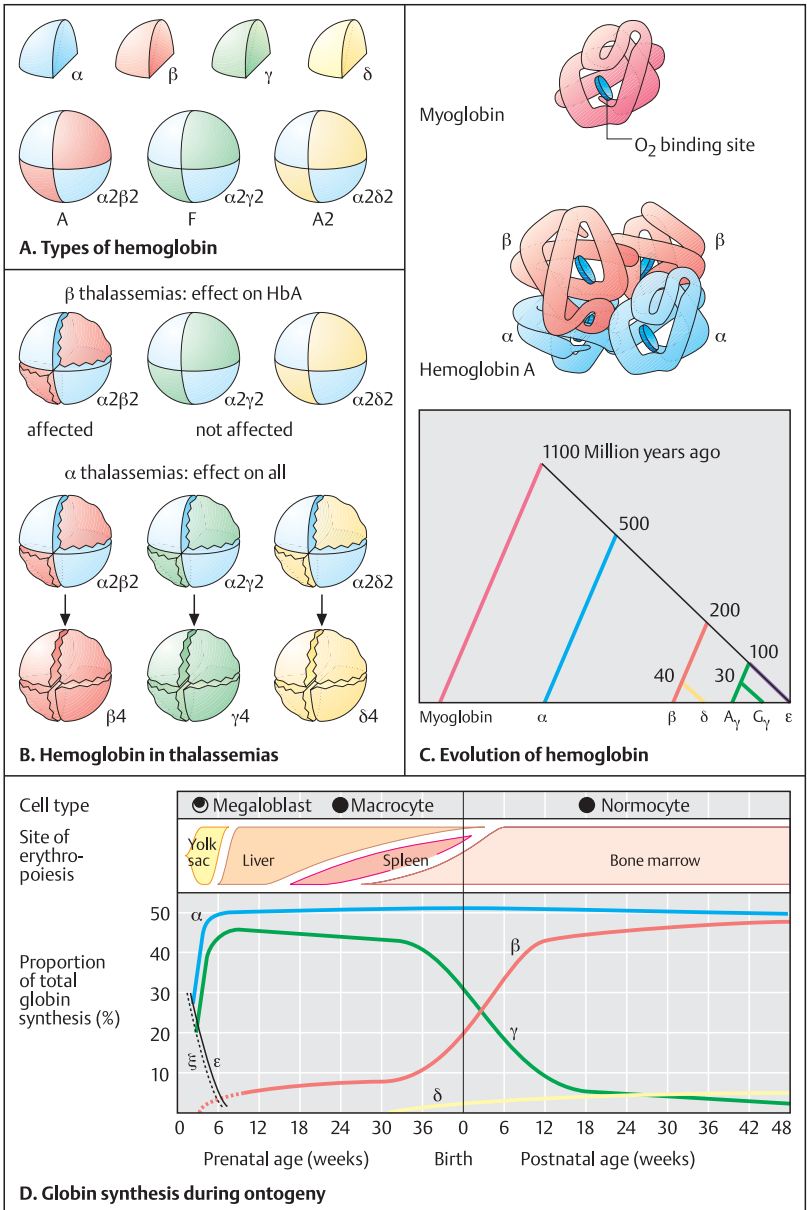
D. Globin formation in ontogeny

Different types of globin chains are formed at different developmental stages: embryonic hemoglobin during the early embryonic period (to about the 12th week), fetal hemoglobin from about the 12th week until birth, and adult hemoglobin thereafter. They differ in oxygen-binding affinity. Thus, oxygen delivery is optimized for different phases of development. The site of synthesis also differs. During the fetal phase, globin chains are synthesized mainly in the liver, whereas after birth they are synthesized in red blood cell precursors in the bone marrow.

(Figures based on Lehmann & Huntsman, 1974).

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Hemoglobin Genes

Each of the globin polypeptide chains is coded for by a gene. In humans and in other mammals, the β -like genes (β , γ , δ) are located together in a cluster on one chromosome, while the α genes are located on another. They are arranged in the order of their activation during ontogeny.

A. The β -globin and α -globin genes

The β -globin-like genes (ϵ , γ G, γ A, δ , β) of man are located on the short arm of chromosome 11 in region 1, band 5.5 (11p15.5). They span about 60 000 base pairs (bp), or 60 kb (kilobases), of DNA.

There are two γ genes, γ A and γ G, which differ only in codon 136. Codon 136 of γ A is alanine, and of γ G, glycine. A pseudogene ($\psi\beta_1$) is located between the γ gene and the δ gene. It is similar to the β gene, but has been permanently altered by deletion and an internal stop codon, so that it cannot code for a functional polypeptide. A region that jointly regulates these genes is located upstream (in the 5' direction) from the β genes (LCR, long-range control region).

In humans, two α -globin genes are located on the short arm of chromosome 16 (16p13.11 to 16p13.33) on a DNA segment of about 30 kb. A ζ gene, which is active only during the embryonic period, lies in the 5' direction. Three pseudogenes: $\psi\zeta$, $\psi\alpha_2$, and $\psi\alpha_1$ are located in between. A further gene, θ , with unknown function, has been identified in this region.

B. Structure of the α -globin and β -globin genes

As a result of their origin from a common ancestral gene, all globin genes have a similar structure. Their coding sequences are arranged in three exons. Each globin transcription unit includes nontranslated sequences at the 5' and the 3' ends (see section on the structure of eukaryotic genes, p. 50). The lengths of the β -globin and α -globin exons are similar (e.g., exon 1 of the β gene has 30 codons; exon 1 of the α gene has 31 codons), whereas the lengths of the introns differ.

C. Tertiary structure of the β -globin chain

The three-dimensional structures of myoglobin and of the hemoglobin α and β chains are very similar, although their amino acid sequences

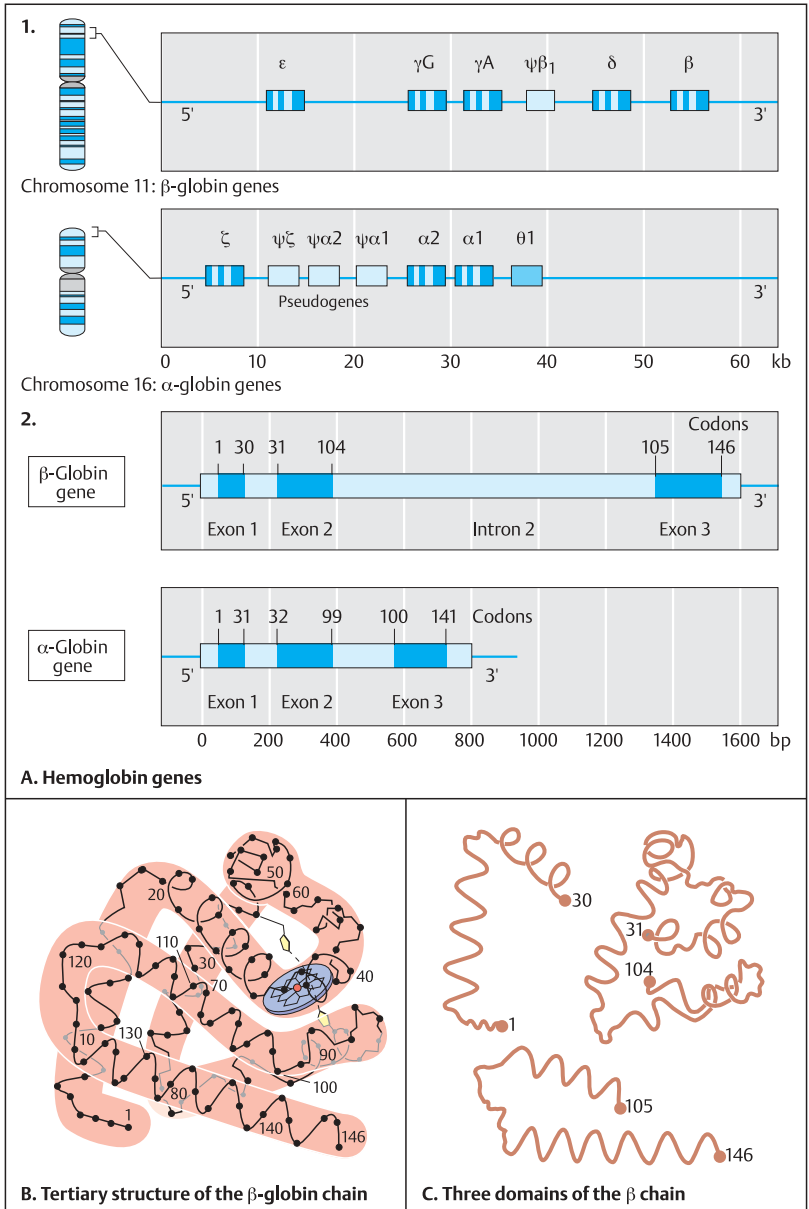
correspond in only 24 of 141 positions. The β chain, with 146 amino acids, is somewhat longer than the α chain, with 141 amino acids. The structural similarity is functionally significant: The oxygen-binding region lies inside the molecule, where it is protected, and oxygen uptake from the aqueous surroundings is reversible.

D. Domains of the β chain

Three functional and structural domains can be distinguished in all globin chains. They correspond to the three exons of the gene. Two domains, consisting of amino acids 1–30 and 105–146 (coded for by exons 1 and 3), are located on the outside. They are mainly formed of hydrophilic amino acids. A third domain, lying inside the molecule (coded for by exon 2), contains the oxygen-binding site and consists mainly of nonpolar hydrophobic amino acids. The amino acid sequences of the hemoglobins of more than 60 investigated species are identical in nine positions. These invariant positions are especially important for the function of the molecule. Changes (mutations) in the invariant positions affect function so severely that they are not tolerated.

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Sickle Cell Anemia

Sickle cell anemia is a severe progressive disease resulting from homozygosity for a mutation in the β -globin gene. It is especially frequent in Africa and in the black population of North America. With a frequency of 1 in 500, it is an important cause of morbidity and mortality in these regions. The disease is transmitted by autosomal recessive inheritance. Heterozygous carriers can readily be identified (see p. 348).

A. Sickle cells: erythrocytes deformed by hemoglobin S

In a normal blood smear under the light microscope (1), erythrocytes (red blood cells) appear as regular round disks of about 7 μ m diameter. Since a normal red blood cell is nonnucleated and biconcave, the center appears paler than the periphery. The erythrocytes of affected persons are deformed and resemble sickles (2). However, even the non-sickle-shaped red cells display unusual sizes and shapes. In the course of the disease, acute crises called sickle crises (3) occur, during which sickle-like cells are greatly increased and completely dominate the blood picture. Heterozygotes show occasional sickle cells but do not suffer from sickle crises, and at the most have only very mild signs and symptoms. (Figure from Lehmann and Huntsman, 1974.)

B. Result of a mutation: sickle cell anemia

All manifestations of sickle cell anemia are due to the substitution of a single nucleotide base in the β -globin gene. The sickle cell mutation is the transversion of the second nucleotide base of codon 6, adenine (A), to thymine (T). This changes the codon GAG, for glutamic acid, to GTG, the codon for valine. During the 1950s, Vernon M. Ingram determined the amino acid sequence of hemoglobin and found that the only difference between sickle cell hemoglobin (HbS) and normal adult hemoglobin (HbA) was this exchange in the β chain. This has far-reaching pathophysiological consequences and explains all manifestations of the disease. Sickle cell hemoglobin (HbS) is less soluble than normal hemoglobin and does not allow normal erythrocyte distortion. It crystallizes in the deoxy state and forms small rods. Thus, the

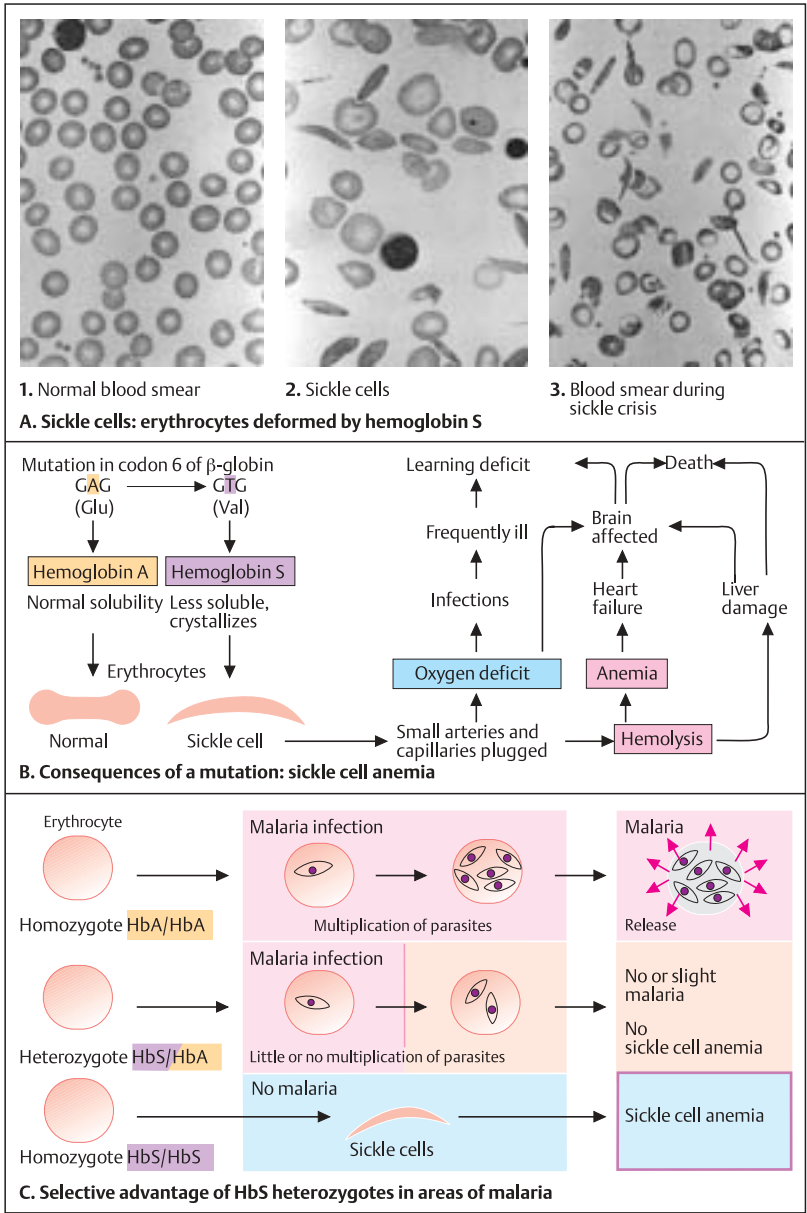
erythrocytes become firm and deform into sickle cells. Unlike normal erythrocytes, sickle cells are unable to pass through small arteries and capillaries. These become clogged and cause local oxygen deficiency in the tissues, followed by infection. As a rule, learning disability due to frequent illness occurs. Defective erythrocytes are destroyed (hemolysis). Chronic anemia and its numerous sequelae such as heart failure, liver damage, and infection are the result.

C. Selective advantage for HbS heterozygotes in areas of malaria

Heterozygotes for the sickle cell mutation are relatively resistant to malarial infection. Erythrocytes of heterozygotes for the sickle cell mutation are a less favorable environment for the malaria parasite than those of normal homozygotes. Thus, heterozygotes develop malaria in a much milder form or not at all. However, this protection is at the expense of the affected homozygotes (HbS/HbS): although they do not contract malaria, they suffer from the severe hemoglobin disorder. The protection against malaria conferred by sickle cell heterozygosity is an advantage in regions where malaria is common. With reduced morbidity and mortality, heterozygotes have a higher probability of survival and of being able to reproduce (selective advantage). This explains the high frequency of the sickle cell gene observed there (see p. 168). The sickle cell mutation has arisen independently in at least four or five different malaria-infested regions and has subsequently spread out in the respective populations. Sickle cell anemia is the best example in humans of a selective advantage in heterozygotes for a mutant allele that leads to severe illness in the homozygous state.

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Mutations in Globin Genes

All types of mutations have been demonstrated in the globin genes. The most frequent are point mutations in a single codon. The functional consequences vary, depending on the electrical charge and size of the substituted amino acid and its position in the polypeptide. If one of the hydrophilic amino acids at the surface is replaced by a hydrophobic amino acid (e.g., valine for glutamic acid in the sickle cell mutation), profound physicochemical changes will result. Mutations may decrease the elasticity of the molecule, alter its oxygen affinity, or cause instability.

A. Point mutations of the β -globin gene

Over 300 point mutations in the β -globin gene and over 100 in one of the α -globin genes have been documented. Two clinically important mutations affect codon 6: the sickle cell mutation, 6 Glu $\rightarrow$ Val (sickle cell hemoglobin, HbS, resulting in the incorporation of valine instead of glutamic acid) and 6 Glu $\rightarrow$ Lys (hemoglobin C, HbC, incorporating lysine instead of glutamic acid in codon 6). Compound heterozygotes with the HbS mutation on one chromosome and the HbC on the other (HbSC) are not rare. The marked methemoglobin formation in Hb Zürich and Hb Saskatoon results from substitutions for histidine (His) in codon 63, which alter the oxygen-binding region of the hemoglobin molecule.

B. Deletion due to unequal crossing-over within a gene

Marked sequence homology of certain regions of the globin genes may lead to nonhomologous pairing and unequal crossing-over during meiosis, e.g., in the regions of codons 90–94 of one DNA strand and codons 95–98 of the other. This explains the deletion of codons 91–95 in hemoglobin Gun Hill.

C. Unequal crossing-over between similar genes

The sequence homology of the β -globin-like genes (explained by their common evolution) may lead to unequal crossover between regions of the two γ -globin genes (γ A and γ G), the δ -globin gene, or the β -globin gene. The best known example is partial deletion of the δ and

of the β loci (δ – β fusion) in hemoglobin Lepore. The corresponding duplication of the δ/β sequence results in hemoglobin anti-Lepore.

D. Unstable hemoglobin due to chain elongation

If one of the globin chains is too long, it will destabilize the tetrameric hemoglobin molecule. Hemoglobin Cranston (Hb^{Cr}) (1) arises from the insertion of two nucleotide bases (adenine and guanine) into positions 1 and 2 of codon 145 (tyrosine) of the β chain, which leads to a shift of the reading frame. This changes the normal stop codon UAA into AGU, the RNA codon for threonine (Thr). As a result, the normally nontranslated sequences that follow the stop codon are now translated, and a polypeptide is formed that is 11 amino acids too long, extending to position 157. With hemoglobin Constant Spring (2), the α chain is lengthened by mutation of the stop codon UAA to CAA, which codes for glutamine (Gln). The sequences that normally follow the stop codon now become translated, and a peptide that is 31 amino acids too long is formed. A number of other chain-elongation mutations due to similar mechanisms, such as with hemoglobin Icaria (an α chain with 172 amino acid residues), have been described.

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Mutant hemoglobin	Codon number in β -globin gene									Important effects in homozygotes
	6	23	26	63	97	98	121	145	146	
	Glu	Val	Glu	His	Glu	Val	Glu	Tyr	His	
HbS	Val									Sickle cell anemia
HbC	Lys									
Hb Freiburg		Deletion								
HbE			Lys							
Hb Zürich				Arg						Methemoglobin formation
Hb Saskatoon				Tyr						
Hb Malmö					His					Polycythemia
Hb Köln						Met				
HbO (Arabia)							Lys			
Hb Osler								Asp		

A. Examples of point mutations in the β -globin gene (10 of 310)

β -globin sequences

Codons 89 90 91 92 93 94 95 96

Strand 1 ... AGT - GTG - CTG - CAC - TGT - GAC - AAG - CTG ...

Strand 2 ... AAG - CTG - CAC - GTG ...

Unequal crossing-over between codon 90 and 96

90 96 97 98

... AGT - GTG - CTG - CAC - GTG ... Hb Gun Hill

Deletion Codon 91-95

β -globin genes

... γ G - γ A - δ - β ...

Unequal crossing-over

... γ G - γ A - $\delta\beta$... Hb Lepore

... γ G - γ A - δ - $\delta\beta$ - β ... Hb Anti-Lepore

B. Deletion by unequal crossing-over within a gene

C. Unequal crossing-over between similar genes

β -Globin

Hb^A 144 Lys 145 Tyr 146 His Stop Not translated

Hb^{Cr} 144 Lys 145 Ser 146 Ile Thr Lys Leu Ala Phe Tyr Stop

Insertion

Frameshift eliminates stop codon after position 146

1. Hemoglobin Cranston: chain elongation by frameshift

α -Globin

Hb^A 141 Arg Stop Not translated

Hb^{Constant Spring} 141 Arg 142 Gln 143 Ala 144 Gly 145 Ala 170 Val 171 Phe 172 Glu Stop

Mutation T $\rightarrow$ C

2. Hemoglobin Constant Spring: chain elongation by mutation in the stop codon

D. Unstable hemoglobin due to chain elongation

The Thalassemias

The thalassemias are a heterogeneous group of diseases caused by decreased or absent formation of a globin chain. The thalassemias occur predominantly in the Mediterranean region, parts of Africa, and Southeast Asia (the word thalassemia is derived from *thalassa*, the Greek word for sea). In these regions, they are a significant cause of morbidity and mortality, but they are frequent because heterozygotes are protected from severe malarial infection (see p. 340).

A. Thalassemia, a chronic anemia

Depending on which globin chain is not formed in sufficient amounts, α -thalassemia, β -thalassemia, or $\delta\beta$ -thalassemia results. This leads to chronic anemia, which causes the various manifestations of thalassemia. Oxygen deficiency in the peripheral tissues leads to increased extramedullary (outside the bone marrow) blood formation. A tendency toward infection, undernourishment, and other signs characterize the severe clinical picture. (Photographs from Weatherall and Clegg, 1981).

B. β -Thalassemia and α -thalassemia

The thalassemias have a wide spectrum of different genotypes and phenotypes (disease manifestations and course). In the β -thalassemias (1), complete absence (β^0) is distinguished from decreased formation (β^+) of the β chain. With the α -thalassemias (2), one, two, three, or all four loci for α -globin may be affected. Altogether, there are 12 principal genotypes. In individuals with two mutations at the α -loci (α -thalassemia), the two can lie either on the same chromosome (thal-1) or on different chromosomes (thal-2). Thal-1 occurs mainly in Southeast Asia; thal-2, mainly in Africa. Each α gene is located within a 4 kb region of homology, interrupted by small, nonhomologous regions. The most frequent mechanism for the origin of a chromosome with only one α -globin gene is nonhomologous crossing-over between two α -globin gene loci after mispairing of the homologous chromosomes during meiosis.

C. β -Thalassemia due to different mutations

Many mutations in the β -globin gene region can lead to β -thalassemia. The mutations may also occur in noncoding sequences (5' to exon 1 and within introns).

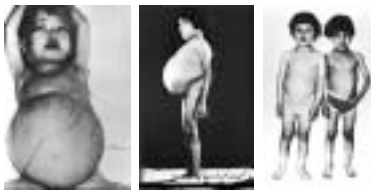
D. Haplotypes resulting from polymorphic restriction sites in the β -globin gene cluster

From the presence or absence of recognition sites of a number of restriction enzymes (restriction fragment length polymorphism, RFLP), different haplotypes can be distinguished in the β -globin-like gene region (β -globin gene cluster). Each haplotype is characterized by the presence or absence of several polymorphic restriction sites. By establishing the haplotypes of the affected and unaffected individuals within a family, the mutation-carrying haplotype can be identified (indirect genotype analysis). Different mutations have occurred on the background of different haplotypes. Frequently, a particular mutation is linked to a distinct haplotype (linkage disequilibrium). This reflects the time elapsed since the mutation first occurred in the population, where it has been maintained by selection. (Data in C and D from Antonarakis et al., 1985.)

α -Thalassemia may be associated with mental retardation. Two different syndromes can be distinguished: One occurs in patients with a large (1–2 Mb) deletion on the tip of chromosome 16 including the α -globin gene cluster (ATR-16 syndrome). The other is an X-linked disorder with a remarkably uniform phenotype and a mild form of HbH disease without α -globin deletion. A *trans*-acting regulatory factor appears to be encoded on the X chromosome.

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Different forms of thalassemia:

- α : Decreased synthesis of α -globin
- β : Decreased synthesis of β -globin
- $\delta\beta$: Decreased synthesis of δ - and β -globin

Unstable hemoglobin $\rightarrow$ Chronic anemia

A. Thalassemia, a chronic anemia

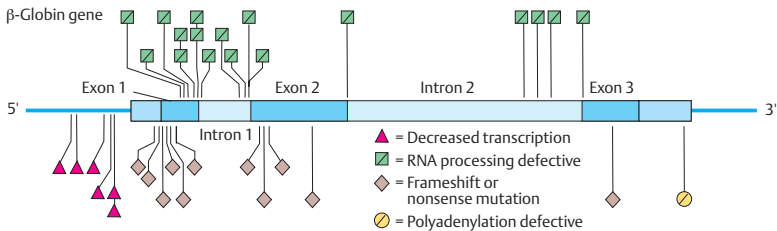
1. β -Thalassemias

Genotype	Phenotype
$\begin{matrix} + \\ - \end{matrix} \quad \beta^+ \text{ heterozygote}$	Thalassemia minor (asymptomatic)
$\begin{pmatrix} + \\ + \end{pmatrix} \quad \beta^+ \text{ heterozygote}$	
$\begin{pmatrix} + \\ + \end{pmatrix} \quad \beta^+ \text{ homozygote}$	Thalassemia intermedia (not transfusion dependent)
$\begin{matrix} + \\ - \end{matrix} \quad \beta^+ \text{ heterozygote}$	
$\begin{matrix} - \\ - \end{matrix} \quad \beta^+ \text{ homozygote} \quad (\beta^0 \text{ Thalassemia})$	Thalassemia major (transfusion dependent)
$\begin{matrix} - \\ + \end{matrix} \quad \beta^+/\beta^0 \text{ homozygote} \quad (\beta^0 \text{ Thalassemia})$	

2. α -Thalassemias

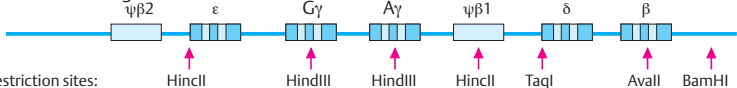
Genotype	Phenotype
$\begin{matrix} \alpha & \alpha \\ \alpha & \alpha \end{matrix}$	Normal
$\begin{matrix} \alpha & - \\ \alpha & \alpha \end{matrix}$	"Silent carrier" (normal)
$\begin{matrix} - & - \\ \alpha & \alpha \end{matrix} \quad (\text{thal-1})$	Thalassemia
$\begin{matrix} \alpha & - \\ \alpha & - \end{matrix} \quad (\text{thal-2})$	
$\begin{matrix} \alpha & - \\ - & - \end{matrix}$	HbH Disease (HbH = β_4)
$\begin{matrix} - & - \\ - & - \end{matrix}$	Hydrops fetalis

B. β -Thalassemia and α -thalassemia



C. β -Thalassemia due to different mutations

β -Globin-related genes



Frequency	Haplotype	HincII	HindIII	HindIII	HincII	TaqI	AclI	BamHI
47%	A	+	-	-	-	-	+	+
17%	B	-	+	+	-	+	+	+
8%	C	-	+	-	+	+	+	-
1%	D	-	+	-	+	+	-	+
12%	E	+	-	-	-	-	+	-

D. Haplotypes resulting from polymorphic restriction sites in the β -globin gene cluster

Hereditary Persistence of Fetal Hemoglobin (HPFH)

Hereditary persistence of fetal hemoglobin (HPFH) refers to a genetically heterogeneous group of diseases in which the temporal expression of the β -globin genes during development has been altered. Individuals with HPFH produce increased amounts of fetal hemoglobin (HbF). Under some conditions, HbF may be the only β -globin-like gene product formed. Clinically, HPFH is relatively benign, although HbF is not optimally adapted to postnatal conditions. Analysis of HPFH has yielded insight into the control of globin gene transcription and the effect of mutations in noncoding sequences.

A. Large deletions in the β -globin gene cluster

A number of very large deletions in the β -globin gene cluster region are known, especially in the 3' direction. The deletions show different distributions in different ethnic populations, reflecting that they originated at different points in time. $\delta\beta$ -Thalassemia and failure of β -globin production have been the result in some cases.

B. Mutations in noncoding sequences of the promoter region

Mutations in the noncoding sequences of the promoter region at the 5' end of the β -globin cluster (on the 5' side of the γ -globin genes) can also lead to hereditary persistence of fetal hemoglobin. Although the highly conserved sequences CACCC, CCAAT, or ATAAA are not affected, the number of observed mutations substantiates the significance of the remaining noncoding sequences (long-range transcription control). They are probably required for the changes in transcription control of the different gene loci that occur during embryonic and fetal development. (Figure after Gelehrter and Collins, 1990.)

C. Frequent mutations of β -thalassemia in different populations

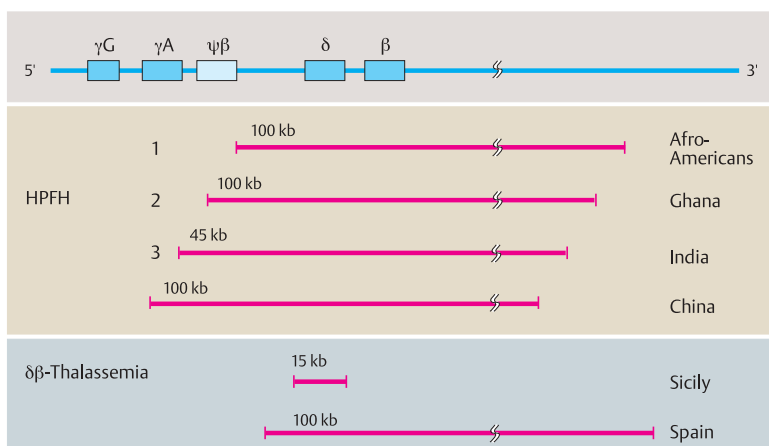
Heterozygotes for β -thalassemia mutations occur in different ethnic populations with different frequencies. Since a few mutations are quite frequent in certain populations, preventive diagnostic programs to determine the risk

of disease are possible. (Data after Antonarakis et al. 1985.)

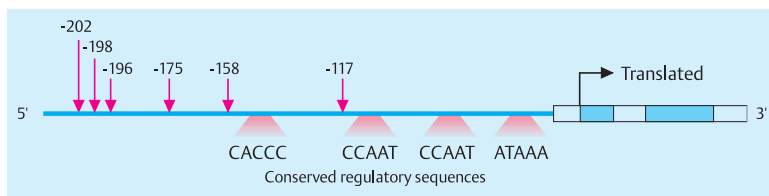
According to estimates of the WHO (Bull World Health Org. 1983) about 275 million persons are heterozygotes for hemoglobin diseases worldwide. Substantial numbers are due to the β -thalassemias in Asia (over 60 million), α^0 -thalassemia in Asia (30 million), HbE/ β -thalassemia in Asia (84 million), and sickle cell heterozygosity in Africa (50 million), India, the Caribbean, and the USA (about 50 million). At least 200 000 severely affected homozygotes are born annually, about 50%, due to sickle cell anemia and 50% to thalassemia (Weatherall, 1991).

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A. Large deletions in the β -globin cluster



B. Mutations in noncoding sequences in the promoter of γ -globin genes cause hereditary persistence of fetal hemoglobin (HPFH)

β -thal mutation	Frequency	Ethnic group	Type
Intron 1 (110 G $\rightarrow$ A)	35%	Mediterranean	β^+
Codon 39 (C $\rightarrow$ T)	27%	Mediterranean	β°
TATA-Box (-29 A $\rightarrow$ G)	39%	Afro-Americans	β^+
Poly A (T $\rightarrow$ C)	26%	Afro-Americans	β^+
Intron 1 (5 G $\rightarrow$ C)	36%	India	β^+
Partial deletion (619 nt)	36%	India	β°
Codon 71-72 frameshift	49%	China	β°
Intron 2 (654 C $\rightarrow$ T)	38%	China	β°

C. Frequent mutations in β -thalassemia in different populations

DNA Analysis in Hemoglobin Disorders

Numerous procedures that do not require direct determination of the altered nucleotide base sequence can be used to demonstrate a mutation. An available probe of the gene or gene region being investigated and knowledge of the normal Southern blot pattern after restriction analysis of the gene are prerequisites (cf. p. 62).

A. Direct demonstration of a deletion

A partial deletion may cause an altered pattern of a Southern blot. Two genes ($\alpha 2$ and $\alpha 1$) of α -globin are presented (1). They are both located on a 14.5 kb restriction fragment. If partial deletion results in loss of a segment of, e.g., about 4.5 kb that is part of both the $\alpha 2$ and the $\alpha 1$ gene, a fragment of 10.0 instead of 14.5 kb will result in this area (2). Three genotypes are possible (3): two normal genes, $\alpha 2$ and $\alpha 1$, represented by a 14.5 kb fragment; a normal DNA segment (14.5 kb) and one with a deletion (10.0 kb); or a deletion in both gene segments (only one fragment, of 10.0 kb). This can be demonstrated directly in the Southern blot pattern with a probe for the $\alpha 1$ gene (4).

B. Indirect evidence for a mutation by RFLP analysis

Indirectly, a mutation can be demonstrated if there is an individual difference (polymorphism) in the base sequences of the mutant and the normal gene segments (restriction fragment length polymorphism, RFLP, see p. 64). For instance, if two of the same DNA segments differ in a polymorphism for the recognition sequence of a restriction enzyme, then DNA fragments of different sizes (here, either 7 kb and 6 kb, or 13 kb) result after cleavage with the enzyme (1). If the mutation has occurred within the 13 kb fragment, then this fragment indicates presence of the mutation. In the given DNA segment, there are three possibilities (genotypes): two fragments of 7 kb without mutation; one fragment of 7 kb (normal) and one fragment of 13 kb (which carries the mutation); and two mutation-carrying fragments of 13 kb (3). The Southern blot (4) shows whether the person being examined is homozygous normal (has no 13 kb fragment), is heterozygous (a

7 kb and a 13 kb fragment), or is homozygous for the mutation (two 13 kb fragments).

The prerequisite for this indirect analysis is previous knowledge of which of the DNA fragments contains the mutation. The observed difference is not the result of the mutation, as in A. If the Southern blot pattern of affected and unaffected individuals does not differ, then this method will not be informative for the disorder.

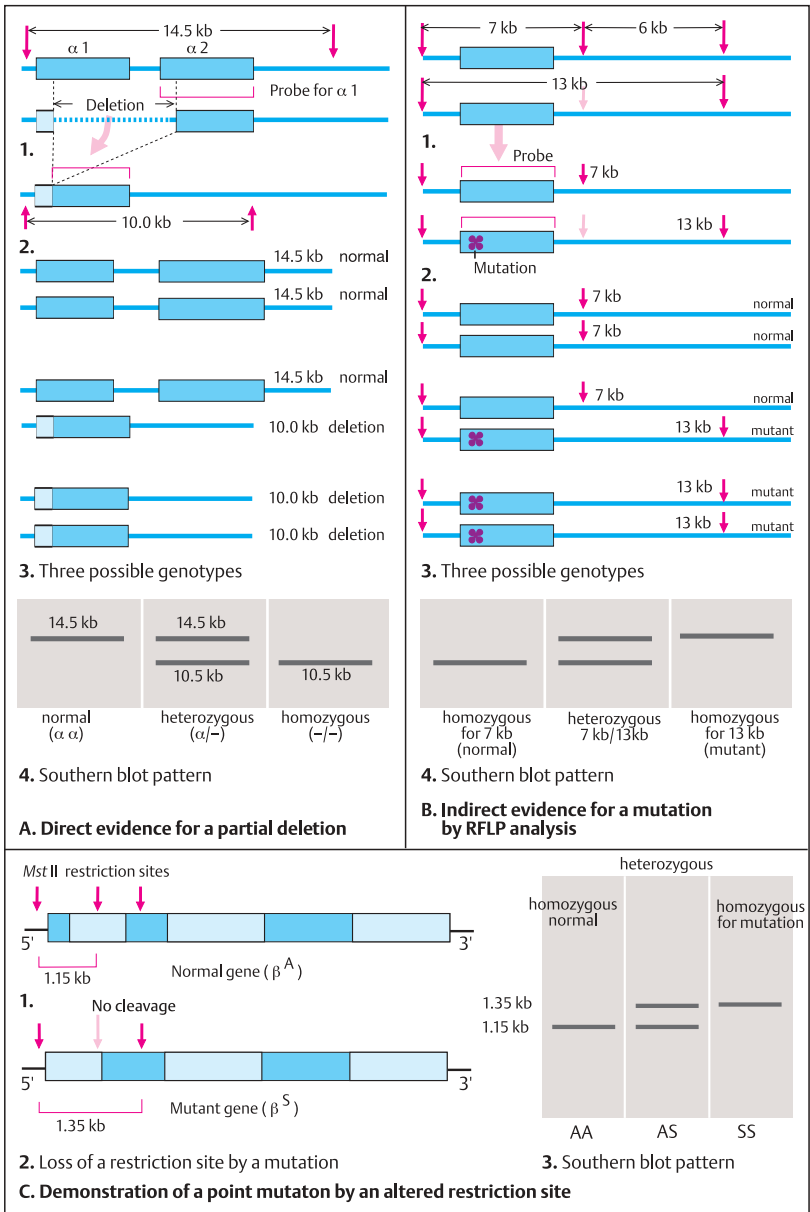
C. Demonstration of a point mutation from an altered restriction site

A restriction site may be altered by a mutation. For example, a sickle cell mutation in codon 6 of the β gene of hemoglobin (see p. 340) (1) causes loss of a restriction site for the enzyme *MstII* (CCTNAGG instead of CCTNTGG) because the A (adenine) has been replaced by a T (thymine) (2). The normal allele (β^A) in this area produces a 1.15 kb fragment after *MstII* digestion, whereas the mutation eliminates the restriction site in the middle so that a 1.35 kb fragment results. The 1.35 kb fragment in the Southern blot indicates (3) the presence of the sickle cell mutation (β^S). Thus, homozygous normal individuals (AA), heterozygotes (AS), and homozygotes for the sickle cell mutation (SS) can be clearly distinguished; each of the three genotypes can be precisely diagnosed.

Increasingly, RFLP analysis is being replaced by the analysis of small polymorphic DNA repeats (microsatellites, see p. 72).

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Peroxisomal Diseases

Peroxisomes are small round organelles about 0.5–1.0 μm diameter (somewhat smaller than mitochondria). They are found mainly in the cytoplasm of kidney and liver cells. They are the site of some important metabolic functions. The name is derived from hydrogen peroxide, which is formed as an intermediary product of oxidative metabolism in the peroxisomes. A number of defects in peroxisome formation or peroxisome enzymes lead to severe diseases in humans (peroxisomal diseases).

A. Biochemical reactions in peroxisomes

The electron micrograph (1) shows peroxisomes in a section of rat liver. The dark striated structures within the organelles consist of urates (peroxisomes contain an enzyme that oxidizes uric acid). Peroxisomes have both catabolic (substances are degraded) and anabolic (substances are synthesized) functions (2). Two biochemical reactions are especially important: a peroxisomal respiratory chain and the β -oxidation of very long-chain fatty acids. In the peroxisomal respiratory chain (3), certain oxidases and catalases act together. Specific substrates of the oxidases are organic metabolites of intermediary metabolism. Very long-chain fatty acids are broken down by β -oxidation (4) in a cycle with four enzymatic reactions. Energy production in peroxisomes is relatively inefficient compared with that of mitochondria. While free energy in mitochondria is mainly preserved in the form of ATP (adenosine triphosphate), in peroxisomes it is mostly converted into heat. Peroxisomes are probably a very early adaption of living organisms to oxygen. (Photograph from de Duve, 1986.)

B. Peroxisomal diseases

Several peroxisomal diseases are known in man; the six most important are listed. All are autosomal recessive hereditary disorders. Patients with neonatal adrenoleukodystrophy do not form sufficient amounts of plasmalogens and cannot adequately degrade phytanic acid and pipecolic acid. When cultured fibroblasts from patients with genetically different types of peroxisomal diseases are fused, the hybrid cells form normal peroxisomes (cells with different defects can correct each other). More than ten

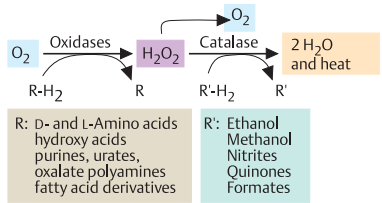
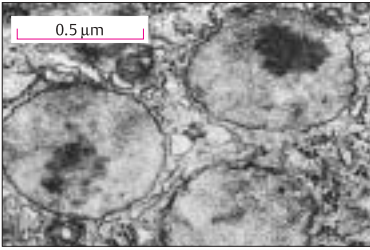
complementation groups are known (Raymond et al., 1999).

C. Cerebro-hepato-renal syndrome type Zellweger

Patients with this autosomal recessive hereditary disease have a characteristic facial appearance (1–4), extreme muscle weakness (5), and a number of accompanying manifestations such as calcified stippling of the joints on radiographs (6), renal cysts (7, 8), and clouding of the lens and cornea. The severe form of the disease (type Zellweger) usually leads to death before the age of one year. (Photographs 1–5 from Passarge and McAdams, 1967.)

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1. Peroxisomes in a rat liver cell

a) Catabolic

H_2O_2 -involving cellular respiration
 β -Oxidation of long-chain fatty acids,
 Prostaglandins, cholesterol side chains and others
 Purines, urates
 Pipecolic acid, dicarboxy acids
 Ethanol, methanol

b) Anabolic

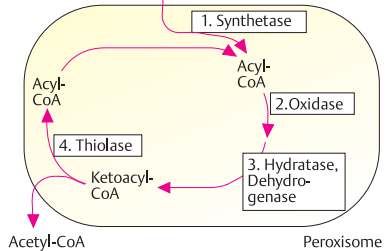
Phospholipids (Plasmalogen)
 Cholesterol, bile acids
 Gluconeogenesis
 Glyoxalate transamination

2. Function of peroxisomes

A. Biochemical reactions in peroxisomes

3. Peroxisomal respiratory chain

Very long-chain fatty acids (more than 12 C)



4. β -Oxidation

214100 Cerebro-hepato-renal syndrome Zellweger
 202370 Neonatal adrenoleukodystrophy
 266510 Infantile Refsum disease

239400 Hyperpipecolic acidemia
 215100 Rhizomelic chondrodysplasia punctata
 259900 Primary hyperoxaluria type I
 and others

B. Examples of peroxisomal diseases



1.



2.



5.



6.



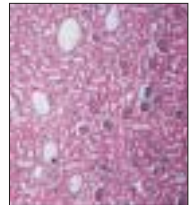
3.



4.



7.



8.

C. Cerebro-hepato-renal syndrome Zellweger

Lysosomes and LDL Receptor

Lysosomes and Endocytosis

Lysosomes are membrane-enclosed cytoplasmic organelles with a diameter of 0.05–0.5 μm . They contain a wide variety of active hydrolytic enzymes (hydrolases) such as glycosidases, sulfatases, phosphatases, lipases, phospholipases, proteases, and nucleases (lysosomal enzymes) in an acid milieu (pH about 5). Lysosomal enzymes enter a lysosome by means of a recognition signal (mannose 6-phosphate) and a corresponding receptor.

A. Receptor-mediated endocytosis and lysosome formation

Extracellular molecules to be degraded are taken into the cell by endocytosis. First, the molecules are bound to specific cell surface receptors (receptor-mediated endocytosis). The loaded receptors are concentrated in an invagination of the plasma membrane (coated pit). This separates from the plasma membrane and forms a membrane-enclosed cytoplasmic compartment (coated vesicle). Hormones, growth factors, energy-delivering proteins, and numerous viruses and toxins also enter cells by receptor-mediated endocytosis (see p. 360). The cytoplasmic lining of the vesicle consists of a network of a trimeric protein, clathrin. The clathrin coat is quickly lost within the cell, and an endosome forms, which fuses with membrane vesicles from the Golgi apparatus to form larger endosomal compartments. Here, the receptors are separated from the ligands and are returned to the cell surface in membrane vesicles (receptor recycling). Parts of the membrane are also reused. The ligands are now within a multivesicular body (endolysosomes). Hydrolases (lysosomal enzymes) are transported from the Golgi apparatus to an endolysosome in clathrin-enclosed vesicles after they become equipped with a recognition signal (mannose-6-phosphate receptor), required for uptake into the endolysosome and for normal functioning of the lysosome.

There are different classes of endolysosomes, defined according to relative acidity, receptor content, biochemical composition, morphological appearance, and other characteristics.

The acid milieu in the lysosomes is maintained by a hydrogen pump in the membrane, which hydrolyzes ATP and uses the energy produced to move H^+ ions into the lysosome. Some of the mannose-6-phosphate receptors are transported back to the Golgi apparatus.

B. Mannose-6-phosphate receptors

There are two types of mannose-6-phosphate receptor molecules, which differ in their binding properties and their cation dependence. They consist of either 2 or 16 extracellular domains with different numbers of amino acids. The cDNA of Ci-MPR (cation-independent mannose-6-phosphate receptor) is identical with insulinlike growth factor II (IGF-2). Thus, Ci-MPR is a multifunctional binding protein.

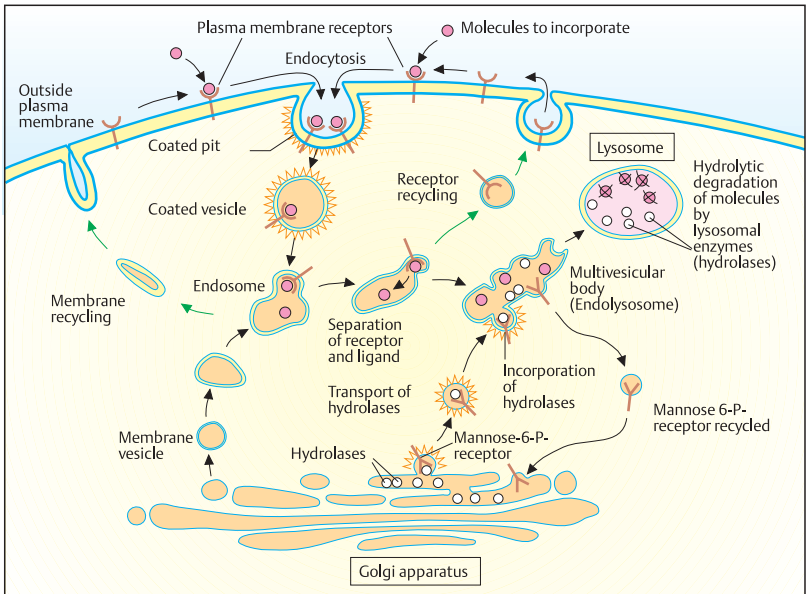
C. Biosynthesis of the recognition signal

Two enzymes are essential for the formation of mannose-6-phosphate recognition signals: a phosphate transferase and a phosphoglycosidase. The phosphate is delivered by uridine-diphosphate-*N*-acetylglucosamine (UDP-GlcNAc) (uridine-5'-diphosphate-*N*-acetylglucosamine-glycoprotein-*N*-acetylglucosaminylphosphotransferase). A second enzyme, (*N*-acetylglucosamine-1-phosphodiester-*N*-acetylglucosaminidase) cleaves off the *N*-acetylglucosamine, leaving the phosphate residue at position 6 of the mannose.

(Figures after Sabatini and Adesnik, 1995, and C. de Duve, 1986; the relative sizes of the individual structures are not to scale.)

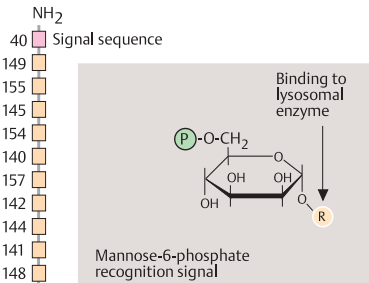
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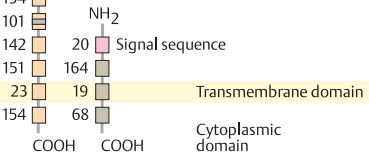


A. Receptor-mediated endocytosis and lysosome formation

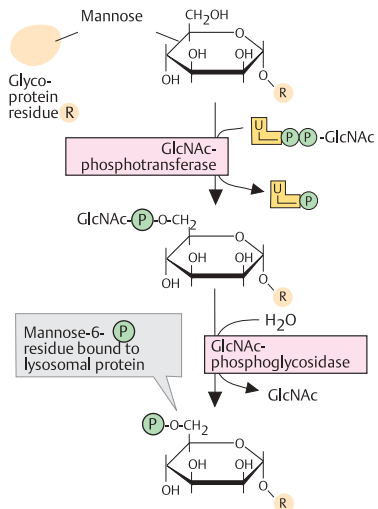
CI-MPR (Cation-independent)



CD-MPR (Cation-dependent)



B. Mannose-6-phosphate receptors (MPR)



C. Biosynthesis of mannose-6-phosphate recognition signal

Diseases Due to Lysosomal Enzyme Defects

Mutations in genes for enzymes that degrade complex macromolecules in lysosomes (lysosomal enzymes) lead to disease. Clinical signs and biochemical and cellular manifestations depend on the altered enzyme's specificity in lysosomal metabolism. With such an enzyme defect, macromolecules that normally would be degraded are stored (storage disease). This occurs at different rates, so that each disease has its own characteristic course. Twelve groups of diseases due to genetically determined disorders of specific lysosomal function are known, each with about three to ten individually defined diseases.

A. Defective uptake of enzymes into lysosomes: I-cell disease (mucopolidiosis type II)

Due to a mutation of the gene on chromosome 4 for the phosphotransferase needed to form the mannose-6-phosphate recognition signal (see p. 352), hydrolases cannot be taken up into lysosomes. Unlike normal cultured fibroblasts (1), those of patients (2) contain numerous densely packed cytoplasmic inclusion bodies (thus, the name I-cell disease). The vesicular inclusions consist of hydrolases that cannot enter the lysosomes because the mannose-6-phosphate recognition signal is absent. Numerous enzymes are missing from the lysosomes, while their concentration in other parts of the cells and in body fluids is increased. Patients (3) show a severe progressive clinical picture, with the first signs usually apparent in the first half-year of life.

B. Degradation of heparan sulfate by eight lysosomal enzymes

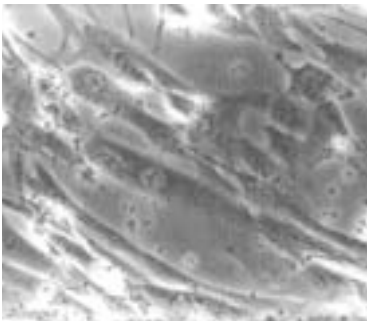
Heparan sulfate is an example of a macromolecule that is degraded stepwise by different lysosomal enzymes. Lysosomal enzymes are bond-specific, not substrate-specific. Thus, they also degrade other glycosaminoglycans, such as dermatan sulfate, keratan sulfate, and chondroitin sulfate (mucopolysaccharides). Specific enzyme defects cause the mucopolysaccharide storage diseases (see next page).

The first step in mucopolysaccharide degradation is the removal of sulfate from the terminal

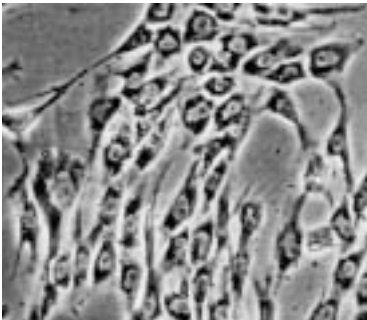
iduronate group by an iduronate sulfatase. A defect in the gene that codes for this enzyme leads to mucopolysaccharide storage disease type II (Hunter). The gene is located on the X chromosome, so that Hunter disease is transmitted by X-chromosomal inheritance. All other mucopolysaccharidoses are autosomal recessive. In the next step (2), the terminal iduronate is split off by an α -L-iduronidase. A mutation of the gene coding for this enzyme in the homozygous state leads to mucopolysaccharidosis (MPS) type I (Hurler/Scheie). In the next three steps a mutation (in the homozygous state) of a gene coding for one of the enzymes causes mucopolysaccharidosis type III (Sanfilippo). The four genetically and enzymatically different types (III-A to III-D) cannot be distinguished clinically. MPS type VII (Sly), due to a defect of β -glucuronidase, has a further characteristic clinical picture.

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1. Normal fibroblast culture

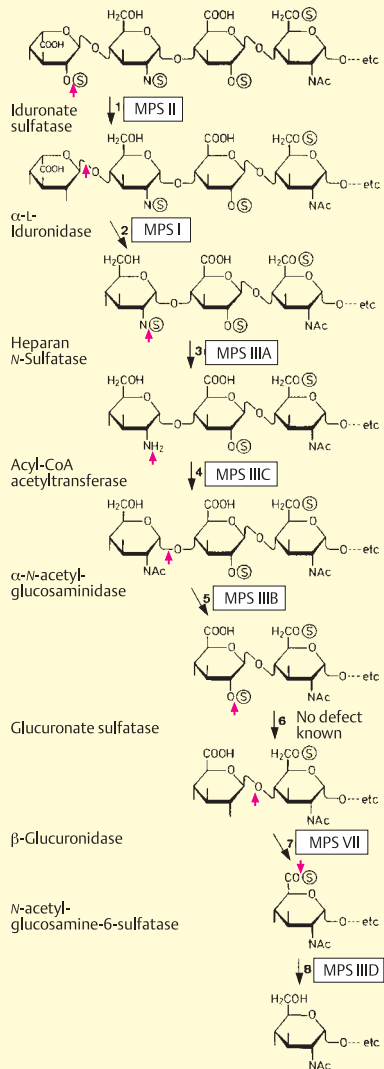


2. Fibroblast culture in I-cell disease



3. Patient with I-cell disease

A. Defective uptake of enzymes in lysosomes: I-cell disease



B. Degradation of heparan sulfate by eight lysosomal enzymes

Mucopolysaccharide Storage Diseases

The mucopolysaccharide storage diseases (the mucopolysaccharidoses) are a clinically and genetically heterogeneous group of lysosomal storage diseases caused by defects in different enzymes for mucopolysaccharide degradation (glycosaminoglycans). Except for mucopolysaccharide storage disease type II (Hunter), all are transmitted by autosomal recessive inheritance.

A. Mucopolysaccharide storage disease type I (Hurler)

At first almost inapparent, the early signs of the disease occur at about 1 – 2 years of age, with increasing coarsening of the facial features, retarded mental development, limited joint mobility, enlarged liver, umbilical hernia, and other

signs. Radiographs show coarsening of skeletal structures (dysostosis multiplex). The photographs show the same patient at different ages (own data).

B. Mucopolysaccharide storage disease type II (Hunter)

This type of mucopolysaccharidosis is transmitted by X-chromosomal inheritance. Four cousins from one pedigree are shown. Clinically, the disease is similar to, but less rapidly progressive than, MPS type I. (Photos from Passarge et al., 1974). Molecular diagnosis is possible in most cases.

References

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Classification of the mucopolysaccharide storage diseases (MPS)

MPS Type	Enzyme Defect	Important Manifestations
IH (Hurler)	α -L-Iduronidase	Dysostosis multiplex, severe developmental disorder, corneal clouding
IS (Scheie)	α -L-Iduronidase	Stiff joints, corneal clouding, normal mental development
II (Hunter) (X-chromosomal)	Iduronate sulfatase	Dysostosis multiplex, no corneal clouding, developmental retardation
III (Sanfilippo) A	Heparan N-sulfatase	Severe psychomotor retardation beginning about age 6 – 8 years, relatively mild somatic signs.
B C	α -N-Acetylglucosaminidase Acetyl-CoA: α -glucosaminide N-acetyltransferase	
D	N-acetylglucosamine-6-sulfate sulfatase	
IV (Morquio) A	Galactose-6-sulfatase	Corneal clouding, severe skeletal changes, short stature, odontoid process hypoplasia, normal mental development
B	β -Galactosidase	
VI (Maroteaux – Lamy)	N-acetylgalactosamine-4-sulfatase (aryl-sulfatase B)	Dysostosis multiplex, corneal clouding, normal mental development
VII (Sly)	β -Glucuronidase	Dysostosis multiplex, corneal clouding

(After McKusick, 1998)
Passarge, Color Atlas of Genetics © 2001 Thieme



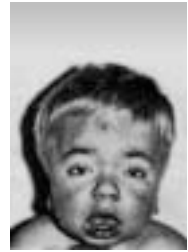
8 weeks



7 months



2 1/4 years



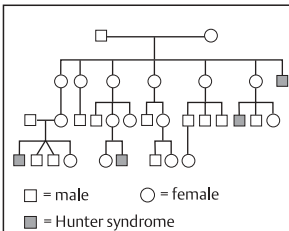
3 3/4 years



5 years



8 years

Dysostosis
multiplexJoint
contractures

X-Chromosomal inheritance



4 1/2 years



10 years



13 years



21 years

A. Mucopolysaccharide storage disease type I (Hurler)

B. Mucopolysaccharide storage disease type II (Hunter)

Familial Hypercholesterolemia

Familial hypercholesterolemia (FH) is a hereditary disorder of intracellular lipid metabolism. Several different genetic forms exist, each characterized by the step of the metabolic pathway involved and the type of mutation. Mixed forms, caused by multigenic and environmental factors, and monogenic forms can be distinguished.

A. The disease phenotype

Familial hypercholesterolemia (1) (McKusick 144440) occurs in about 1 in 500 persons in the heterozygous state. In the rare homozygous state (1 in 10^6) it is a devastating disease usually leading to death during the first or second decade of life. The heterozygous form is characterized by early signs of atherosclerosis (2). The number of functional LDL receptors per cell is decreased by about 50% (3). Deposits of cholesterol esters in the tendons, especially the achilles tendon, and the skin (xanthomas) are common in heterozygotes (4). A characteristic sign is lipid deposits in the eye in front of the iris (arcus senilis, 5).

B. The LDL receptor

The LDL receptor is a cell surface receptor for low-density lipoprotein (LDL), which contains apoB-100, the protein that carries most of the cholesterol ester in human plasma. This receptor mediates endocytosis of the extracellular ligand. The LDL receptor is a membrane-bound protein of 839 amino acids with five domains: three extracellular domains, one transmembrane domain, and one intracellular, with the carboxyl end. The extracellular domains consist of one domain with seven cysteine-rich units of 40 amino acids each, the ligand-binding region; a domain with epidermal growth factor (EGF) precursor homology; and a small serine- and threonine-rich domain linked to oligosaccharides. The transmembrane domain contains 22 hydrophobic amino acids. The fifth domain with the intracellular COOH terminus consists of 50 amino acids. It controls the interaction of the receptor with the coated pit during endocytosis. The corresponding gene consists of 18 exons that span 45 kb genomic DNA on human chromosome 19p13.1–13.3. In addition to the main locus on 19p, two additional loci for autosomal dominant hypercholesterolemia exist

(Varret et al., 1999; Haddad et al., 1999). (Figure adapted from H. Schuster, Berlin).

C. LDL receptor-mediated endocytosis

The LDL receptor mediates the endocytosis of LDL. The receptors loaded with LDL accumulate in a coated pit (a), which separates from the plasma membrane and forms an endocytotic vesicle (b). This transports LDL molecules to a lysosome. (Photograph from Anderson et al., 1977).

D. Homology with other proteins

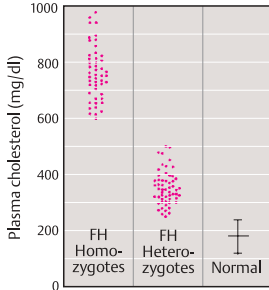
The mammalian LDL receptor is one of a five-member family including the LDL receptor itself, the VLDL receptor (very low density lipoprotein), the ApoE receptor 2 (ApoER2), the LDL receptor-related protein (LRP), and megalin. LRP and megalin are multifunctional and bind diverse ligands such as lipoproteins, protease and their inhibitors, peptide hormones, and carrier proteins of vitamins (Krieger and Herz, 1994). The proximal halves of the extracellular domains of the LDL receptor family are structurally related to the epidermal growth factor family (EGF). These are related to protease of the blood coagulation system, factors IX and X, protein C, and complement C9.

References

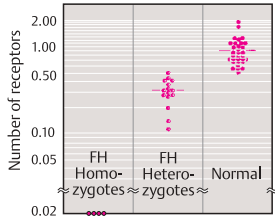
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- Low-density lipoprotein (LDL) and cholesterol elevated in blood plasma
- Premature arteriosclerosis
- Xanthoma in skin and tendons
- Decreased life expectancy
- Autosomal dominant
- Mutation in LDL receptor gene

1. General features



2. Hypercholesterolemia



3. LDL receptors decreased

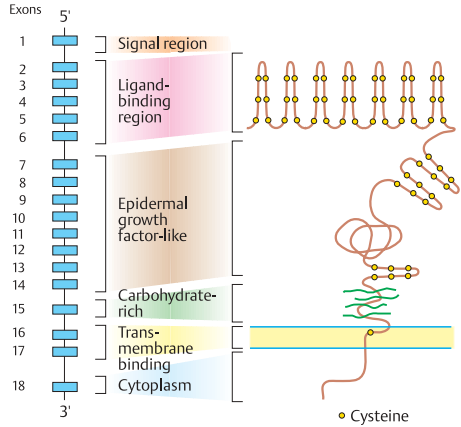


4. Xanthoma formation

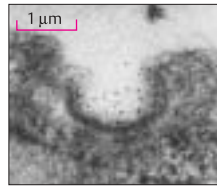


5. Arcus lipoides

A. Familial hypercholesterolemia



B. LDL Receptor



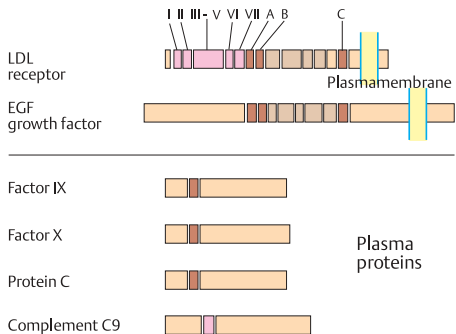
a Coated pit



b Endocytotic vesicle

Electron micrographs of fibroblasts in culture that have taken up LDL molecules (black dots, made visible by binding to ferritin).

C. Receptor-mediated endocytosis of LDL



D. Homology with other proteins

Mutations in the LDL Receptor

Low-density lipoprotein (LDL) is the main carrier of cholesterol in the blood. An LDL particle has a diameter of 22 nm and a molecular mass of about 3000 kDa. Its hydrophobic core contains about 1500 esterified cholesterol molecules surrounded by an outer layer of phospholipids and unesterified cholesterol containing a single apoB-100 lipoprotein molecule. LDL delivers cholesterol to peripheral tissues and regulates *de novo* cholesterol synthesis there. Mutations in the LDL receptor gene or in the ligand apoB-100 lipoprotein result in hypercholesterolemia.

A. Intracellular LDL receptor metabolism and classes of mutation

Five principal classes of LDL receptor mutations can be distinguished: (1) receptor null mutations (R^-) due to lack of receptor protein synthesis in the endoplasmic reticulum (ER); (2) defective intracellular transport to the Golgi apparatus; (3) defective extracellular ligand binding; (4) defective endocytosis (R^+ mutations); and (5) failure to release the LDL molecules inside the endosome (recycling-defective mutations). The receptor-LDL complex enters the cell by endocytosis. In the endosome the LDL including of apoB-100 is separated from the receptor. In the lysosome the LDL is broken down into amino acids and cholesterol. Free cholesterol activates the enzyme acetyl-CoA cholesterol transferase (ACAT), which catalyzes the esterification. The LDL receptor is recycled to the cell surface in a recycling vesicle. The key enzyme for endogenous cholesterol synthesis is 3-hydroxy-3-methylglutaryl-CoA reductase (HMG-CoA reductase). This enzyme is downregulated by exogenous LDL uptake. LDL receptor mutations interrupt this control feedback mechanism and result in increased endogenous cholesterol synthesis. HMG-CoA reductase also downregulates LDL receptor protein synthesis to prevent overloading with cholesterol.

B. Mutational spectrum in the LDL receptor gene

About 350 mutations have been recorded in the LDL receptor gene (Varret et al., 1998). Of these, 63% are missense mutations. Mutations occur in all parts of the gene, but there is a relative

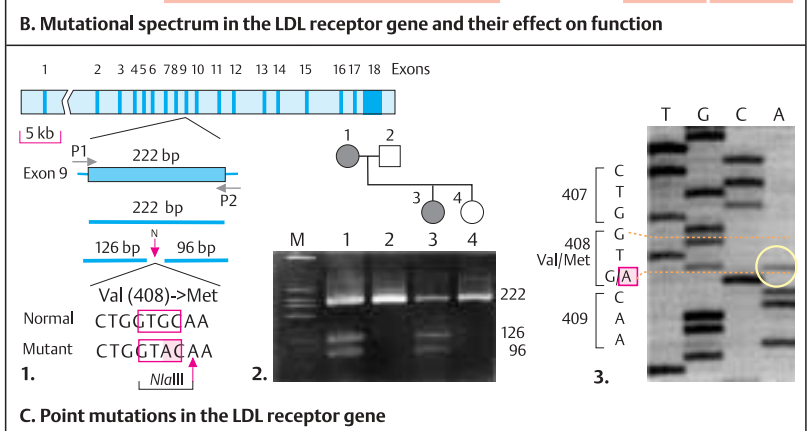
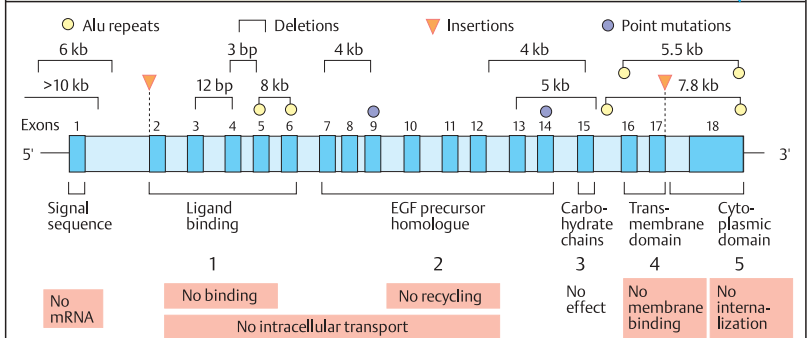
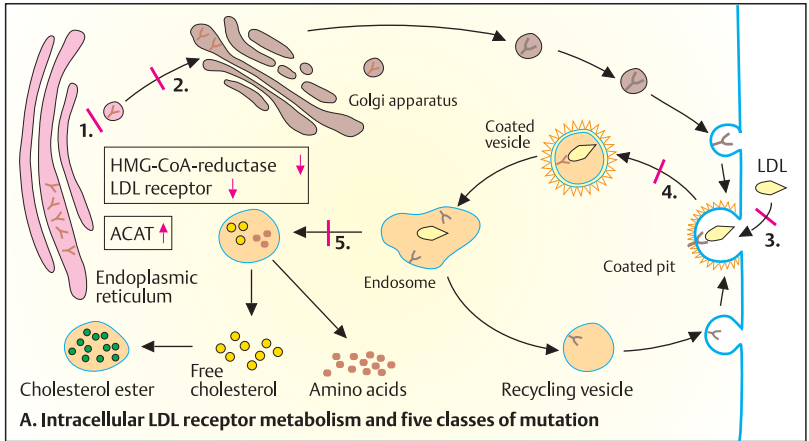
excess of mutations in exons 4 and 9. Exons 13 and 15 are involved less often than expected. A high proportion of mutations (74%) located in the ligand-binding domain (exons 2–6) involve amino acids conserved in evolution. (Varret et al., 1998; data also accessible at <http://www.umd.necker.fr>). In addition to point mutations, several deletions of various sizes and locations, and insertions have been described. Depending on the intragenic location of a mutation, different effects can be observed, including absent mRNA synthesis, defective intracellular transport due to abolished binding (1) or receptor recycling (2), reduced membrane anchorage (4), and defective internalization (5). Alu repeats may be involved as a cause of intragenic deletions.

C. Diagnosis of a point mutation in the LDL receptor gene

Direct sequencing demonstrates a mutation in exon 9. First, exon 9 is amplified by PCR (P1 and P2 = primers 1 and 2). The mutation in codon 408, GTG (valine) to GTA (methionine), produces a recognition site (N) for *Nla*III (GATC) that is not normally present. This results in two fragments of 126 and 96 base pairs (bp) instead of the usual 222 bp fragment. Thus, affected individuals (1 and 3 in the pedigree) have two smaller fragments of 126 and 96 kb (2) in addition to the 222 kb fragment. Sequence analysis of the patient (individual 1 in the pedigree) demonstrates the mutation by the presence of an additional adenine (A) next to the normal guanine (3). With this knowledge about the mutation, the latter can be indirectly demonstrated within a family by the additional recognition site for a restriction enzyme. (Data from H. Schuster, Berlin).

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Homeostasis

Insulin and Diabetes Mellitus

Diabetes mellitus is one of the most common diseases of the Western world, occurring in about 1–2% of the population. The blood sugar is abnormally elevated due to a variety of causes, including genetic factors. With time, this leads to numerous complications, such as myocardial infarction, stroke, renal failure, vascular damage leading to amputation, and blindness.

A. Insulin production

In humans, the gene for insulin is located on the short arm of chromosome 11 in region 1, band 5.5. With 1430 base pairs, it is a small gene. It consists of a signal sequence (L, leader) and two exons. The gene is expressed exclusively in β -cells of the islands of Langerhans of the pancreas. A β -cell-specific enhancer is located at the 5' end of the gene, and a variable number of tandem repeats (VNTR) are located further upstream. The primary transcript is spliced to produce the mRNA template for preproinsulin. The signal peptide (24 amino acids) is removed, and the B chain and A chain are joined by two disulfide bridges. Proper binding and three-dimensional structuring require the presence of a connecting peptide (C peptide). The complete insulin molecule consists of an A chain of 21 amino acids and a B chain of 30 amino acids. The signal peptide of the insulin molecule is required for secretion.

B. Insulin receptor

Insulin initiates its physiological effect by binding to a receptor (insulin receptor). When bound to insulin, the insulin receptor functions as an enzyme and phosphorylates tyrosine in the target proteins. This is the intracellular signal for the metabolic processes induced by insulin.

C. Diabetes mellitus (simplified model)

Diabetes mellitus is classified into two basic types: type I (insulin-dependent diabetes mellitus, IDDM) and type II (non-insulin-dependent, NIDDM). The majority of diabetes type I cases are caused by external factors, such as certain viral infections, on a background of genetic susceptibility. Diabetes type II is mainly due to

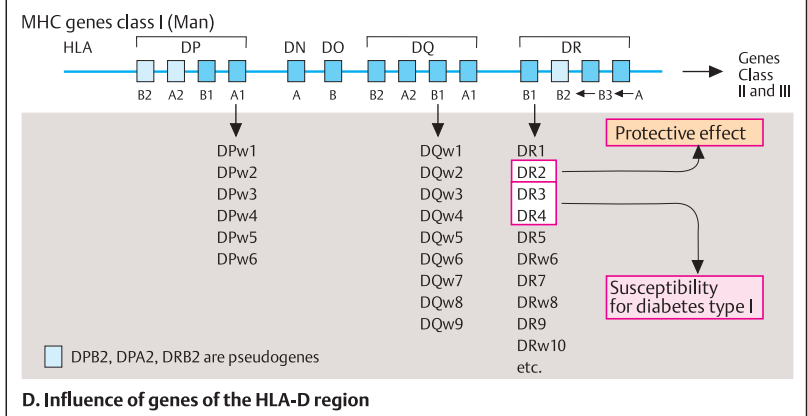
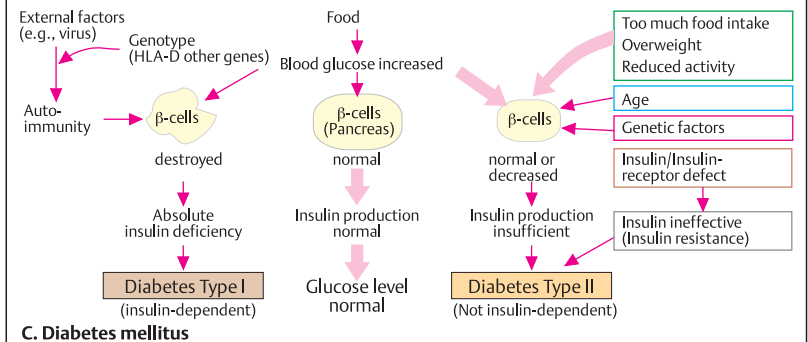
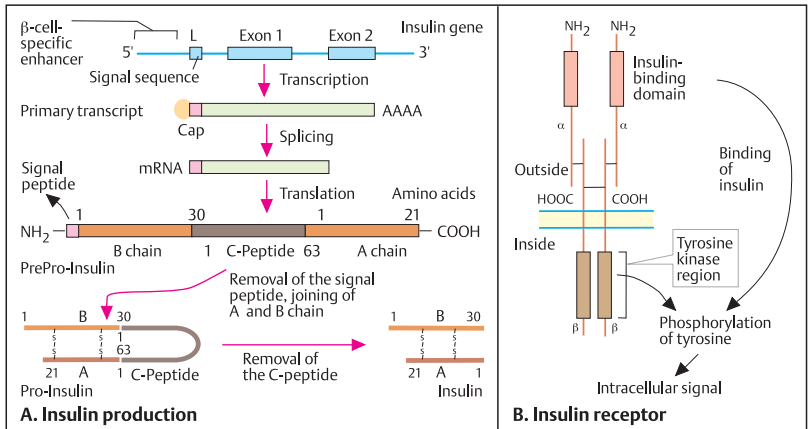
genetic factors, but also in part to overnourishment. Apart from an autosomal dominant hereditary form with onset in young adults, it is not a monogenic disorder. Monozygotic twins are concordant for type II in about 40–50% of cases and for type I in about 25%, as opposed to a risk of less than 10% for type I in first-degree relatives (about 2–7% according to family relationship and age at onset of disease). Diabetes mellitus is a secondary manifestation of a number of genetically determined diseases, e.g., insulin receptor defect (insulin resistance syndrome).

D. Influence of genes of the HLA-D region

Genetic susceptibility to diabetes type I is especially influenced by certain alleles of class I MHC genes (see p. 308). The presence of alleles DR3 and DR4, especially in DR3/DR4 heterozygotes, is associated with susceptibility to diabetes type I. DR2 confers relative resistance to diabetes. Genes conferring susceptibility to diabetes have been located at several sites of the genome.

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Protease Inhibitor

α_1 -Antitrypsin

α_1 -Antitrypsin (α_1 -AT) is an essential protease inhibitor in blood plasma. It binds to a wide range of proteases, such as elastase, trypsin, chymotrypsin, thrombin, and bacterial proteases. Its most important physiological effect is the inhibition of leukocyte elastase, a protease that breaks down the elastin of the pulmonary alveolar walls. Deficiency of α_1 -antitrypsin leads to increasing destruction of the pulmonary alveoli, obstructive emphysema of the lungs, and in newborns, a form of hepatitis.

A. α_1 -Antitrypsin

α_1 -Antitrypsin (1) in humans is a glycoprotein composed of 394 amino acids and 12% carbohydrate. It is coded for by a 10.2 kb gene with five exons on chromosome 14 (14q32.1).

B. α_1 -Antitrypsin deficiency

The uninhibited action of leukocyte elastase on the elastin of the pulmonary alveoli leads to chronic obstructive pulmonary emphysema (1). (Radiograph from N. Konietzko, Essen.) The most frequent deficiency allele is Pi(Z). The plasma concentration of α_1 -AT with genotype PiZZ (homozygote) is usually about 12–15% of normal (with the normal allele M). MZ heterozygotes have 64%, and MS heterozygotes 86% of MM homozygote activity. α_1 -Antitrypsin deficiency in the lung can be corrected by intravenous administration of α_1 -antitrypsin. (After R. H. Ingram.)

C. α_1 -Antitrypsin: protein, gene, and important mutations

The α_1 -antitrypsin protein has three oligosaccharide side chains at positions 46, 83, and 247. The protein is highly polymorphic because of differences in the amino acid sequence and in carbohydrate side chains. The reactive site is located at position 358/359 (methionine/serine). Clinically, the most important mutations affect codons 213 (PiZ), 256 (PiP), 264 (PiS), 342 (PiZ), and 357 (Pi[Pittsburgh]). The gene contains variant restriction enzyme sites, which can be used for reliable diagnosis. Today the diagnosis is often made using the PCR reaction.

D. Synthesis of α_1 -antitrypsin

The α_1 -AT gene is expressed in liver cells (hepatocytes). The gene product is channeled through the Golgi apparatus and released from the cell (secreted). The Z mutation leads to aggregation of the enzyme in the liver cells, with too little of it being secreted. The S mutation leads to premature degradation. About 2–4% of the population in Central and Northern Europe are MZ heterozygotes.

E. Reactive center of protease inhibitors

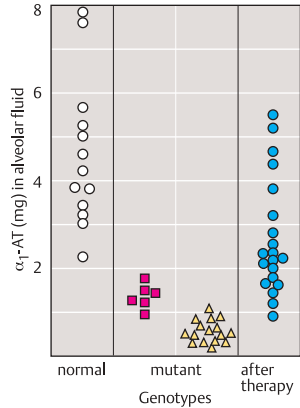
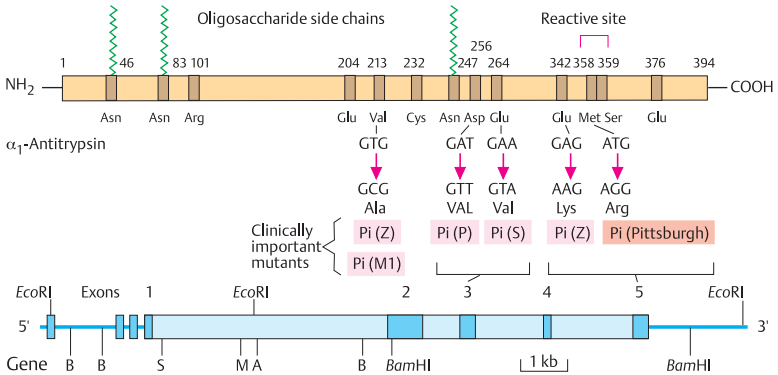
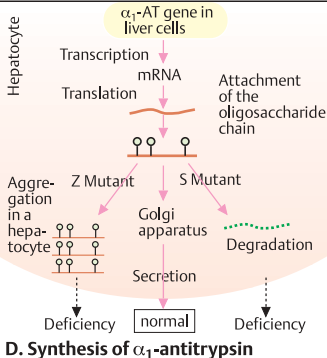
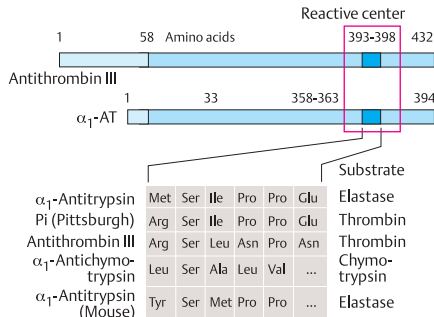
α_1 -Antitrypsin is one member of a family of protease inhibitors that show marked homology, especially at their reactive centers. Oxidizing substances have an inhibitory effect and inactivate the molecule. Smokers have a much more rapid course of α_1 -AT deficiency disease (onset of dyspnea at 35 years of age instead of 45–50).

(Figures in C–E adapted from Cox, 1995, and Owen et al., 1983).

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Function:
main
plasma protease
inhibitor,
especially of
elastase in the
bronchial system
of the lung
Gene product:
52 kDa glycoprotein
12% carbohydrate
content
Gene:
10.2 kb gene,
5 exons
gene locus 14q32.1

**A. α_1 -Antitrypsin****B. α_1 -Antitrypsin deficiency****C. α_1 -Antitrypsin: Protein, gene, and important mutants****D. Synthesis of α_1 -antitrypsin****E. Reactive center of protease inhibitors**

Blood Coagulation Factor VIII (Hemophilia A)

Hemophilia was the first major disease recognized to be genetically determined. The Talmud refers to its increased occurrence in males in certain families, corresponding with X-chromosomal inheritance. Hemophilia A results from the deficiency of blood coagulation factor VIII; hemophilia B results from a deficiency of factor IX. Factor VIII functions as a cofactor in the activation of factor X to factor Xa during the intermediate phase of the coagulation cascade.

A. X-chromosomal inheritance of hemophilia A

A classic example of the X-chromosomal inheritance of hemophilia A was seen in some royal families in Europe in the nineteenth century and the first half of the twentieth century.

B. Blood coagulation factor VIII

When activated by thrombin, factor VIII protein consists of five subunits (A1, A2, A3, C1, C2) held together by calcium ions (1). The inactive factor VIII protein (2) contains three domains (A, B, C). Domain A occurs in three homologous copies (A1, A2, A3), domain C in two (C1, C2), and domain B in one copy. In humans, the gene for factor VIII (3) maps to the distal long arm of the X chromosome in region 2, band 8 (Xq28). It consists of 26 exons and spans 186 000 base pairs (186 kb), corresponding to about 0.1% of the whole X chromosome. Noteworthy in this gene are the large exon 14 (3106 base pairs), which codes for the B domain, and a large intron of 32 000 base pairs between exons 22 and 23. Most point mutations occur in DNA sequences involving TCGA, the recognition sequence for the restriction enzyme *TaqI*. It contains the dinucleotide CG, which is easily mutated. Since the cytosine of this dinucleotide is frequently methylated and deamination of methyl cytosine leads to C-to-T transition, mutations in CG dinucleotide regions are frequent. Mutation of TCGA to TTGA creates a stop codon (TGA), resulting in a truncated factor VIII protein. Even a stop codon at position 2307 leads to severe hemophilia, although only the last 26 amino acids are missing (Gitschier et al., 1985). Polymorphic restriction sites (RFLPs, restriction fragment length polymorphisms) can be utilized

for molecular genetic diagnosis of hemophilia A (4). When present, a variant recognition sequence (B*) for the restriction enzyme *BclI* in the region of exons 17 and 18 produces a fragment of 879 base pairs and a fragment of 286 base pairs; when it is absent, a single fragment of 1165 base pairs results. This can be used in RFLP diagnosis (5): The index patient (II-1) with hemophilia A carries the 879 bp fragment. This fragment indicates the mutation. His sister (II-2) has an affected son (III-2) who also carries the 879 bp fragment, inherited from his mother. A brother (III-1) carries the 1165-bp fragment and thus is not at risk for the disease, because this is not linked to the mutation.

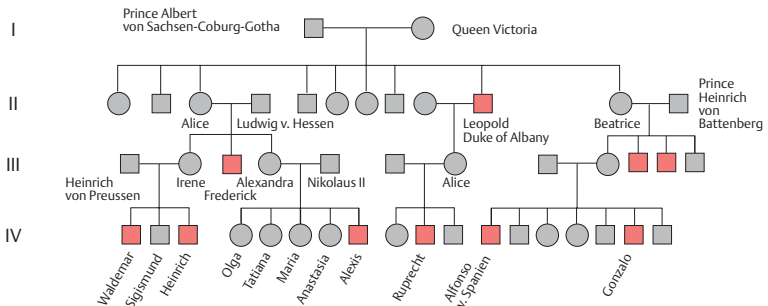
In addition to point mutations, factor VIII gene rearrangements involving the long intron 22 are frequent.

C. Severity and factor VIII activity

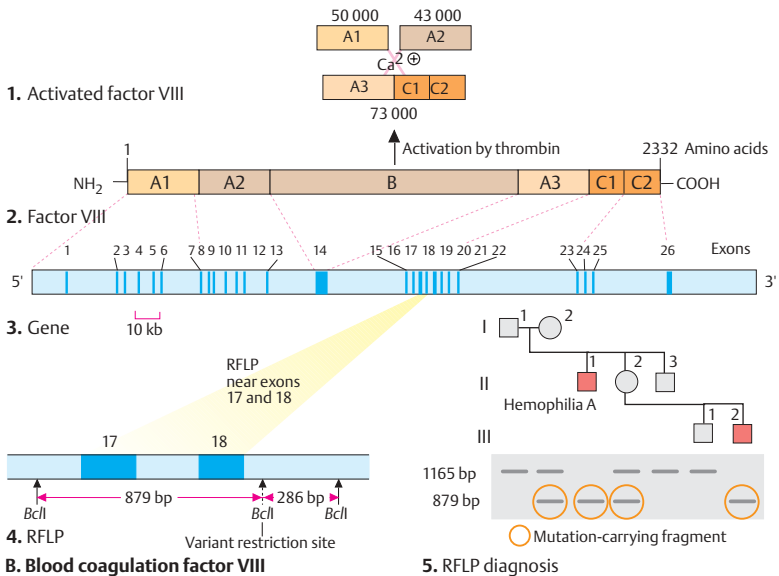
Hemophilia occurs with a frequency of about 1 in 10 000 male newborns. Severity and frequency of bleeding are dependent on the degree of residual factor VIII activity.

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A. X-Chromosomal inheritance of hemophilia A



B. Blood coagulation factor VIII

Factor VIII activity			
Hemophilia A	under 2%	2 – 10%	10 – 30%
Severity	Spontaneous bleeding into joints, muscle, internal organs	Bleeding after light trauma, sometimes spontaneously	Bleeding after trauma
Proportion of patients	48%	31%	21%

C. Severity of hemophilia A and factor VIII activity

Von Willebrand Factors

Von Willebrand factor (vWF) is a complex multimeric protein found in plasma, platelets, and subendothelial connective tissue. It has two basic biological functions: it binds to specific receptors on the surface of platelets and subendothelial connective tissue, and it forms bridges between platelets and damaged regions of a vessel. Furthermore, it binds to clotting factor VIII and stabilizes it. Deficiency of vWF leads to decreased or absent platelet adhesion and to secondary deficiency of factor VIII (von Willebrand disease or von Willebrand–Jürgens syndrome). Hereditary deficiency of vWF is the most common bleeding disorder in man, with a frequency of about 1 : 250 for all forms, including the mild ones, and about 1 : 8000 for severe forms.

A. Von Willebrand cDNA and prepropeptide

Von Willebrand factor is formed in endothelial cells, in megakaryocytes, and possibly in some other tissues and is coded for by a large (178 kb) gene with 52 exons of various sizes on chromosome 12 (12p12-pter). Several polymorphic restriction sites (red arrows) exist. The cDNA of vWF is about 8.7 kb long. The corresponding mRNA codes for a primary peptide (prepro-vWF) of 2813 amino acids, including a signal peptide of 22 amino acids, a segment of 741 amino acids (vW antigen II), and a subunit of four different domains (A–D), which together make up more than 90% of the sequence. The three A domains (A1–A3) in the middle contain binding sites for collagen, heparin, and thrombocytes. Three small B domains are on the carboxy side of the D4 domain, before the two C domains. vWF contains 8.3% cysteine (234 of 2813 amino acids), concentrated at the amino and carboxy ends, whereas the three A domains are cysteine-poor. After posttranslational modification, the mature plasma vWF contains 12 oligosaccharide side chains (19% of the total weight is carbohydrate).

B. Biosynthesis of the von Willebrand factors (vWF)

vWF is first formed as a prepropeptide. After the signal peptide is removed, two pro-vWF units attach to each other at their carboxy ends by means of numerous disulfide bridges to form a

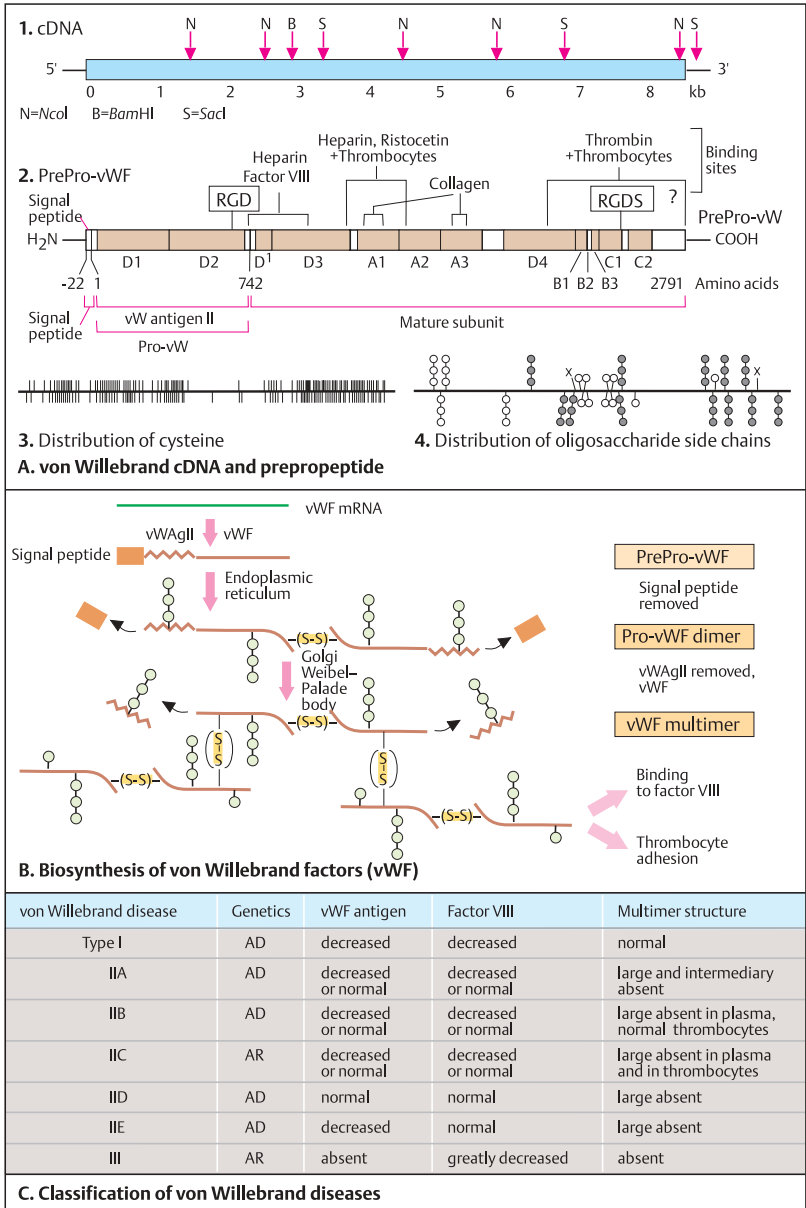
dimer. The dimers represent the repetitive units, or protomers, of mature vWF. The pro-vWF dimers are transported to the Golgi apparatus, where the pro-vWF (vW antigen II or vWagII) is removed. Mature vWF and vWagII are stored in Weibel–Palade bodies in epithelial cells. The mature subunits and vWagII contain binding sites for factor VIII, heparin, collagen, ristocetin + platelets, and thrombin-activated platelets.

C. Classification of von Willebrand diseases

Von Willebrand disease is a heterogeneous group of disorders divided into several subtypes. In types I and III, the defect is quantitative; in type II, qualitative. Dominant and recessive phenotypes with vWF deficiency often cannot be readily distinguished because heterozygosity may not be manifest and can only be determined by laboratory tests. Type I with subtypes A and B is the most frequent group (70% of all patients). vWF deficiency may simulate platelet dysfunction or hemophilia. (Figures adapted from Sadler, 1995).

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Cytochrome P450 Genes

Complex chemical substances, such as drugs or plant toxins, are degraded by an oxidation system (monooxygenases) in the endoplasmic reticulum of liver cells. These enzymes (collectively referred to as cytochrome P450) absorb light maximally at 450 nm after binding to CO. Cytochrome P450 is the last enzyme in the essential electron-transporting chain in microsomes of the liver and mitochondria of the adrenal cortex. A large system of evolutionarily related genes code for the different P450 proteins in mammals.

A. Cytochrome P450 system

The cytochrome P450 system (1) consists of oxidizing enzymes (mixed monooxygenases). They represent the first phase of detoxification: a substrate (RH) is oxidized to ROH utilizing atmospheric oxygen (O_2), with water (H_2O) being formed as a byproduct. A reductase delivers hydrogen ions (H^+) either from NADPH or NADH. A characteristic feature of P450 enzymes (2) is that a single chemical substrate can frequently be degraded by several P450 enzymes and that a single P450 protein can oxidize a number of structurally different chemical substances. The capacity to metabolize and detoxify a wide range of chemical substances is considerable. However, the enzyme activities of phase I and phase II must be well coordinated, since toxic intermediates with undesirable side effects occasionally arise in the initial stages of phase II.

B. Debrisoquin metabolism

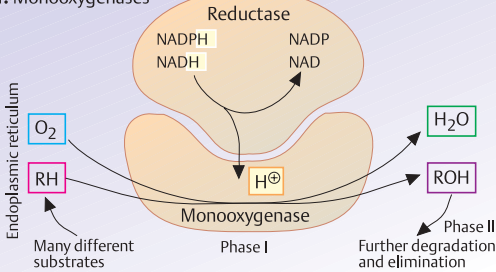
Debrisoquin is an isoquinoline-carboxamidine. It was used to treat high blood pressure until it was found to cause severe side effects in 5–10% of the population. These persons have reduced activity of a degrading enzyme, debrisoquin-4-hydroxylase. A number of other medications, including β -adrenergic blockers, antiarrhythmics, and antidepressives, are also degraded by this enzyme and may also cause untoward reactions in persons with low activity. Individuals with a slow rate of degradation show an increased ratio of debrisoquin/4-hydrodebrisoquin (1). The enzyme is coded for by the *450-db1* gene, a member of the cytochrome P450-IID family (CYP2D). Mutations may cause aberrant splicing and produce a variant pre-mRNA containing an additional intron (Gonzalez et al., 1988).

C. CYP gene superfamily (cytochrome P450 genes)

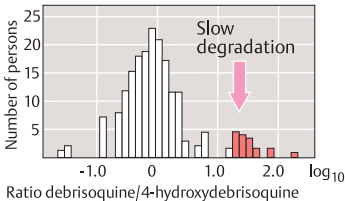
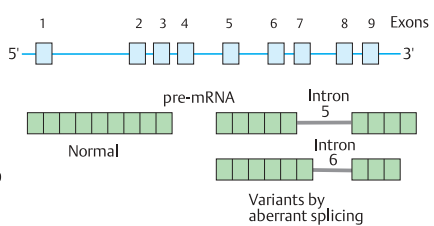
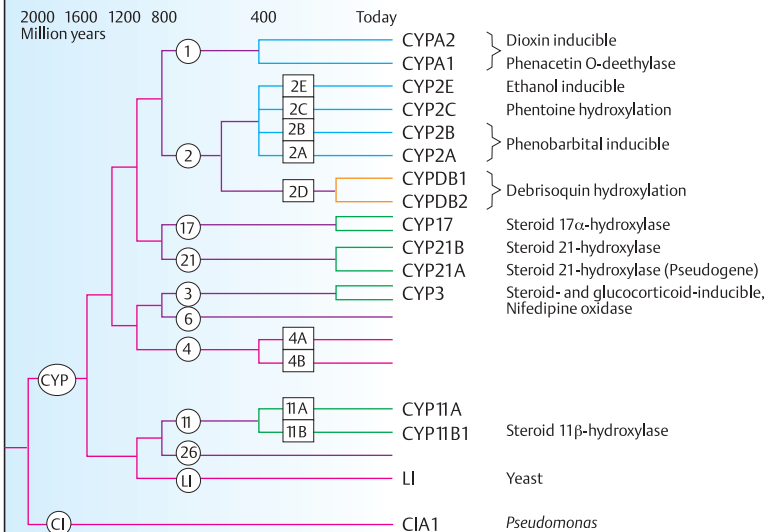
The cytochrome P450 genes in mammals are designated CYP genes. They make up a superfamily of genes that resemble each other in exon/intron structure and that code for similar gene products. An evolutionary pedigree has been derived based on comparisons of their cDNA sequences. According to this pedigree, the CYP gene family arose during the last 1500–2000 million years. It is assumed that the CYP-2 family in particular developed in response to toxic substances in plants that had to be detoxified by animal organisms. At least 30 gene duplications and gene conversions have led to an unusually diverse repertoire of CYP genes. (Figures adapted from Gonzales et al, 1988, and Gonzales & Nebert, 1990).

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1. Monooxygenases**2. P450 enzymes**

Aliphatic oxidation
Aromatic hydroxylation
N-Dealkylation
O-Dealkylation
S-Dealkylation
Oxidative deamination
Sulfoxide formation
N-Oxidation
N-Hydroxylation
Oxidative dehalogenation
Reductive dehalogenation

A. Cytochrome P450 system**1. Debrisoquin polymorphism****2. P450 *db1* gene (CYP2D6)****B. Debrisoquin metabolism****C. CYP gene superfamily (cytochrome P450 genes)**

Pharmacogenetics

Many medications are degraded at different rates in different individuals. This has a genetic basis. Enzymes coded for by genes with different alleles may have different catabolic rates, which in turn can result in genetically determined differences in the reaction to drugs (pharmacogenetics).

A. Malignant hyperthermia due to abnormal regulation of a calcium channel in muscle cells

Malignant hyperthermia is a severe, life-threatening complication of anesthesia that may occur in persons with extreme hypersensitivity to halothane and similar agents used in general anesthesia. Normally, a nerve impulse depolarizes the plasma membrane of a nerve ending at the nerve-muscle endplate (1) (motor endplate), and the volt-gated calcium channel in the plasma membrane of the nerve ending is temporarily opened. The massive influx of calcium into the cell (the extracellular Ca^{2+} concentration is about 1000 times higher than the intracellular) triggers the release of acetylcholine. Binding of the latter to the acetylcholine receptor of the muscle cell temporarily opens the receptor-controlled cation (Na^+) channels. This opens calcium channels located in the sarcoplasmic reticulum of the muscle cell. The resulting rapid increase in Ca^{2+} concentration in the cytosol causes the myofibrils in the muscle cell to contract. The calcium channels in the sarcoplasmic reticulum are regulated by a receptor (ryanodin receptor) (2). Ryanodin (an alkaloid) binds to the calcium channel. The ryanodin receptor is a protein with four transmembrane domains. Mutations in the ryanodin receptor lead to greatly increased sensitivity to halothane and other anesthetic agents (3), which cause muscle spasm, drastic elevation of temperature (hyperthermia), acidosis, and cardiac arrest (4). Malignant hyperthermia is inherited as an autosomal dominant trait (5). One gene in man lies on chromosome 19 at 19q13.1 (MacLennan and Phillips, 1992). Additional loci are on 7q, 17q, and 3q13.1 (Subrak et al., 1995). The mutant haplotype of a given family can be determined by segregation analysis. A ryanodin receptor mutation has been demonstrated in porcine malignant hyperthermia.

B. Serum pseudocholinesterase deficiency (butyrylcholinesterase)

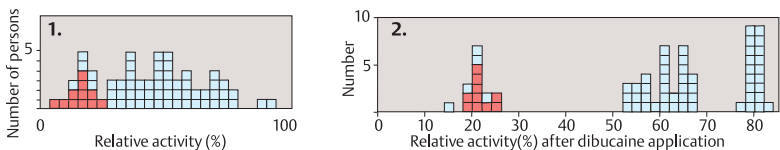
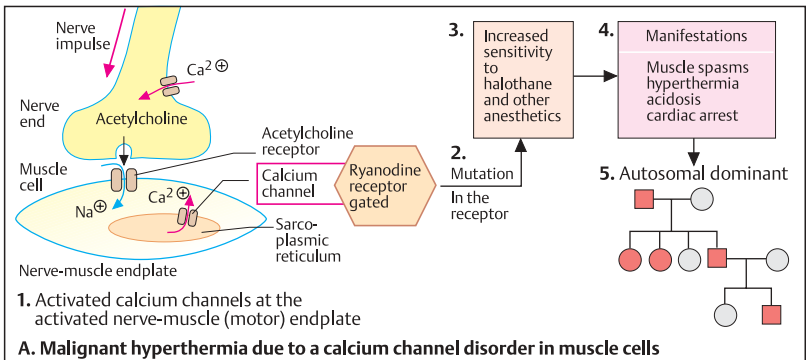
About 1 in 200 individuals reacts to muscle relaxants, such as suxamethonium (succinylcholine), with prolonged muscle relaxation and respiratory arrest. In such persons, serum pseudocholinesterase activity is decreased. Persons at risk cannot be identified by determining their pseudocholinesterase activity alone (1), but by determining dibucaine inhibition of their enzyme activity. Whereas homozygous normal persons show 80% enzyme activity after dibucaine administration, persons at risk show only 20%. Individuals with intermediate values of 60% are regarded as heterozygotes (2). A number of different alleles can lead to different degrees of reduced enzyme activity. (Figure after Harris, 1975.) This enzyme is now referred to as butyrylcholinesterase because it hydrolyzes butyrylcholine more readily than acetylcholine.

C. Examples of genetically determined adverse reactions to drugs

See table opposite. (Note: Hemoglobin H involves deletions of the α loci, not a loci.)

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B. Serum pseudocholinesterase deficiency

Defect	Relevant chemical substance	Clinical consequence	Frequency	Pathogenesis	Genetics
Coumarin resistance	Coumarin (warfarin)	Ineffective anticoagulation therapy	Rarer than 1:80 000	Increased vitamin K affinity due to enzyme or receptor defect	Autosomal dominant
Increased sensitivity to isoniazid	Isoniazid, Sulfamethazine, Phenelzine, Hydralazine, etc.	Polyneuritis, lupus-like reaction	In about 50%	Decreased activity of liver Isoniazid acetylase	Autosomal recessive
Isoniazid ineffective	Isoniazid, Sulfamethazine, Phenelzine, Hydralazine	Reduced antituberculous effect		Increased INH excretion	Autosomal dominant
Glucose-6-phosphate dehydrogenase (G6PD) deficiency	Sulfonamides, Antimalarial drugs, Nitrofurantoin, Vicia faba	Hemolysis	Rare in Europeans, frequent in Africa	G6PD deficiency in erythrocytes	X-chromosomal (many mutant forms)
Hemoglobin Zürich	Sulfonamides	Hemolysis	Rare	Unstable hemoglobin due to point mutation in β -globin (arginine instead of histidine in position 63)	Autosomal dominant
Hemoglobin H	Sulfonamides	Hemolysis	Rare	Unstable hemoglobin of 4 β chains due to deletion of the α loci	Autosomal dominant
Glaucoma in adults (some forms)	Corticoids	Glaucoma	Frequent	Unknown	Possibly autosomal dominant

C. Examples of genetically determined reactions to pharmaceuticals

Maintaining Cell and Tissue Shape

Cytoskeletal Proteins in Erythrocytes

Eukaryotic cells contain supportive proteins that confer stability while allowing flexibility (cytoskeletal proteins). The membrane skeleton is a network of structural proteins underlying the plasma membrane and partly associated with it. Erythrocytes must meet extreme requirements: about a half million times during a 4-month lifespan, they traverse small capillaries with diameters less than that of the erythrocytes themselves. Membrane flexibility is also essential for muscle cell function. Thus, it is not surprising that the cytoskeletal proteins of erythrocytes and muscle cells are similar.

A. Erythrocytes

A normal erythrocyte is maintained in a characteristic biconcave discoid form by the cytoskeletal proteins. Genetic defects in different cytoskeletal proteins lead to characteristic erythrocyte deformations: as ellipses (elliptocytes), as spheres (spherocytes), or as cells with a mouthlike area (stomatocytes) or thornlike projections (acanthocytes). The various forms are the result of defects of different proteins. (Scanning electron micrographs from Davies and Lux, 1989.)

B. Skeletal proteins in erythrocytes

SDS polyacrylamide gel electrophoresis differentiates numerous membrane-associated erythrocyte proteins. Each band of the gel is numbered, and the individual proteins are assigned to them. The main proteins include α - and β -spectrin, ankyrin, an anion-channel protein (band-3 protein), proteins 4.1 and 4.2, actin, and others. The chromosomal localization of their genes and associated diseases due to mutations are known for man and mouse.

C. α - and β -Spectrin

The main component of cytoskeletal proteins is spectrin, a long protein composed of a 260 kDa α chain and a 225 kDa β chain. The chains consist of 20 (α chain) and 18 (β chain) subunits, each with 106 amino acids. Each subunit is

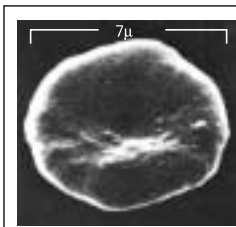
composed of three α -helical protein strands running counter to one another. Subunit 10 and subunit 20 of the α chain consist of five, instead of three, parallel chains. The individual subunits are assigned to different domains (I–V in the α chain and I–IV in the β chain).

D. Proteins of the erythrocyte membrane

The rod-shaped spectrin proteins, which run parallel to the erythrocyte plasma membrane, are attached to the anion channels by ankyrin and to the glycophorin molecules by protein 4.1. The anion channels in erythrocytes are important for CO₂ transport. Glycophorins (A, B, C) are transmembrane proteins with several carbohydrate units. Actin is the main protein for muscle contraction and cell flexibility. (Figures after Luna and Hitt, 1992).

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Normal erythrocyte



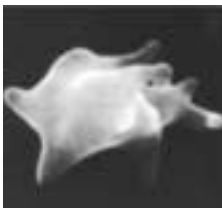
Elliptocyte



Spherocyte



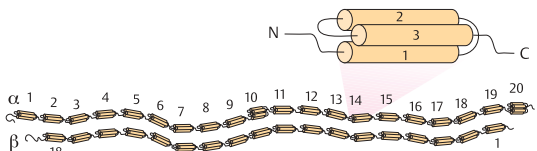
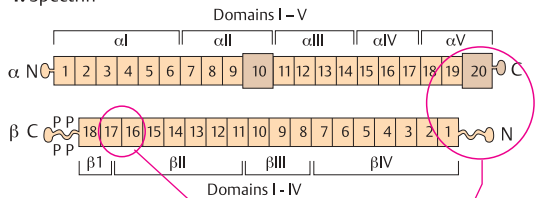
Stomatocyte



Acanthocyte

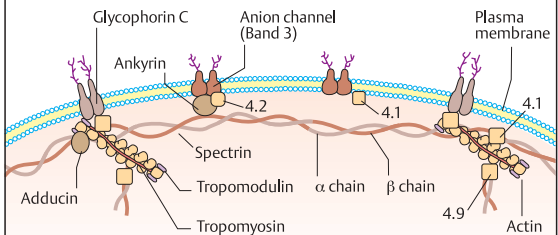
A. Erythrocytes

Band	SDS gel	Protein	Chromosomal localization	Disease
1		α -Spectrin	1q22 – 25	Elliptocytosis-1
2		β -Spectrin	14q23 – 24	Spherocytosis-2
		Ankyrin	8p11 – 21	Spherocytosis-1
3		Anion channel	17	Acanthocytosis
4.1		Protein 4.1	1q22 – 25	Elliptocytosis-2
4.2		Protein 4.2		
5		Actin	7pter – q22	
6		Glycerol-aldehyde-3-P dehydrogenase	12p13	
7		Tropomyosin (non-muscle)	1q31 – 41	Stomatocytosis

B. Erythrocyte skeleton proteins**1. Spectrin****2. Spectrin subunits**

Ankyrin binding site

Actin- and protein 4.1-binding sites

C. α - and β -spectrin**D. Proteins in the erythrocyte membrane**

Hereditary Muscle Diseases

Spontaneous degeneration of muscle fibers and death of muscle cells (muscular dystrophy) is a common cause of muscle disease in infants, children, and adults. Muscular dystrophies are genetically heterogeneous and clinically variable. About 50 different forms are listed in McKusicks's catalogue Mendelian Inheritance of Man Online (OMIM at <http://www3.ncbi.nlm.nih.gov/Omim/>).

A. The dystrophin-glycan complex

A complex system of interconnected non-covalently bound proteins in the sarcolemma (plasma membrane) of muscle cells lends the cell stability under the extreme exertion of contraction and relaxation. They connect the extracellular matrix and the intracellular myofibrils, elongated protein molecules aligned in parallel chains (myofilaments). The largest of the interconnected proteins, α -dystroglycan (156 kDa), is located outside the cell. It is connected to the extracellular matrix by a heterotrimeric protein, laminin-2. β -Dystroglycan (43 kDa) is embedded in the sarcolemma and connected to a series of other cytoskeletal proteins, which are divided into the sarcoglycan and syntrophin subcomplexes. Several members of the sarcoglycan complex are related to specific types of muscular dystrophies due to mutations in the corresponding genes.

Dystrophin, a large, elongated protein, provides a bridge between the intracellular cytoskeleton involved in the contractile myofilaments and the extracellular matrix. Two dystrophin molecules connect neighboring dystrophin-glycan complexes. The N-terminal end of dystrophin is connected to the thin myofilament F-actin (filamentous actin). The C-terminal end of dystrophin is connected to β -dystroglycan and the syntrophins. (This highly schematic illustration is adapted from Pasternak, 1998; Duggan et al., 1997; and Nova-Castra Immunohistochemistry Laboratories, Newcastle upon Tyne, UK.)

B. Model of the dystrophin molecule

Dystrophin, the largest member of the spectrin superfamily, is composed of 3685 amino acids (molecular mass 427 kDa) which form four functional domains: (1) the N-terminal actin-binding domain of 336 amino acids; (2) 24 long repeating units, each consisting of 88- to 126-

amino-acid triple-helix segments as in spectrin; (3) a 135-amino-acid cysteine-rich domain, which binds to the sarcolemma proteins; and (4) the C-terminal domain of 320 amino acids with binding sites to syntrophin and dystrobrevin. The triple helix segments form the central rod domain, which is 100–125 nm long. (Diagram adapted from Koenig et al., 1988).

C. The dystrophin gene

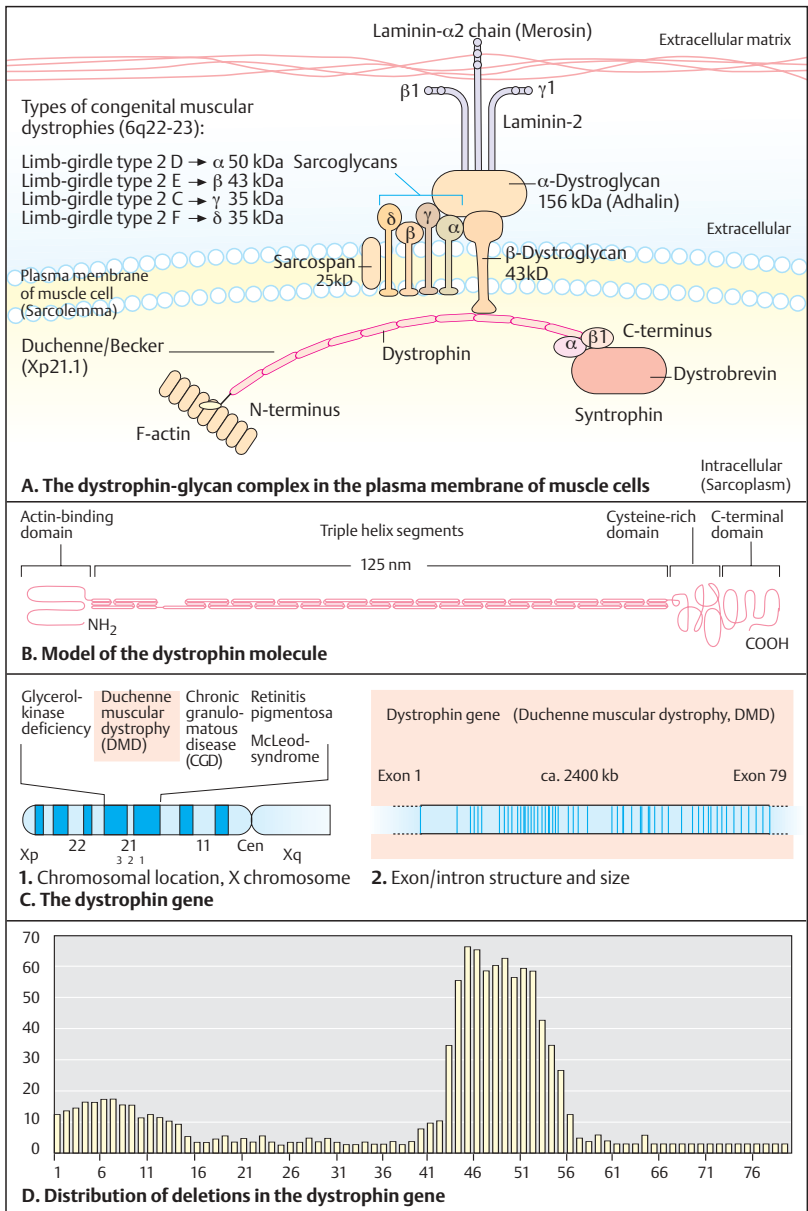
The human *dystrophin* gene (*DMD*) is located on the short arm of the X chromosome in region 2, band 1.1 (Xp21.1). *Dystrophin* is by far the largest known gene in man, spanning 2.4 million base pairs (2.4 Mb or 2400 kb) in 79 exons (2). The large *DMD* transcript has 14 kb. The dystrophin gene contains at least seven intragenic promoters. The primary transcript is alternatively spliced into a variety of different mRNAs that encode smaller proteins expressed in other tissues than muscle cells, especially in the central nervous system.

D. Distributions of deletions in the dystrophin gene

The frequent deletions in the *DMD* gene (60% of patients) are unevenly distributed. Most frequently involved are exons 43–55 and exons 1–15, roughly corresponding to the F-actin binding site and the dystroglycan-binding site. Duplications (in 6% of patients) and point mutations also occur. (Figure kindly provided by Professor C. R. Müller-Reible, Würzburg).

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Duchenne Muscular Dystrophy

Duchenne muscular dystrophy (DMD, McKusick 310200) is the most common of the more than 10 clinically and genetically distinct muscular dystrophies. It is caused by mutations in the *DMD* gene. It occurs in 1 of 3500 live born males either by a new mutation or by transmission of the mutation from a heterozygous mother or a mother with germline mosaicism (i.e., the mother carries a *DMD* mutation in a variable proportion of her germ cells). The mutation rate is high, probably because the gene is unusually large and has a high rate (10%) of recombination within it. The French neurologist Guillaume Duchenne (1806–1875) was the first to report this disease, in 1861.

A clinically milder variant, Becker muscular dystrophy (BMD), is an allelic disorder due to in-frame mutations in the same gene that allow residual function of the dystrophin protein.

A. Clinical signs

DMD is the most distinctive progressive proximal muscular dystrophy. The age of onset is usually less than 3 years and signs are evident at 4–5 years; the patient requires a wheelchair by 12 years and usually succumbs to the disorder by age 20 years. Progressive muscular weakness of the hips, thighs, and back cause difficulties in walking and using steps. Lumbar lordosis and enlarged but weak calves (pseudohypertrophy) are visible (1). The affected child performs a characteristic series of maneuvers to rise from a kneeling position (Gower's sign, 2). (Drawings Duchenne, 1861, and Gowers, 1879, from Emery, 1987).

In the Becker type, the age of onset is later and the disease progresses much more slowly than in DMD.

B. Dystrophin analysis in muscle cells

Dystrophin, which normally lies along the plasma membrane (sarcolemma) of muscle cells (1), is absent in patients (2). Female heterozygotes show a patchy distribution of groups of normal and defective muscle cells (3) as a result of X inactivation (see p. 228). (Photographs kindly provided by Dr. R. Gold, Department of Neurology, University of Würzburg).

C. Investigation of a family with DMD

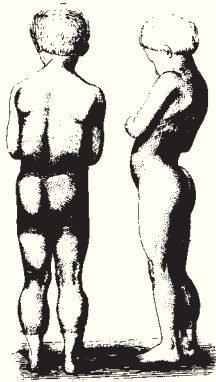
Deletions occur in certain regions in 55% of cases and duplications in 5%. However, point mutations (in 40%) are not always detectable. In this situation, indirect DNA analysis can be performed. The panel shows a simplified example of a two-allele system (marker DXS7). (Data kindly provided by Dr. C. R. Müller-Reible, Institute of Human Genetics, University of Würzburg). Since the affected patients III-1 and III-2 carry the marker allele 1 at the *DMD* locus, allele 1 indicates the presence of the mutation. The unaffected male II-4 confirms that allele 2 does not represent the mutation. The females I-2, II-1, and II-2 are obligate heterozygotes (2–1). In this example the males III-3 and III-4 are not affected. This can be explained by recombination in their mother, II-5. In current practice one uses a set of several linked markers flanking the disease locus to avoid an erroneous diagnosis due to recombination. Female heterozygotes show mild clinical signs in 2–3%. About 23% of mothers of isolated patients are noncarriers.

D. Other forms of muscular dystrophy

Several other forms of genetically determined muscular dystrophy are known in man. Course, diagnosis, and molecular genetic analysis depend on the basic disorder. Selected examples are listed.

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1. Calf hypertrophy and lordosis



2. Difficulty in rising (Gower's sign)



1. Normal dystrophin

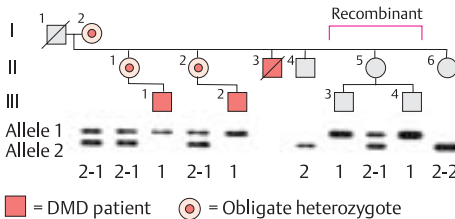


2. Dystrophin absent



3. Areas lacking dystrophin in heterozygotes

A. Clinical signs of Duchenne muscular dystrophy



C. Investigation of a family with DMD by DNA markers

B. Dystrophin analysis in muscle cells

Disease	Chromosomal location	McKusick No.
X-chromosomal:		
Muscular dystrophy Duchenne	Xp21.2	310200
Muscular dystrophy Becker (allelic with DMD)	Xp21.2	310200
Muscular dystrophy Emery–Dreifuss	Xq28	310300
Autosomal dominant:		
Myotonic dystrophy	19q13	160900
Facioscapulo-humeral dystrophy	4q35 – qter	158900
Oculo-pharyngeal muscular dystrophy	Unknown	164300
Autosomal recessive:		
Duchenne-like muscular dystrophy	13q12 – 13	253700
Congenital muscular dystrophy-type Fukuyama	9q31 – 33	253800
Limb-girdle muscular dystrophy (several types)	15q15 – q22, other loci	253600

D. Important forms of hereditary muscular dystrophy in man